TRANSDUCER ENGINEERING

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PREFACE

This textbook has been written as per the latest syllabus of Anna University to meet the requirements for the syllabus of B.E., E.I.E., and I.C.E.

The primary aim of this book is to acquaint the students with the basic principles of Sensors and Transducer systems and their applications for the measurement of various variables.

To illustrate the concepts, a large number of diagrams have been provided in this book.

This book uses a very simple everyday language to explain the subject and it will be very useful not only to the students but also to the teachers.

We are very much grateful to our beloved Principal Dr.K.Arulmozhi, Ph.D., Kamaraj College of Engineering and Technology, Virudhunagar, who have been a constant source of inspiration and guidance to all our efforts.

We wish to express our profound thanks to Mr. M. Sethuraaman, M/s. Anuradha Publications, the most leading technical book publisher for publishing this book in such a short span of time with great enthusiasm and effort.

We are indebted to Mr. J. Gnanavadivel, M.E., Mepco Schlenk Engineering College, Sivakasi, for his timely help that motivated and encouraged us to write this book.

Our sincere thanks to our family members for much needed moral support and encouragement provided by them.

Any comments and suggestions for this book will be thankfully acknowledged and incorporated in the next edition.

Authors

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UNIT - I

Science of Measurements and Instrumentation of Transducers

1.1 INTRODUCTION

The study of any subject matter in engineering should be motivated by an appreciation of the uses to which the material might be put in the every day practice of the profession. Measurement systems are used for many detailed purposes in a wide variety of application areas. The easiest way to assess the amount of use of science and technology is to examine the number of measurements that are being made and how they are being used.

All the successful achievements in science and technology are entirely due to the ability to measure the state, condition or characteristics of the physical systems, in quantitative terms with sufficient accuracy.

Lord Kelvin stressed the importance of measurement in this context, by saying: "When you can measure what you are speaking about, and express it in numbers, you know something about it".

1.2 MEASUREMENT

The measurement is usually undertaken to ascertain and present the state, condition or characteristic of a system in quantitative terms. To reveal the performance of a physical or chemical system, the first operation carried out on it is measurement. The process or the act of measurement consists of obtaining a quantitative comparison between a pre defined standard and a measurand. The word measurand is used to designate the particular physical parameter being observed and quantified that is, the input quantity to the measuring process.

Measurements are generally made

- to understand an event or an operation.

- to monitor an event or an operation.
- to control an event or an operation.
- to collect data for future analysis and
- to validate an engineering design.

Fig. 1.1 shows the fundamental measuring process

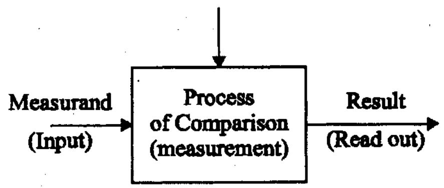


Fig. 1.1 Fundamental measuring process

1.2.1 Fundamental methods of measurement

There are two basic methods of measurement

1. Direct comparison with either a primary or a secondary standard.
2. Indirect comparison through the use of a calibrated system.

Direct comparison

To measure the length of a bar, we compare the length of the bar with a standard, and find that the bar is so many inches long because that many inch-units on the standard has the same length as the bar. Thus we have determined the length by direct comparison. The standard that we have used is called a secondary standard. Measurement by direct comparison is less common than the measurement by indirect comparison.

Indirect comparison

Indirect comparison makes use of some form of transducing device. This device converts the basic form of input in to an analogous form, which it then processes and presents at the output as a known function of the input.

1.2.2 Functional elements of a measurement system

Fig. (1.2) shows the functional elements of an instrument or a measurement system.

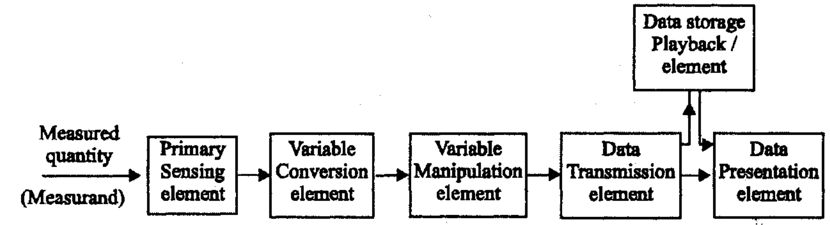


Fig. 1.2 functional elements of an instrument or a measurement system.

(i) Primary sensing element

The primary sensing element is the one which first receives energy from the measured medium and produces an output depending in some way on the measured quantity (measurand).

(ii) Variable conversion element

The output signal of the primary sensing element is some physical variable, such as displacement or voltage. For the instrument to perform the desired function, it may be necessary to convert this variable to another more suitable variable while preserving the information content of the original signal. An element that performs such a function is called a variable conversion element.

(iii) Variable manipulation element

The element that performs "manipulation" by which the numerical value of the variable is changed according to some definite rule but the physical nature of the variable is preserved is called a variable-manipulation element.

(iv) Data-transmission element

When the functional elements of an instrument are actually physically separated, it becomes necessary to transmit the data from one to another. An element performing this function is called a data-transmission element.

(v) Data-presentation element

If the information about the measured quantity is to be communicated to a human being for monitoring, control, or analysis purposes, it must be put in to a form recognizable by one of the human senses. An element that performs this "translation" function is called a data-presentation element. This function includes the simple indication of a pointer moving over a scale and the recording of a pen moving over a chart.

(vi) Data storage/playback element

Although data storage in the form of pen/ink recording is often employed, some applications require a distinct data storage/play back function which can easily recreate the stored data upon command. The magnetic tape recorder/reproducer is the example.

Example for measurement system

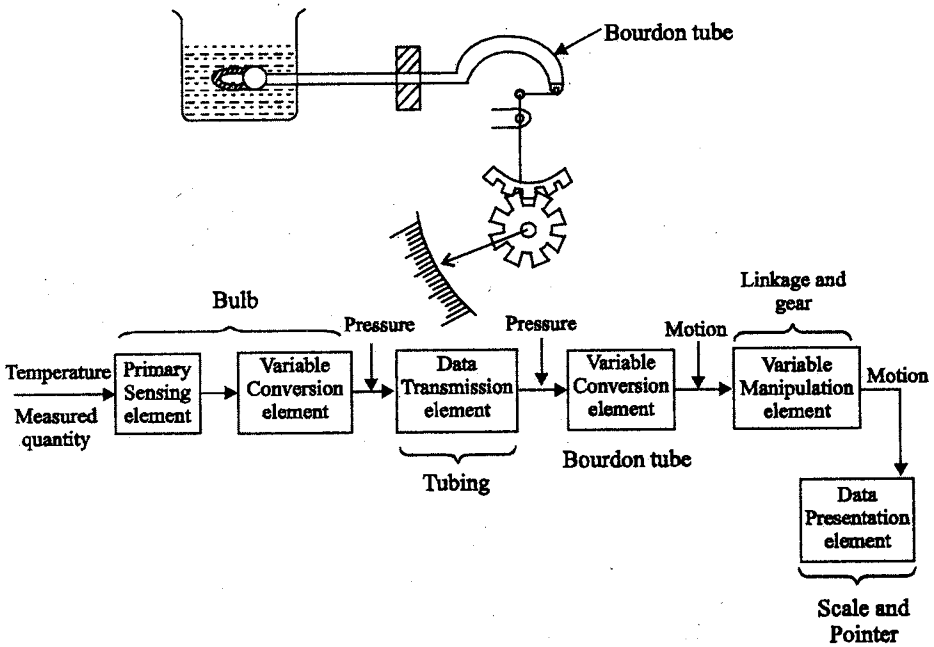


Fig. 1.3 Pressure thermometer

As an example of the above concepts, consider a pressure type thermometer [see fig (1.3)]. The liquid-filled bulb acts as a primary sensor and variable-conversion element since a temperature change results in a pressure build up with in the bulb, because of the constrained thermal expansion of the filled fluid.

This pressure is transmitted through the tube to a Bourdon-type pressure gauge, which converts pressure to displacement.

This displacement is manipulated by the linkage and gearing to give a larger pointer motion. A scale and pointer again serve for data presentation.

1.3 STANDARDS, DIMENSIONS AND UNITS OF MEASUREMENT

- The term "dimension" connotes the defining characteristics of an entity.
- The "unit" is a basis for quantification of the entity.

For example, length is a dimension where as centimeter is a unit of length, time is a dimension and the second is a unit of time.

1.3.1 Units and standards

For the past years, a considerable number of systems of Units have been used at various time periods. However, there are some systems of units which have been accepted through out the world.

Unit

We measure a physical quantity by the measurement system. The result of the measurement of the physical quantity must be defined both in kind and magnitude. The standard measure of each kind of physical quantity is called a "Unit". In general, we can write:

$$\text{Magnitude of a physical quantity} = (\text{Numerical ratio}) \times (\text{Unit}) \qquad (1.1)$$

The Numerical Ratio is the number of times the unit occurs in any given amount of the same quantity and therefore, is called the number of measures. This may be otherwise called a numerical multiplier.

For e.g., if we measure a distance of 10 metre, its magnitude may be,

$$\text{Distance} = (10) \times (\text{m})$$

- Here metre (m) is the unit of length and
- 10 is the number of units in the length.
- The physical quantity, distance, in this case is defined by the unit, metre.
- Without unit, the numerical ratio has no physical meaning.

Types of Units

- Fundamental units
- Derived units

Units which are fundamental to most other physical quantities are called fundamental units.

Fundamental units are measures of length, mass and time. Since length, mass and time are fundamental to most other physical quantities, they are called the "Primary Fundamental Units".

Measures of certain physical quantities in the thermal, electrical, illumination fields are also represented by fundamental units. These units are used only where these particular disciplines are involved and therefore they are called Auxiliary Fundamental Units.

All other units which can be expressed in terms of fundamental units with the help of physical equations are called Derived Units. Every derived unit originates from some physical law or equation which defines that unit. For e.g., the area, A, of a room is equal to the product of its length l, and breadth, b. Therefore, $A = l \times b$.

If metre is chosen as the unit of length, then the area of a room 6m × 4m is 24 m^2. Note that the number of measures $(6 \times 4 = 24)$ as well as the units $(m \times m = m^2)$ are multiplied. The derived unit of area is m^2.

1.3.2 Dimensions

Every quantity has a quality which distinguishes it from all other quantities. This unique quality is called Dimension. The dimension is written in a characteristics notation, For eg., $[L]$ for length, $[T]$ for time etc.

A derived unit is always recognized by its Dimensions, which can be defined as the complete algebraic formula for the derived unit. Thus when quantity such as area A of a rectangle is measured in terms of other quantities (i.e) length, l and breadth, b then the relationship is expressed as,

$$\text{Area, } A = \text{ a constant } \times l \times b \rightarrow \tag{1.2}$$

Since l and b each have the dimensions of a length, $[L]$, the dimensions of area are

$$[A] = [L]\,[L] = [L^2].$$

- Since the constant is a pure numerical ratio and is, therefore, dimensionless.
- The three fundamental units are length, mass and time. Their dimensions are: Length $= [L]$; Mass $= [M]$; Time $= [T]$

Dimension of Mechanical Quantities

All mechanical quantities can be expressed in terms of the three fundamental quantities like length, mass and time.

1.	Velocity $= \dfrac{\text{length}}{\text{time}}$	$[v] = \dfrac{[L]}{[T]} = [LT^{-1}]$
2.	Acceleration $= \dfrac{\text{velocity}}{\text{time}}$	$[a] = \dfrac{[LT^{-1}]}{[T]} = [LT^{-2}]$
3.	Force = mass × acceleration	$F = [M]\,[LT^{-2}] = [MLT^{-2}]$
4.	Work = force × distance	$[w] = [MLT^{-2}]\,[L] = [ML^2T^{-2}]$
5.	Power $= \dfrac{\text{work}}{\text{time}}$	$[P] = \dfrac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$
6.	Energy = power × time	$[ML^2T^{-3}\,[T] = [= ML^2T^{-2}]$
7.	Momentum = mass × velocity	$= [M]\,[ML^{-1}] = [MLT^{-1}]$
8.	Torque = force × distance	$= [MLT^{-2}]\,[L] = [ML^2T^{-2}]$
9.	Stiffness $= \dfrac{\text{torque}}{\text{angle}}$	$[K] = [ML^2T^{-2}]$
10.	Surface Tension $= \dfrac{\text{force}}{\text{length}}$	$[\sigma] = \dfrac{[MLT^{-2}]}{[L]} = [MT^{-2}]$

Table 1.1 Dimension of mechanical quantities

1.3.3 System of Units

A number of systems of units are in use since 16th century. The important systems of units are

1. FPS system (foot, pound, second)
2. CGS system (centimeter, gram, second)
3. MKS system (meter, kilogram, second)
4. Rationalised MKSA system (meter, kilogram, second, ampere)
5. SI system (six fundamental units, two supplementary units and twenty seven derived units)

1. CGS system of units

The most commonly used units in electrical work were CGS units. These units involve the use of unit of a fourth quantity in addition to units of mass, length and time. Two systems of CGS units are

(i) Electromagnetic Units (e.m. units)

(ii) Electrostatic Units (e.s. units)

Electromagnetic Units

Units based on electromagnetic effects are known as electromagnetic units and the system is known as electromagnetic system of units. This system involves the units of four quantities: permeability (μ) of the medium and the units of length, mass and time. The value of permeability of free space (vacuum) is taken as unity in this system.

Electrostatic Units

Units based on electrostatic effects are known as electrostatic units and the system is electrostatic system. This system involves the units of four quantities: permittivity ($\in$) of the medium and the units of length, mass and time. The value of permittivity of free space is taken as unity in this system.

Absolute units

An absolute system of units is defined as a system in which the various units are all expressed in terms of a small number of fundamental units. Absolute measurements do not compare the measured quantity with arbitrary units of the same type but are made in terms of Fundamental Units.

Practical units

Practical units are derived either from the absolute units or by reference to arbitrary standards. Table (1.2) shows the symbols and magnitudes of practical units.

Table 1.2 Practical Units

No.	Quantity	Practical unit	Symbol
1.	Charge	coulomb	Q
2.	Current	ampere	I
3.	Potential difference	volt	E
4.	Resistance	ohm	R
5.	Inductance	henry	L
6.	Capacitance	farad	C
7.	Power	watt	P
8.	Energy	joule	W

Dimensions in Electrostatic system

In this system the dimension of permittivity ε is taken as the fourth fundamental dimension.

1. Charge

According to coulomb's law, the force exerted between two charges Q_1 and Q_2 is

$$F = \frac{Q_1 Q_2}{\varepsilon d^2}$$

where d is the distance between charges Q_1 and Q_2.

The dimension is $[MLT^{-2}] = \frac{[Q^2]}{[\varepsilon]\,[L^2]}$

$\therefore$ Dimension of charge, $[Q] = [\varepsilon^{1/2} M^{1/2} L^{3/2} T^{-1}]$

2. *Current*

Current is charge per unit time

$$[I]=\frac{[Q]}{[T]}=\frac{[\varepsilon^{1/2}M^{1/2}L^{3/2}T^{-1}]}{[T]}$$

$$[I]=[\varepsilon^{1/2}M^{1/2}L^{3/2}T^{-2}]$$

3. *Potential difference or Emf*

Potential difference is work done per unit charge

$$[E]=\frac{[W]}{[Q]}=\frac{[ML^2T^{-2}]}{[\varepsilon^{1/2}M^{1/2}L^{3/2}T^{-1}]}$$

$$=[\varepsilon^{-1/2}M^{1/2}L^{1/2}T^{-1}]$$

4. *Capacitance*

$$\text{Capacitance } C=\frac{Q}{E}$$

$$\text{Dimension of capacitance } [C]=\frac{[Q]}{[E]}$$

$$=\frac{[\varepsilon^{1/2}M^{1/2}L^{3/2}T^{-1}]}{\varepsilon^{-1/2}M^{1/2}L^{1/2}T^{-1}]}=[\varepsilon L]$$

5. *Resistance*

$$\text{Resistance } R=\frac{E}{I}$$

$$\text{Dimension of resistance}\quad [R]=\frac{[E]}{[I]}$$

$$=\frac{[\varepsilon^{-1/2}M^{1/2}L^{1/2}T^{-1}]}{[\varepsilon^{1/2}M^{1/2}L^{3/2}T^{-2}]}=[\varepsilon^{-1}L^{-1}T]$$

6. *Inductance*

$$\text{Inductance L}=\frac{\text{emf}}{\text{rate of change of current}}$$

$$=\frac{E}{dI/dt}$$

$$\text{Dimension of inductance}\quad [L]=\frac{[E]}{[I]/[T]}$$

$$=\frac{[E]\,[T]}{[I]}$$

$$=\frac{[\varepsilon^{-1/2}M^{1/2}L^{1/2}T^{-1}]\,[T]}{[\varepsilon^{1/2}M^{1/2}L^{3/2}T^{-2}]}=[\varepsilon^{-1}L^{-1}T^2]$$

Dimensions in Electromagnetic system

The permeability, μ is the fourth dimension in this system.

1. *Pole strength*

$$\text{Force F}=\frac{m_1\,m_2}{\mu\,d^2}$$

where d is the distance between poles of strengths m_1 and m_2.

$$\therefore\ [MLT^{-2}]=\frac{[m^2]}{[\mu]\,[L^2]}$$

$$\text{Dimensions of pole strength, } [m]=[\mu^{1/2}M^{1/2}L^{3/2}T^{-1}]$$

2. *Magnetizing force*

Magnetizing force H is measured by force exerted on a unit pole.

Dimensions of magnetizing force

$$[H]=\frac{[F]}{[m]}=\frac{[MLT^{-2}]}{[\mu^{1/2}M^{1/2}L^{3/2}T^{-1}]}$$

$$=[\mu^{-1/2}M^{1/2}L^{-1/2}T^{-1}]$$

3. *Current*

The magnetizing force at the centre of a loop of radius r is

$$H = \frac{2\pi \cdot I}{r}$$

$$[H] = \frac{[I]}{[L]}$$

Dimensions of current $[I] = [H]\,[L]$

$$= [\mu^{-1/2} M^{1/2} L^{1/2} T^{-1}]$$

4. Charge

Charge = current × time

Dimensions of charge, $[Q] = [I]\,[T]$

$$= \mu^{-1/2} M^{1/2} L^{1/2} T^{-1}]\,[T] = [\mu^{-1/2} M^{1/2} L^{1/2}]$$

5. Potential difference

Potential difference is work done per unit charge. The dimensions of potential difference are

$$[E] = \frac{[W]}{[Q]} = \frac{[ML^2 T^{-2}]}{[\mu^{-1/2} M^{1/2} L^{1/2}]} = [\mu^{1/2} M^{1/2} L^{3/2} T^{-2}]$$

6. Capacitance

The dimensions of capacitance are

$$[C] = \frac{[Q]}{[E]} = \frac{[\mu^{-1/2} M^{1/2} L^{1/2}]}{[\mu^{1/2} M^{1/2} L^{3/2} T^{-2}]} = [\mu^{-1} L^{-1} T^{2}]$$

7. Resistance

The dimensions of resistance are

$$[R] = \frac{[E]}{[I]} = \frac{[\mu^{1/2} M^{3/2} L^{1/2} T^{-2}]}{[\mu^{-1/2} M^{1/2} L^{1/2} T^{-1}]} = [\mu\, L\, T^{-1}]$$

8. Inductance

Dimensions of inductance are

$$[L] = \frac{[E]}{[I]/[T]} = \frac{[E]\,[T]}{[I]}$$

$$= \frac{[\mu^{1/2} M^{1/2} L^{3/2} T^{-2}]\,[T]}{[\mu^{-1/2} M^{1/2} L^{1/2} T^{-1}]} = [\mu\, L]$$

2. M.K.S system (Giorgi system)

The C.G.S system suffers from the following disadvantages

(i) There are two systems of units (e.m.u and e.s.u) for fundamental theoretical work and a third (practical units) for practical engineering work.

(ii) There are two sets of dimensional equations for the same quantity.

In M.K.S system, metre, kilogramme and second are the three fundamental mechanical units. In order to connect the electrical and mechanical quantities, a fourth fundamental quantity has to be used. This fourth quantity is usually permeability. The permeability of free space is taken as $\mu_o = 10^{-7}$. The permeability of μ of any other medium is given by $\mu = \mu_r \mu_o$, where μ_r is the relative permeability. The permeability of free space in C.G.S system is unity.

$\therefore$ M.K.S unit of permeability $= 10^7 \times$ C.G.S. unit of permeability

1. Charge

The dimensions of charge in e.m.u system are $[\mu^{-1/2} M^{1/2} L^{1/2}]$

M.K.S. unit of length, metre = 100 centimetre

= 100 × C.G.S units of length

M.K.S. unit of mass, kilogramme = 1000 gm = 1000 × C.G.S units of mass

M.K.S unit of time, second = C.G.S unit of time, second

M.K.S unit of charge $= 10^{-1} \times$ C.G.S e.m unit of charge

= practical unit of charge

= 1 coulomb

2. *Current*

The dimensions of current in e.m.u system are

$$[\mu^{-1/2} M^{1/2} L^{1/2} T^{-1}]$$

M.K.S unit of current $= 10^{-1} \times$ C.G.S e.m units of current

= practical unit of current = 1 ampere

3. *Potential difference (EMF)*

The dimensions of potential difference are

$[\mu^{1/2} M^{1/2} L^{3/2} T^{-2}]$

M.K.S unit of emf $= 10^{8} \times$ C.G.S. e.m unit of emf

= practical unit of emf = 1 volt

4. *Resistance*

The dimensions of resistance are $[\mu LT^{-1}]$

M.K.S unit of resistance $= 10^{9} \times$ C.G.S e.m units of resistance

= practical unit of resistance = 1 ohm

5. *Inductance*

The dimensions of inductance are $[\mu L]$

M.K.S unit of inductance $= 10^{9} \times$ C.G.S e.m units of inductance

6. *Capacitance*

The dimensions of capacitance are $[\mu^{-1} L^{-1} T^{2}]$

M.K.S unit of capacitance $= 10^{-9} \times$ C.G.S e.m units of capacitance

= practical unit of capacitance = 1 farad

7. *Power*

The dimensions of power are $[ML^{2} T^{-3}]$

M.K.S unit of power $= 10^{7} \times$ C.G.S e.m units of power

= practical unit of power = 1 watt

8. *Energy*

The dimensions of energy are $[ML^{2} T^{-2}]$

M.K.S unit of energy $= 10^{7} \times$ C.G.S e.m unit of energy

= practical unit of energy

= 1 joule

Advantages of M.K.S system of units are

(i) This system connects the practical units directly, with the fundamental laws of electricity and magnetism.

(ii) This system gives specified formulae for expressions of electromagnetism involving only practical units.

Rationalised M.K.S.A system

The M.K.S system in its rationalised form, utilizes four fundamental units. They are metre, kilogram, second and ampere.

Table (1.1) shows rationalised M.K.S.A system

Table 1.3 Rationalised M.K.S.A system

No.	Quantity	Symbol	Dimension
1.	Current	I	$[I]$
2.	Charge	Q	$[TI]$
3.	Emf	E	$[ML^{2} T^{-3} I^{-1}]$
4.	Resistance	R	$[ML^{2} T^{-3} I^{-1}]$
5.	Flux (magnetic)	ϕ	$[ML^{2} T^{-2} I^{-1}]$
6.	Flux density	B	$[MT^{-2} I^{-1}]$
7.	MMF	$\Im$	$[I]$

No.	Quantity	Symbol	Dimension
8.	Magnetizing force	H	$[L^{-1} I]$
9.	Reluctance	R	$[M^{-1} L^{-2} T^2 I^2]$
10.	Inductance	L	$[ML^2 T^{-2} I^{-2}]$
11.	Electric flux	Ψ	$[TI]$
12.	Electric flux density	D	$[L^{-2} TI]$
13.	Electric field strength	ε	$[MLT^{-3} I^{-1}]$
14.	Capacitance	C	$[M^{-1} L^{-2} T^4 I^2]$

3. S.I Units

An international organization of which most of the advanced and developing countries, including India are members, called the General Conference of Weights and Measures (CGPM).

The Eleventh General conference of Weights and Measures which met in October, 1960 recommended a unified systematically constituted, coherent system of fundamental supplementary and derived units for international use. This system, called the International system of Units and designated by the abbreviation, *SI*, Systems International *d'* Units has been accepted internationally.

1.3.4 Standards

Standards of mass, length and such other physical quantities are physical devices and systems representing the fundamental unit of the particular quantity.

Standards have been developed for all the fundamental units as well as some of the derived mechanical and electrical units. They are classified as follows:

1. International standards
2. Primary standards
3. Secondary standards
4. Working standards

1. International standards

These standards are those defined and agreed upon internationally. They are maintained at the International Bureau of Weights and Measures and are not accessible outside for calibration of instruments.

2. Primary standards

These standards are those maintained by national standards laboratories in different parts of the world and they are also not accessible outside for calibration. The primary standards established for the fundamental and some derived units are independently calibrated by absolute measurements at each of the national standards laboratories and an average value for the primary standard is obtained with the highest accuracy possible. These are used for verification and calibration of the secondary standards.

Secondary standards

These standards are usually fixed standards for use in industrial laboratories, where as working standards are for day-to-day use in measurement laboratories.

Working standards

Working standards may be lower in accuracy in comparison to secondary standards. The accuracy of secondary standards is maintained by periodic comparison with the primary standards, where as working standards may be checked against secondary standards.

1.4 CALIBRATION

Calibration is an essential process to be undertaken for each instrument and measuring system frequently. A reference standard atleast ten times more accurate than the instrument under test is normally used. Calibration is the process where the test instrument (the instrument to be calibrated) is compared with the standard instrument. It consists of reading the standard and test instruments simultaneously when the input quantity is held constant at several values over the range of the test instrument. The calibration is better carried out under the stipulated environmental conditions. All industrial grade instruments can be checked for accuracy in the laboratory by using the working standards.

Generally, certification of an instrument manufactured by an industry is undertaken by the National Physical Laboratory and other authorized laboratories where the secondary standards and the working standards are kept.

1.4.1 Generalized performance characteristics of Instruments

The instrument performance characteristics are generally broken down in to two areas

(i) Static characteristics

(ii) Dynamic characteristics

(i) Static characteristics

- Some applications involve the measurement of quantities that are constant or vary only slowly.
- Under these conditions, it is possible to define a set of performance criteria that give a meaningful description of the quality of measurement. So "Static characteristics are a set of performance criteria that give a meaningful description of the quality of measurement while the measured quantities are either constant or vary slowly.

(ii) Dynamic characteristics

- Dynamic characteristics describe the quality of measurement when the measured quantities are rapidly varying quantities.

Let us study in detail about the characteristics in the Unit II.

1.4.2 Static calibration

The static performance characteristics are obtained by one form or another of the process of static calibration.

- In general, static calibration refers to a situation in which all inputs except one are kept at some constant values.
- Then the one input under study is varied over some range of constant values, which causes the outputs to vary over some range of constant values.
- The input-output relations developed in this way comprise a static calibration valid under the stated constant conditions of all the other inputs.
- This procedure may be repeated, by varying in turn each input considered to be of interest and thus developing a family of static input-output relations.

1.4.3 Procedure for calibration

1. Examine the construction of the instrument, and identify and list all the possible inputs.
2. Decide, which of the inputs will be significant in the application for which the instrument is to be calibrated.
3. Select the apparatus that will allow you to vary all the significant inputs over the ranges considered necessary. Select standards to measure each input.
4. By holding some inputs constant, varying others and recording the outputs develop the desired static input-output relations.

1.5 ERRORS IN MEASUREMENT

A measurement can not be made without errors. These errors can only be minimized but not eliminated completely. It is important to find out the accuracy of measurement and how different errors have entered in to the measurement. Before that it is essential to know the different errors that can possibly enter in to the measurement.

1.5.1 Classification of errors

1. Gross errors
2. Systematic errors
3. Random errors

1. *Gross errors*

This type of errors mainly covers human mistakes in reading the instruments (misreading the instruments) making adjustments (incorrect adjustments) and application of instruments (improper application). The computational errors are also grouped under this type of error.

The human being may grossly misread the scale. For eg., due to an oversight, he may read the temperature as 31.5°C while the actual reading may be 21.5°C. He may transpose the reading while recording. For eg., he may read 25.8°C and record 28.5°C.

When human beings are involved in measurement, gross errors may be committed. Although complete elimination of gross errors is probably impossible, one should try to anticipate and correct them.

One common gross error frequently encountered involves the improper selection of the instrument. When a voltmeter is used to measure the potential difference across two points in a circuit, the input impedance of the voltmeter chosen should be atleast 10 times greater than the output impedance of the measuring circuit. As the output impedance of a circuit is normally not known before hand, the selection of the voltmeter may not be made correctly, leading to a gross error. The error caused by the improper selection of a voltmeter is shown by the following example.

Example 1.1:

A voltmeter reads 20 V in its 40 V scale when connected across an unknown resistor as shown in fig (1.4). The resistance of the voltmeter coil is 2000 ohms/volt. If the milliammeter reads 2 mA, calculate (a) apparent value of the unknown resistor (b) actual value of the unknown resistor (c) gross error.

Solution

(a) Apparent value of resistance

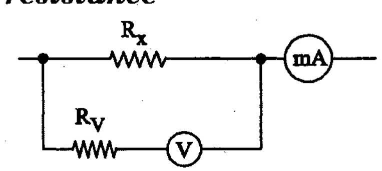


Fig. 1.4 Example (1.1)

$$R_A = \frac{V}{I} = \frac{20}{2} = 10 \text{ k}\,\Omega$$

(b) Voltmeter resistance

$R_V = 2000 \times 40 = 80 \text{ k}\,\Omega$

Since the voltmeter is connected in parallel with the unknown resistor,

$$R_A = \frac{R_x \cdot R_V}{R_x + R_V}$$

where R_x is the unknown resistance value

$$R_x = \frac{R_A \cdot R_V}{R_V - R_A}$$

$$= \frac{10 \times 10^3 \times 80 \times 10^3}{10^3\,[80 - 10]} = 11.43 \text{ k}\,\Omega$$

(c) $\% \text{ error} = \dfrac{\text{Apparent} - \text{Actual}}{\text{Actual}} \times 100$

$$= \frac{10 - 11.43}{11.43} \times 100 = -\,12.5\%$$

This error is due to the appreciable current drawn by the voltmeter which is known as loading effect.

Gross errors may be avoided by two means. They are

1. Great care should be taken in reading and recording the data.
2. Two, there or even more readings should be taken for the quantity under measurement.

2. Systematic errors

Systematic errors are due to shortcomings of the instrument and changes in external conditions affecting the measurement. These type of errors are divided in to three categories:

(i) Instrumental errors

(ii) Environmental errors

(iii) Observational errors

(i) Instrumental errors

These errors arise due to the following:

(a) Due to inherent shortcomings of the instrument.

(b) Due to misuse of the instruments and

(c) Due to loading effects of instruments.

(a) Inherent shortcomings of instruments

These errors are inherent in instruments because of their mechanical structure. They may be due to construction, calibration or operation of the instruments or measuring devices.

(b) Misuse of instruments

Often, the errors caused in measurements are due to the fault of the operator than that of the instrument. A good instrument misused may cause errors. There are some ill practices like using the instrument contrary to manufacturer's instructions and specifications which in addition to producing errors cause permanent damage to the instruments as a result of overloading and overheating.

(c) Loading effects

Errors occur when we use the instrument in an improper manner. For eg., a well calibrated voltmeter may give incorrect reading when connected across a high resistance circuit. The same voltmeter, when connected in a low resistance circuit, may give correct reading. This is due to the loading effect of voltmeter.

(ii) Environmental errors

Environmental errors are due to changes in the environmental conditions such as temperature, humidity, pressure, electrostatic and magnetic fields. For eg., the resistance of a strain gauge changes with variation in temperature.

(iii) Observational errors

The observational error may be caused due to parallax. For eg., the pointer of a voltmeter rests slightly above the surface of the scale. Thus an error on account of parallax will occur unless the line of vision of the observer is exactly above the pointer. This may be minimized by mirrored scales in the meters.

3. Random (Residual) errors

Random errors are unpredictable errors and occur even when all systematic errors are accounted for, although the instrument is used under controlled environment and accurately pre-calibrated before measurement. Over a period of observation, the readings may vary slightly. The happenings or disturbances about which we are unaware are lumped together and called "Random" or "Residual". Hence the errors caused by these happenings are called Random (or Residual) errors.

4. Limiting errors (Guarantee errors)

In most instruments the accuracy is guaranteed to be with in certain percentage of full scale reading. The manufacturer has to specify the deviations from the nominal value of a particular quantity.

The limits of these deviations from the specified value are defined as limiting errors or Guarantee errors. In general,

Actual value of quantity,

$$Q_a = Q_s \pm \delta Q$$

where, Q_s - nominal value of quantity

For eg., the nominal magnitude of resistor is 10 Ω with a limiting error of ± 1 Ω. The magnitude of the resistance will be between the limits:

$$Q_a = 10 \pm 1\ \Omega \text{ or}$$

$$Q_a \geq 9\ \Omega \text{ and}$$

$$Q_a \leq 11\ \Omega$$

- The manufacturer guarantees that the value of resistance of the resistor lies between 9 Ω and 11 Ω.

1.5.2 Error analysis

The analysis of the measurement data is necessary to obtain the probable true value of the measured quantity. Any measurement is associated with a certain amount of uncertainty. The best method of analysis is the statistical method. For the statistical analysis, a large number of measurements is required. Also the systematic errors should be small compared with random errors. When temperature of liquid in a tank is to be measured, 10 readings are taken over a period of time by means of a thermocouple. Each of these 10 readings may be different from the others. We can not find which reading is correct. Here the statistical methods will give the most probable true value of temperature. For statistical methods the terms like arithmetic mean, deviation, mode & median are to be determined.

1. Arithmetic mean

The most probable value of measured variable is the arithmetic mean of the number of readings taken. The best approximation is made when the number of readings of the same quantity are very large. Theoretically, an infinite number of readings would give the best result. But practically, only a finite number of measurements can be made.

The arithmetic mean is given by

$$\overline{X} = \frac{x_1 + x_2 + x_3 + x_4 + \ldots + x_n}{n}$$

$$= \frac{\sum_{a=1}^{n} X_a}{n}$$

$\overline{\text{X}} \rightarrow$ arithmetic mean

$x_1, x_2, \ldots x_n \rightarrow$ readings or variates or samples.

$n \rightarrow$ number of readings

2. Deviation

Deviation is departure of the observed reading from the arithmetic mean of the group of readings. Let the deviation of reading x_1 be d_1 and that of reading x_2 be d_2, etc.

Then

$$d_1 = x_1 - \overline{X}$$

$$d_2 = x_2 - \overline{X}$$

$$\vdots \quad \vdots$$

$$\vdots \quad \vdots$$

$$d_n = x_n - \overline{X}$$

Average deviation is defined as the average of the modulus of the individual deviations and is given by

$$D = \frac{|d_1| + |d_2| + \ldots + |d_n|}{n}$$

$$= \frac{\Sigma\,|d|}{n}$$

$$= \frac{\sum_{a=1}^{n} |x_a - \overline{X}|}{n}$$

3. Standard deviation

Another term in the statistical analysis of random errors in the standard deviation or the root mean square deviation. The standard deviation of an infinite number of data is defined as the square root of the sum of individual deviations squared, divided by the number of readings.

Standard deviation,

$$S.D = \sigma = \sqrt{\frac{d_1^2 + d_2^2 + \ldots + d_n^2}{n}}$$

$$= \sqrt{\frac{\sum_{a=1}^{n} d_a^2}{n}}$$

4. Variance

Variance is another term which is sometimes used in statistical analysis. This is the square of the standard deviation and is given by

$$V = \sigma^2 = \frac{d_1^2 + d_2^2 + \ldots + d_n^2}{n}$$

$$= \frac{\sum_{a=1}^{n} d_a^2}{n} \text{ for } n > 20$$

$$= \frac{\sum_{a=1}^{n} d_a^2}{n-1} \text{ for } n \le 20$$

5. Median

Median is also used to indicate the most probable value of the measured quantity when a set of readings are taken. When the readings are arranged in the ascending or descending order of magnitude, the middle value of the set is taken as the median. For example, the temperature of a bath is noted by ten observers as follows:

75.5°C, 73.7°C, 77.5°C, 75.7°C, 74.8°C, 77.0°C, 75.9°C, 75.3°C, 73.9°C, 77.5°C.

It is rearranged in ascending order as follows:

73.7°C, 73.9°C, 74.8°C, 75.3°C, 75.5°C, 75.7°C, 75.9°C, 77.0°C, 77.5°C

Now the median is the 75.5°C

6. Mode

Mode is the value which occurs most frequently in a set of observations and around which other items of the set cluster.

For example, the frequency distribution of a set of 100 observations is given below

Temperature readings in °C	30	31	32	33	34	35	36	37
No. of readings	9	4	15	25	15	22	7	3

The value of temperature reading 33 has occurred 25 times (maximum). Hence mode is 33°C.

1.5.2.1 Statistical methods of error analysis

1. Probability of errors

By the nature of the random errors, the uncertainty associated with any measurement cannot be predetermined. Only the probable error can be specified using statistical error analysis. The following are some of the statistical methods of analysing the errors.

(i) Normal distribution of errors

Histogram

When a number of multi sample observations are taken experimentally there is a scatter of the data about some central value. One method of presenting test results is in the form of a "Histogram". The technique is illustrated in fig.(1.5) representing the data given in table (1.4). This table (1.4) shows a set of fifty readings of a length measurement. The most probable or central value of length is 90 mm.

Table 1.4

Length (mm)	Number of readings
89.7	1
89.8	3
89.9	10
90.0	19
90.1	12
90.2	4
90.3	1

Total number of readings = 50

Fig (1.5) shows the histogram which represents these data where the ordinate indicates the number of observed readings (frequency of occurrence) of a particular value. The histogram is also called a "frequency distribution curve".

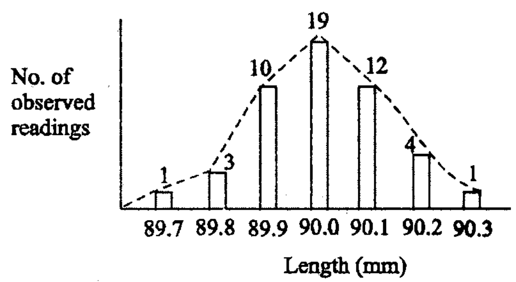


Fig. 1.5 Histogram

(ii) Normal or Gaussian curve of Errors

The normal or Gaussian law of errors is the basis for the major part of study of random errors.

The law of probability states the normal occurrence of deviations from average value of an infinite number of measurements or observations can be expressed by:

$$y = \frac{h}{\sqrt{\pi}} \exp(-h^2 x^2)$$

where,

$x \rightarrow$ magnitude of deviation from mean

$y \rightarrow$ number of readings at any deviation x (the probability of occurrence of deviation x)

$h \rightarrow$ a constant called precision index

Fig. (1.6) shows the Normal probability curve

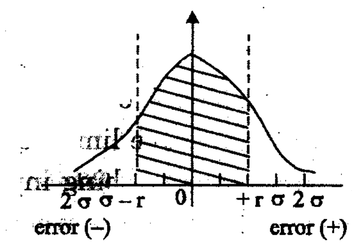


Fig. 1.6 Normal probability curve

(iii) Probable error

The most probable or best value of a Gaussian distribution is obtained by taking arithmetic mean of the various values of the variate. The confidence in the best value (most probable value) is connected with the sharpness of the distribution curve.

1.6 ODDS AND UNCERTAINTY

1.6.1 Specifying Odds

The probability of occurrence can be stated in terms of Odds. Odds is the number of chances that a particular reading will occur when the error limit is specified. For example, if the error limits are specified as $\pm 0.6745\ \sigma$, the chances are that 50% of the observations will lie between the above limits or in other words we can say that odds are 1 to 1.

The odds can be calculated by the following formula,

$$\text{Probability of occurence} = \frac{\text{odds}}{\text{odds} + 1}$$

The table (1.5) shows the corresponding values of Deviation and probability.

Probability (%)	Deviation d	Odds
50.0	$\pm 0.6745\,\sigma$	1 to 1
68.3	$\pm \sigma$	2.15 to 1
95.4	$\pm 2\,\sigma$	21 to 1
99.7	$\pm 3\,\sigma$	256 to 1

1.6.2 Uncertainty

Uncertainty is expressive of the range of variation of the indicated value from the true value. It indicates the probable limits of [illegible] which the indicated value may have due to the influence of disturbing inputs. It is bipolar where as error may be positive or negative depending on whether the indicated value is higher or lower than the true value. Statement of uncertainty signifies the quality of the measuring instrument and hence its accuracy, it is incumbent on the part of every instrumentation engineer to express the uncertainty attendant on each measured value.

(i) Uncertainty Analysis

Many times the data available is a single sample data and therefore the statistical methods discussed earlier cannot be applied directly.

Hence, Kline and McClintock have proposed a method based upon probability and statistics which analyses the data employing uncertainty distribution rather than frequency distribution.

Kline and MCClintock suggest that a single sample result may be expressed in terms of a mean value and an uncertainty interval based upon stated odds.

The result may be written as follows:

$$X = \overline{X} \pm W \text{ (b to 1)}$$

where

$\overline{X}$ = the value if only one reading is available on the arithmetic mean of several readings

W = uncertainty interval

b = odds or the chance that the true value lies with in the stated range, based upon the opinion of the experimenter

For example, the results of a temperature measurement may be expressed as $\theta = 90°C \pm 1°C$.

This means that there is an uncertainty of $\pm 1°C$ in the result. Kline and McClintock proposed that the experimenter specify certain odds for the uncertainty.

So, $\theta = 90°C \pm 1°C$ (20 to 1)

The experimenter is willing to bet 20 to 1 odds that the temperature measurement which he has made are with in $\pm 1°C$ of $90°C$

(ii) Propagation of Uncertainties

The uncertainty analysis in measurements when many variates are involved is done on the same basis as is done for error analysis when the results are expressed as standard deviations or probable errors.

Suppose X is a function of several variables,

$$X = f(x_1, x_2, x_3, \ldots, x_n)$$

where $x_1, x_2, x_3 \ldots x_n \rightarrow$ independent variables with the same degree of odds.

The uncertainty in the result is

$$W_x = \sqrt{\left(\frac{\partial X}{\partial x_1}\right)^2 W_{x_1}^2 + \left(\frac{\partial X}{\partial x_2}\right)^2 W_{x_2}^2 + \ldots + \left(\frac{\partial X}{\partial x_n}\right)^2 W_{x_n}^2}$$

where, W_x = resultant uncertainty

$W_{x_1}, W_{x_2}, W_{x_3} \ldots W_{xn} \rightarrow$ uncertainties in the independent variables $x_1, x_2, x_3 \ldots x_n$ respectively.

1.7 SENSORS AND TRANSDUCERS

Instrument Society of America defines a sensor or transducer as a device which provides a usable output in response to a specified measurand. Here the measured is a physical quantity and the output may be an electrical quantity, mechanical and optical.

(i) Sensor

An element that senses a variation in input energy to produce a variation in another or same form of energy is called a sensor.

(ii) Transducer

Transducer converts a specified measurand into usable output using transduction principle. For example, a properly cut piezo electric crystal can be called a sensor where as it becomes a transducer with appropriate electrodes and input/output mechanisms attached to it. So. the sensor is the primary element of a transducer.

Table (1.6) shows the energy types and corresponding measurands.

Table 1.6 Energy types and corresponding measurands

Energy	Measurands
Mechanical	Length, area, volume, force, pressure, acceleration, torque, mass flow, acoustic intensity and so on.
Thermal	Temperature, heat flow, entropy, state of matter.
Electrical	Charge, current, voltage, resistance, inductance, capacitance, dielectric constant, polarization, frequency, electric field, dipole moment, and so on.
Magnetic	Field intensity, flux density, permeability, magnetic moment, and so on.
Radiant	Intensity, phase, refractive index, reflectance, transmittance, absorbance, wavelength, polarization, and so on.
Chemical	Concentration, composition, oxidation/reduction potential, reaction rate, pH and the like.

1.7.1 Classification of transducers

The transducers may be classified based on

1. The physical effect employed
2. The physical quantity measured
3. The source of energy

1. Classification based on physical effect

The physical quantity applied as measurand (quantity to be measured) to the transducer causes some physical changes in its element. By this physical effect the transducer converts the physical quantity in to electrical quantity. For example, a change in temperature to be measured causes variation of resistance (physical change) in a copper wire (element) and this effect could be used to convert temperature in to an electrical output.

The physical effects commonly employed are

(a) Variation of resistance

(b) Variation of inductance

(c) Variation of capacitance

(d) Piezo electric effect

(e) Magnetostrictive effect

(f) Elastic effect

(g) Hall effect

(a) Variation of resistance

The resistance of a length of metallic wire is given by

$$R = \frac{\rho l}{a}$$

where,

$R \rightarrow$ Resistance in ohm.

$p \rightarrow$ Resistivity (or specific resistance) of the material in ohm-m.

$l \to$ length of wire in m.

$a \to$ Area of cross-section in m^2

As resistance is a function of ρ, l, a (i.e) $R = f(\rho, l, a)$, with any change in any one of the physical quantities ρ, a or l due to variation in resistance, a variable resistance transducer can be designed to convert physical quantity.

Some of the transducers based on this principle are potentiometer, strain gauge, resistance thermometer, carbon microphone, and photoconductive cell.

- The resistance thermometer is based upon thermo resistive effect which is the change in electrical resistivity of a metal or semiconductor due to change in temperature co-efficient of resistivity.
- Carbon microphone works on the principle of change in contact resistance due to applied pressure.
- Photoconductive cell is based on photoconductive effect which is the change in electrical conductivity due to incident light.
- Potentiometer works on the principle of change in resistance due to linear or rotational motion.
- Strain gauge works on the principle of change in resistance due to applied pressure.

(b) Variation of inductance

The inductance of a coil is given by

$$L = N \frac{d\phi}{dt}$$

$$= \frac{N^2 \mu_o \mu_r A}{e}$$

where, $L \to$ inductance in henry

$N \to$ No., of turns

$\mu_o \to$ absolute permeability

$\mu_r \to$ relative permeability

$A \to$ area of cross section of the core

$l \to$ length of magnetic path

$\frac{d\phi}{dt} \to$ rate of change of magnetic flux

As L is a function of N, μ_r, A, l,

(i.e) $L = f(N, \mu_r, A, l)$, when anyone of these quantities changes, the inductance changes. This leads to the design of a variable inductance transducer.

Some of the transducers based on variation of inductance are induction potentiometer, linear variable differential transformer (LVDT) and synchros.

(c) Variation of capacitance

The capacitance between two conductor plates is given by

$$C = \frac{\varepsilon_o \varepsilon_r A}{d}$$

$C \to$ capacitance in farad

$\varepsilon_o \to$ absolute permittivity

$\varepsilon_r \to$ relative permittivity of the separating medium

$A \to$ area of cross-section of the plates

As C is a function of ε_r, A, d (i.e) $C = f(\varepsilon_r, A, d)$, when anyone of these quantities changes, the capacitance varies. This leads to the design of a variable capacitance transducer.

(d) Piezoelectric effect

When a piezoelectric crystal like quartz or Rochelle salt is subjected to mechanical stress, an electric charge is generated. This is known as piezoelectric effect. The transducer based on this effect is piezoelectric transducer.

(e) Magnetostrictive effect

When a magnetic material is subjected to mechanical stress, its permeability changes. This effect is magnetostrictive effect and the transducer based on this effect is magnetostrictive transducer.

(f) *Elastic effect*

When an elastic member is subjected to mechanical stress it is deformed. The transducer based on this effect is called elastic transducer.

(g) Hall effect

When a magnetic field is applied to a current carrying conductor at right angles to the direction of current, a transverse electric potential gradient is developed in the conductor. This effect is called as Hall effect and the transducer based on this effect is called as Hall effect transducer.

2. Classification based on physical quantity measured

The transducers may be classified based on the physical quantity they measure as follows:

- Temperature transducers → Transducers used to measure temperature
- Pressure transducers → To measure pressure
- Flow transducers → To measure flow
- Liquid level transducers → To measure liquid level
- Force/Torque transducers → To measure force & Torque
- Velocity/Speed transducers → To measure velocity & speed
- Humidity transducers → To measure humidity
- Acceleration/vibration transducers → To measure acceleration & vibration
- Displacement transducers → To measure displacement

3. Classification based on source of energy

Transducers may be classified based on source of energy into two types.

- Active transducer
- Passive transducer

(i) *Passive transducer*

A component whose output energy is supplied entirely or almost entirely by its input signal is called a passive transducer. A passive transducer is the

one which absorbs energy from the input medium and converts it directly into the output signal.

Example

A Thermocouple extracts heat energy from the input medium and converts it into electrical energy (voltage).

(ii) *Active Transducer*

An active transducer has an auxiliary source of power which supplies a major part of the output power while the input signal supplies only an insignificant portion (i.e) this transducer uses the energy it absorbs from the input medium as a control signal to transfer energy from the power supply to produce a proportional output.

Example

strain gauge

The energy extracted from the strained member is very small. The energy for the output signal is supplied by an external power source.

Selection of Transducers

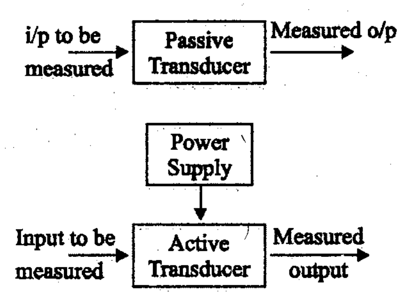


Fig. 1.7 Active and passive transducers

Transducers are used for the measurement of physical quantities. The selection of transducers for particular measurand is very important. The selection of transducers may be based on the following factors for effective measurement.

1. The physical quantity to be measured (measurand).
2. The range of input quantity.

1. Based on physical quantity to be measured

The correct type of transducer should be selected for measuring the physical quantity. The following table (1.7) shows the physical quantity and the corresponding transducer types available.

Table 1.7 Transducer types

Sl.No.	Physical quantity	Transducers available
1.	Temperature	Bimetallic element
		Fluid expansion systems
		(i) Liquid-in-steel bulb thermometers
		(ii) Liquid-in-glass thermometers
		(iii) Vapour pressure thermometers
		Thermoresistive elements
		(i) Resistance Temperature detector (RTD)
		(ii) Thermistor
		Thermocouple
		Linear-Quartz thermometer
		Pyrometry
2.	Pressure	U-tube and ball type manometers
		Ring balance manometer
		Metallic diaphragms
		Capsules and bellows
		Bourdon tubes
		Membranes
3.	Force (weight)	Spring balance
		Cantilever
		Diaphragms
		Pneumatic and hydraulic load cells
		Column and proving ring load cells

Sl.No.	Physical quantity	Transducers available
4.	Torque	Torsion bar
		Flat spiral springs
		Dynamometer
		Gyroscope
5.	Density of liquids	Hydrometer
		Air bubbler system
		U-tube weighing system
6.	Liquid level	Float elements
		Manometer system
		Diaphragms
		Container weight
7.	Viscosity	Capillary tube
		Concentric cylinder system
8.	Flow rate of fluids	Pitot static tube
		Flow-obstruction elements
		Rotating vane system
		Rotameter float system
9.	Displacement	Flapper nozzle system
10.	Absolute displacement, velocity and acceleration	Seismic system
11.	Vehicle attitude	Gyroscope

TWO MARK QUESTIONS AND ANSWERS

1. What is instrument?

Instrument is a device for determining the value or magnitude of a quantity or variable.

2. Add 826 ± 5 to 628 ± 3

$$N_1 = 826 \pm 5 \ (= \pm 0.605\%)$$

$$N_2 = 628 \pm 3 \ (= \pm 0.477\%)$$

$$\text{Sum} = 1.454 \pm 8 \ (= \pm 0.55\%)$$

3. Subtract 628 ± 3 from 826 ± 5

$$N_1 = 826 \pm 5 \ (= \pm 0.605)$$

$$N_2 = 628 \pm 5 \ (= \pm 0.477\%)$$

$$\text{Difference} = 198 \pm 8 \ (= \pm 4.04\%)$$

4. List three sources of possible errors in instruments.

Gross, systematic and random errors are produced in instruments.

5. Define instrumental error.

These are the errors inherent in measuring instrument because of their mechanical structure. It is usually divided into,

(a) Instrumental errors

(b) Environmental errors

6. Define limiting error.

Components are guaranteed to be within a certain percentage of rated value. Thus the manufacturer has to specify the deviations from the nominal value of a particular quantity.

7. Define probable error.

Probable error is defined as $r = \pm 0.6754\,\sigma$ where σ is standard deviation.

Probable error has been used in experimental work to some extent in past, but standard deviation is more convenient in statistical work.

8. Define environmental error.

Environmental errors are due to conditions in the measuring device, including conditions in the area surrounding the instrument, such as the effects of changes in temperature, humidity.

9. Define arithmetic mean.

The best approximation method will be made when the number of readings would give the best result,

$$X = \frac{x_1 + x_2 + x_3 + \dots x_n}{n}$$

$$= \frac{\Sigma x}{n}$$

where,

X	–	Arithmetic mean
x_1, x_2, x_n	–	Readings taken
n	–	Number of readings

10. Define average deviation.

$$\text{Average deviation } D = \frac{|d_1| + |d_2| + |d_3| + \dots + |d_n|}{n}$$

$$= \frac{\Sigma\, |d|}{n}$$

By definition, average deviation is the sum of absolute values of the value deviations divided by the number of readings.

11. Define units.

It is necessary to define a physical quantity both in kind and magnitude in order to use this information for further proceedings. The standard measure of each kind of physical quantity is named as the unit.

12. Define standards.

The physical embodiment of a unit of measurement is a standard. For example, the fundamental unit of mass in the International System is the kilogram and defined as the mass of a cubic decimeter of water at its temperature of maximum density of 4°C.

13. Mention the purposes of the measurement.

Measurement is used,

- To understand an event or an operation.
- To monitor an event or an operation.
- To control an event or an operation.
- To collect data for future analysis.
- To validate an engineer design.

14. What are the methods of measurement?

The methods of measurement are,

- Direct comparison method
- Indirect comparison method

15. Classify standards

Standards are classified as,

- International standards
- Primary standards
- Secondary standards
- Working standards

UNIT II

Characteristics of Transducers

2.1 INTRODUCTION

- The selection of most suitable transducer from commercially available instruments is very important in designing an Instrumentation system.
- For the proper selection of transducer, knowledge of the performance characteristics of them are essential.
- The performance characteristics can be classified into two namely

(i) Static characteristics

(ii) Dynamic characteristics

- Static characteristics are a set of performance criteria that give a meaningful description of the quality of measurement without becoming concerned with dynamic descriptions involving differential equations.
- Dynamic characteristics describe the quality of measurement when the measured quantities vary rapidly with time. Here the dynamic relations between the instrument input and output must be examined, generally by the use of differential equations.

2.2. STATIC CHARACTERISTICS AND STATIC CALIBRATION

- The most important static characteristics of a transducer are
    1. Static sensitivity
    2. Linearity
    3. Precision / Accuracy
    4. Resolution

5. Hysteresis
6. Range and span
7. Input impedance and loading effect.

2.2.1 Static calibration

- All these static characteristics are obtained by one form or another of the process of static calibration.
- In general, static calibration refers to a situation in which all inputs except the desired one are kept at some constant values.
- The desired input is varied over some range in steps and the output values are noted.
- The input - output relationship thus developed is called the static calibration valid under the stated constant conditions of all the other inputs.

2.2.2 Static sensitivity

- Static sensitivity of a transducer can be defined as the slope of the static calibration curve.

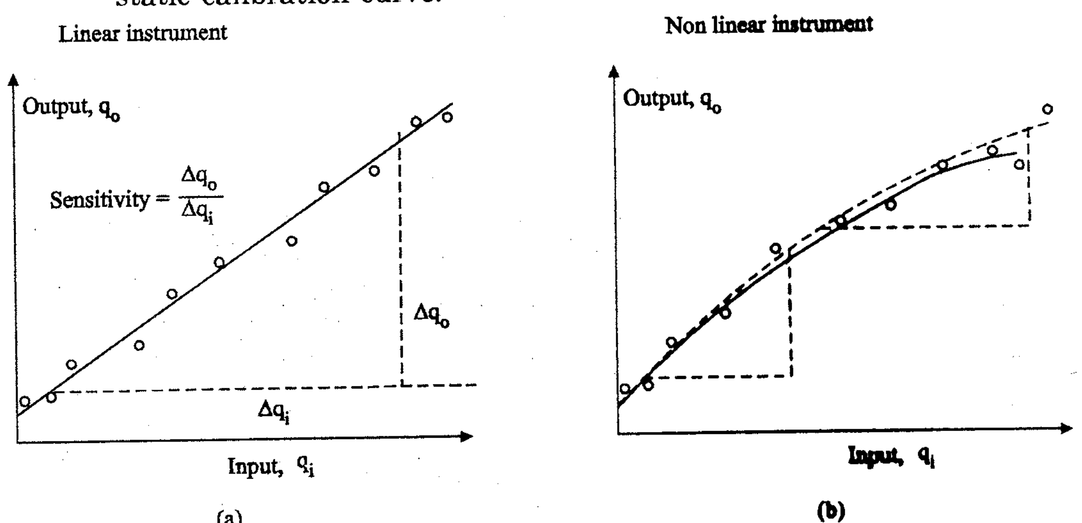


Fig. 2.1 (a) & (b) Definition of sensitivity

- If the curve is a straight line for a linear instrument, the sensitivity will vary with the input value, as shown in fig. (2.1) a.
- If the curve is not a straight line for a non-linear instrument, the sensitivity will vary with the input value, as shown in fig. (2.1) b. Hence the sensitivity should be taken depending on the operating point.
- The sensitivity is expressed in output unit / input unit.

Zero and Sensitivity drift

- When the sensitivity of instrument to its desired input is concerned, its sensitivity to interfering and/or modifying inputs is also to be considered.
- For example, consider temperature as an input to the pressure gauge.
- Temperature can cause a relative expansion and contraction that will result in a change in output reading even though the pressure has not changed. Here, the temperature is an interfering input. This effect is called a zero drift.
- Also, temperature can alter the modulus of elasticity of the pressure-gauge spring, thereby affecting the pressure sensitivity. Here, it is a modifying input. This effect is a sensitivity drift or scale-factor drift.

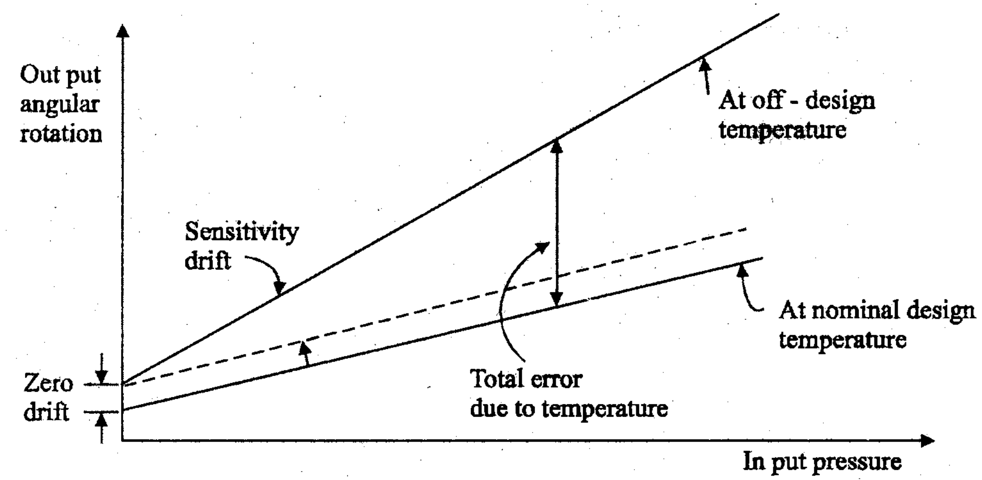


Fig. 2.1 (c) Zero and sensitivity drift

- Fig. 2.1 (c) shows the zero and sensitivity drift.
- Sensitivity $= \dfrac{\Delta q_o}{\Delta q_i}$

where,

Δq_o = change in output quantity

Δq_i = change in input quantity

2.2.3 Linearity

- The calibration curve of a transducer may not be linear in many cases.
- If it is so, the transducer may still be highly accurate.
- However, linear behaviour is most desirable in many applications.
- The conversion from a scale reading to the corresponding measured value of input quantity is most convenient if it is to be multiplied by a fixed constant rather than looking into a calibration chart or a graph.
- Linearity is a measure of the maximum deviation of the plotted transducer response from a specified straight line.
- To select a straight line for a plotted calibration curve there are a number of ways. Some of them are

1. The straight line connecting the calibration point at zero input to that at full-scale input.
2. The straight line may be drawn through as many calibration points as possible.
3. The straight line may be determined by the least squares fit method mathematically. The input-output relationship of a transducer is generally given by the equation

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_n x^n \quad \ldots (2.1)$$

where $x \rightarrow$ input quantity

$y \rightarrow$ output quantity

$a_0, a_1, \ldots a_n \rightarrow$ calibration factors.

The best-fit straight line is mathematically determined by evaluating the deviation of the response curve from the straight line at a number of calibration points and choosing the one that gives the minimum of the sum of the squares of the deviations.

- This procedure is described as least squares fit.

2.2.4 Method of least squares

- Assume that the input to a transducer 'x' is varied over its full range and output 'y' is measured.
- Let the total number of measurements be n.
- The linearised relation between x and y can be expressed as

$$y = ax + b \quad \ldots (2.2)$$

where

a & $b \rightarrow$ constants

- The constants 'a' and 'b' are determined using least-square fit.
- The deviation of the i^{th} reading from the straight line specified by $y = ax + b = y_i - (ax_i + b)$ ... (2.3)
- Sum of the squares of the derivation

$$S = \sum_{i=1}^{n} [y_i - (ax_i + b)]^2 \quad \ldots (2.4)$$

- S would be minimised by setting the following derivatives equal to zero.

$$\frac{\partial S}{\partial a} = 0 = \sum_{i=1}^{n} (bx_i + ax_i^2 - x_i y_i) \quad \ldots (2.5)$$

$$\frac{\partial S}{\partial b} = 0 = \sum_{i=1}^{n} (b + ax_i - y_i) \quad \ldots (2.6)$$

- Solving the above two equations, we get

$$a = \frac{n\Sigma x_i y_i - \Sigma(x_i)\,\Sigma(y_i)}{n\Sigma x_i^2 - (\Sigma x_i)^2} \quad \ldots (2.7)$$

$$b = \frac{(\Sigma y_i)(\Sigma x_i)^2 - \Sigma(x_i y_i)\,\Sigma(x_i)}{n\Sigma x_i^2 - (\Sigma x_i)^2} \quad \ldots (2.8)$$

- This method of least squares can also be used for determining higher - order polynomial for a data set.
- Linearity can be expressed as a percentage of the actual reading or a percentage of full-scale reading or a combination of both.
- The most realistic method of expressing linearity is the combination of both actual and full scale reading, which is known as the independent linearity.
- Independent linearity = $\pm A$ % of reading
  or $\pm B$ % of full-scale,

whichever is greater.

- The specification of independent linearity is illustrated in fig. (2.2).
- In commercial transducers, linearity is specified as the percentage of full-scale reading only.

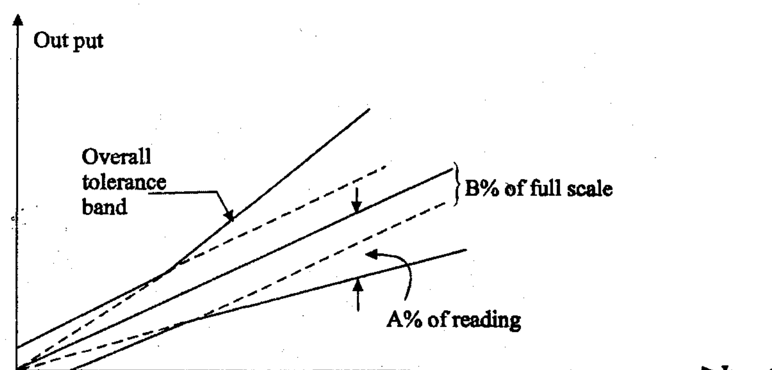


Fig. 2.2 Linearity specification

- In such cases, the transducer gives more accurate result only for readings above 50% of the full-scale value.

- For transducers which are considered linear, the specification of linearity is the specification of overall accuracy.
- Hence if only linearity specification is given by the manufacturer it may be taken as the accuracy specification.

2.2.5 Accuracy

- It is the closeness with which an instrument reading approaches the true value of the quantity being measured.
- Thus accuracy of a measurement means conformity to truth.
- The accuracy may be specified in terms of inaccuracy or limits of error.
- The accuracy can be expressed in the following ways.

1. Point accuracy

- This is the accuracy of the instrument only at one point on its scale.
- The specification of this accuracy does not give any information about the accuracy at other points on the scale. In other words, this accuracy does not give any information about the general accuracy of the instrument.

2. Accuracy as 'percentage of scale range'

- When an instrument has uniform scale, its accuracy may be expressed in terms of scale range.
- For example, the accuracy of a thermometer having a range of 500°C may be expressed as ± 0.5 percent of scale range.
- This means that the accuracy of the thermometer when the reading is 500°C is ± 0.5 percent.

3. Accuracy as 'percentage of true value'

- The best way to express the accuracy is to specify it in terms of the true value of the quantity being measured i.e., within ± 0.5 percent of true value.
- This statement means that the errors are smaller as the readings get smaller.

- Thus at 5% of full scale the accuracy of the instrument would be 20% better than that of an instrument which is accurate to + 0.5% of scale range.

2.2.6 Precision

- It is a measure of the reproducibility of the measurements.
- precision is the degree of closeness with which a given value may be repeatedly measured.
- When a transducer is used to measure the same input at different instances, the output may not be same.
- The deviation from the nominal output in absolute units or a fraction of full-scale is called th precision error or repeatability error.
- The term 'precise' means clearly or sharply defined.
- precision is composed of two characteristics:

(i) Conformity and (ii) Number of significant figures.

- precision is used in measurements to describe the consistency or the reproducibility of results.
- A quantity called precision index describes the spread, or dispersion of repeated result about some central value.
- High precision means a tight cluster of repeated results while low precision indicates a broad scattering of results.

2.2.7 Significant figures

- An indication of the precision of the measurement is obtained from the number of significant figures in which it is expressed.
- Significant figures convey actual information regarding the magnitude and the measurement precision of a quantity.
- The more the significant figures, the greater the precision of measurement.
- For example, if a voltage is specified as 230 V its value should be taken as closer to 230 V than to either 231 V or 229 V.
- If the value of voltage is specified as 230.0 V, it means that the voltage is closer to 230.0 V than it is to 230.1 V or 229.9 V.

- In 230 there are three significant figures while in 230.0 V there are four.
- The latter, with more significant figures, expresses a measurement of greater precision than the former.

2.2.8 Hysteresis

- Hysteresis is a phenomenon which depicts different output effects when loading and unloading whether it is a mechanical system or an electrical system.
- Hysteresis is non-coincidence of loading and unloading curves.
- When the input to a transducer which is initially at rest is increased from zero to full-scale and then decreased back to zero, there may be two output values for the same input (see fig. 2.3 (a))
- This mismatching of the input-output curves is mainly due to internal friction and change in damping of the spring elements in the transducer.
- In a system, it arises due to the fact that all the energy put into the stressed parts when loading is not recoverable upon unloading.
- Hysteresis effects can be minimised by taking readings corresponding to ascending and descending values of the input and then taking their arithmetic average.
- In case of instruments which are used on both sides of zero i.e. input applied on both positive and negative side, the variation of output is as shown in fig. (2.3 (b)).

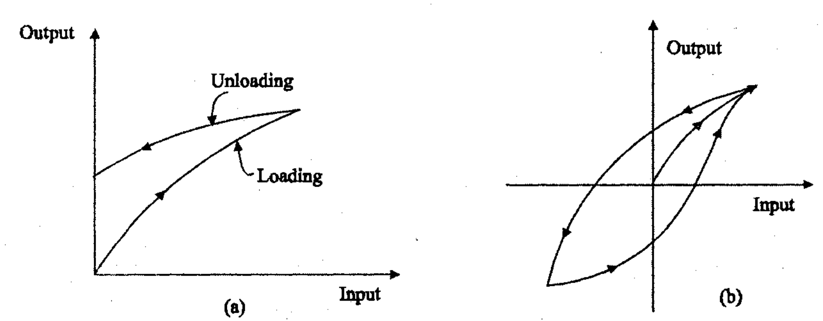


Fig. 2.3 Hysteresis effects

2.2.9 Threshold

- When the input to a transducer is increased gradually from zero, there is a minimum value below which no output can be detected.
- This minimum value of the input is defined as the threshold of the transducers.
- This phenomenon is due to input hysteresis. In mechanical instruments, the first noticeable measurable change may not occur on account of backlash.
- In fig (2.4) which shows a gear train, the driven gear will not move i.e. there will be no noticeable change in the movement of the driven gear unless the driving gear moves through a distance x which is the backlash between the gears.

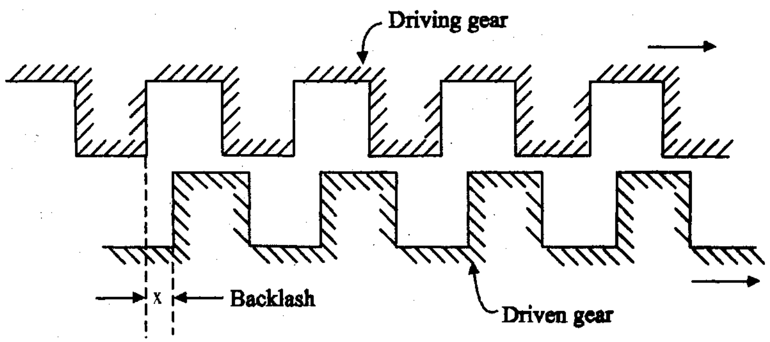


Fig. (2.4) Threshold because of Backlash

2.2.10 Dead time

- Dead time is defined as the time required by a measurement system to begin to respond to a change in the measurand.
- Fig (2.5) shows the measured quantity and its value as indicated by an instrument.
- Dead time is the time before the instrument **begins to respond** after the measured quantity has been changed.

2.2.11 Dead zone

- It is defined as the largest change of input quantity for which there is no output of the instrument. (see fig. 2.5)
- For example if the input applied to the instrument is insufficient to overcome the friction, it will not move at all.
- It will only move when the input is such that it produces a driving force which can overcome friction forces.
- Dead zone is used to backlash and hysteresis in the instrument.

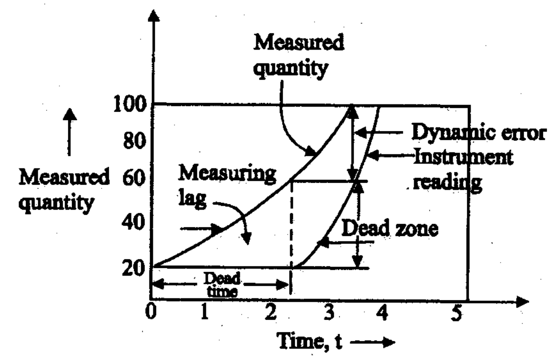


Fig. 2.5 Dead time and Dead zone

2.2.12 Resolution or Discrimination

- When the input to a transducer is slowly increased from some arbitrary (non-zero) value, the change in output is not detected at all until a certain input increment is exceeded.
- This increment is called resolution or discrimination of the instrument.
- Thus the smallest increment in input (the quantity being measured) which can be detected with certainty by an instrument is its resolution or discrimination.
- So resolution defines the smallest measurable input change while the threshold defines the smallest measurable input.
- The resolution of digital instruments is decided by the number of digits used for display.

- For example, the resolution of a four-digit voltmeter with a range of 999.9 volts is 0.1 volt. Whereas for a five-digit voltmeter of the same range, the resolution would be 0.01 volt.

2.2.13 Range and span

- Generally a transducer is recommended to be used between a high and a low values of input.
- The range of the transducer is specified as from the low value of input to the high value of input.
- The span of the transducer is specified as the difference between the high and the low limits of recommended input values.
- For example, if a temperature transducer is recommended to be used between 100°C and 500°C, its range is specified as 100°C to 500°C, whereas its span is 400°C (i.e. 500°C – 100°C = 400°C).
- When an ammeter is specified to be used between 0 and 100 mA, its range is 0 to 100 mA and its span is 100 mA (i.e. 100 mA – 0 mA = 100 mA).

2.2.14 Input Impedance

- A transducer used for any measurement normally extracts some energy from the measuring medium and thereby disturbs the value of the measured quantity.
- This property is known as the loading effect of the transducer.
- An ideal transducer is one which does not absorb any energy and hence does not disturb the prevailing state of the measured quantity.
- The loading effect of a transducer gives a measure of its disturbance on the measuring quantity.
- The loading effect is usually expressed in terms of input impedance and stiffness.
- The fig. (2.6) shows a voltage signal source and input device connected across it.
- The magnitude of the impedance of element connected across the signal source is called "Input Impedance".

- The magnitude of the input impedance is given by

$$Z_i = \frac{e_i}{i_i} \qquad \text{... (2.9)}$$

- The instantaneous power extracted by the input device from the signal source is,

$$p = e_i i_i = \frac{e_i^2}{z_i} \qquad \text{... (2.10)}$$

- From equations (2.9) & (2.10), it is clear that a low input impedance device connected across the voltage signal source draws more current and drains more power from signal source than a high input impedance device.
- In other words a low input impedance device connected across a voltage signal source loads the source more heavily than a high input impedance device.

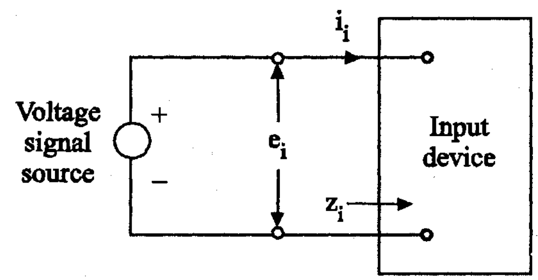


Fig. (2.6) voltage source and input device

2.2.15 Input admittance

- When the signal is of the form of current then series input devices are used.
- Consider a constant current source and an input device connected across it as shown in fig. (2.7)
- The magnitude of input admittance is given by:

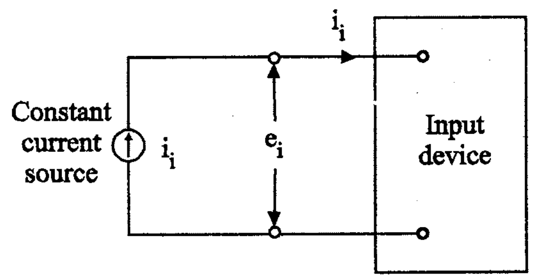


Fig. 2.7 current source and input device

$$y_i = \frac{i_i}{e_i} \quad \text{... (2.11)}$$

- Input impedance, $Z_i = \frac{e_i}{i_i} = \frac{1}{y_i}$ ... (2.12)
- The instantaneous power extracted from signal source is:

$$p = i_i e_i = \frac{i_i^2}{y_i} = i_i^2 z_i \quad \text{... (2.13)}$$

- From the above equations, it is clear that if the input admittance of the device is high, then the power drawn from the current signal source is small in case of series elements (i.e) input impedance is low.
- Therefore, the loading effects are small when their input admittance is large (i.e. when their input impedance is small).

2.3 DYNAMIC CHARACTERISTICS OF TRANSDUCERS

- The dynamic characteristics of a transducer refers to the performance of the transducer when it is subjected to time-varying input.
- The number of parameters required to define the dynamic behaviour of a transducer is decided by the group to which the transducer belongs.
- The transducers can be categorized into
    1. Zero-order transducers
    2. first-order transducers
    3. Second-order transducers
    4. Higher-order transducers

Order of a transducer

The order of a transducer is the highest derivative of the differential equation which describes the dynamic behaviour of a transducer for a specified input.

For example,

If the differential equation relating the input and output of a transducer is

$$\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + 4y(t) = x(t) \quad \text{... (2.14)}$$

where,

$y(t) \rightarrow$ output

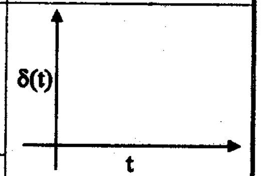
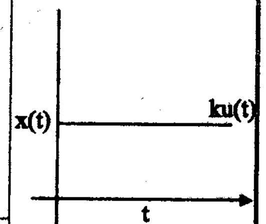
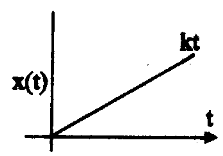
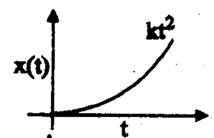
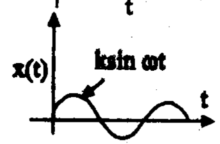
$x(t) \rightarrow$ input

- The highest derivative of the output is 3.
- The order of the transducer is the same as the highest derivative of the output.

Test inputs

- The transducers are normally subjected to inputs of random nature.
- The following test inputs are applied to the transducer to determine its dynamic behaviour.
    1. impulse input
    2. step input
    3. ramp input
    4. Parabolic input
    5. Sinusoidal input
- The various test inputs are represented in the following table (2.1).

Table (2.1): Test inputs

Sl.No.	Name of the input	Time function	Laplace function	Pictorial representation
1.	Impulse input	$x(t) = \delta(t)$ $= 1$ for $t = 0$ $= 0$ for $t \neq 0$	1	
2.	Step input	$x(t) = K$ for $t > 0$ $= 0$ for $t < 0$ If $K = 1$ $x(t) = u(t)$ = unit step	$\frac{K}{S}$	
3.	Ramp input	$x(t) = Kt$ for $t \geq 0$ $= 0$ for $t \leq 0$	$\frac{K}{S^2}$	
4.	Parabolic input	$x(t) = Kt^2$ for $t \geq 0$ $= 0$ for $t \leq 0$	$\frac{2K}{S^3}$	
5.	Sinusoidal input	$x(t) = K \sin wt$ for $t > 0$ $= 0$ for $t \leq 0$	$\frac{Kw}{s^2 + \omega^2}$	

2.3.1 Zero-order transducer

- The input - output relationship of a zero-order transducer is given by

$$y(t) = Kx(t) \qquad \text{... (2.15)}$$

where,

$x(t) \rightarrow$ input

$y(t) \rightarrow$ output

$K \rightarrow$ Static - sensitivity of the transducer

- The transfer function of the zero-order transducer is given by

$$\frac{Y(s)}{X(s)} = K \qquad \text{... (2.16)}$$

- This equation shows that the output varies in the same way as the input.
- Hence, a zero-order transducer response, represents ideal dynamic performance.

Example

- Potentiometer used for displacement measurements is an example for zero-order transducer.
- The output of a potentiometer is given by

$$e_0 = E_b \cdot \frac{x_i}{L}$$

$$= K \cdot x_i$$

where,

$x_i \rightarrow$ displacement of the slider

$L \rightarrow$ total length of the potentiometer

$E_b \rightarrow$ excitation voltage

$e_o \rightarrow$ output in volts

- The static sensitivity of the potentiometer is

$$K = \frac{E_b}{L} \text{ volts/cm.}$$

- The potentiometer behaves as a zero-order instrument when it is a pure resistance.
- The response of zero-order transducers for step input is given in figure (2.9).

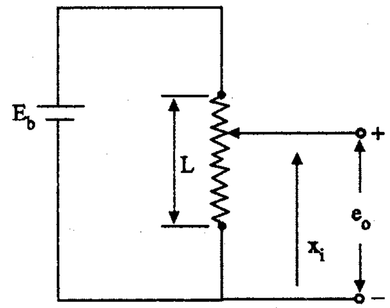


Fig. (2.8) potentiometer (zero-order instrument)

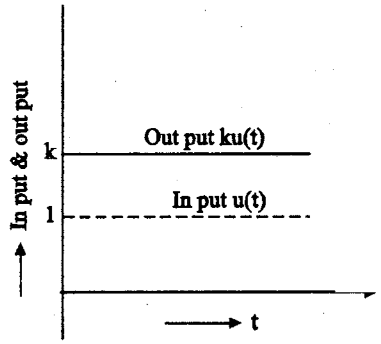


Fig. (2.9) step response of zero-order transducer

2.3.2 First - order transducer

- The differential equation relating the input and output of a first-order transducer is

$$a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_0 u(t) \qquad \text{... (2.17)}$$

where,

a_1, a_0 and $b_0 \rightarrow$ Transducer parameters

- The transfer function of the first-order transducer is given by

$$\frac{y(s)}{x(s)} = \frac{\frac{b_0}{a_0}}{\left[\frac{a_1}{a_0}s + 1\right]} = \frac{K}{(\tau s + 1)} \qquad \text{... (2.18)}$$

where,

$$K = \frac{b_0}{a_0} = \text{static sensitivity}$$

$$T = \frac{a_1}{a_0} = \text{time constant}$$

Example

Thermocouple used for temperature measurements is an example for first-order transducer.

- Let us consider a thermocouple immersed in fluid in a bath (see fig. 2.10).
- The heat balance equation is

$$QA(T_2 - T_1) = MS\frac{dT_1}{dt} \qquad \text{... (2.19)}$$

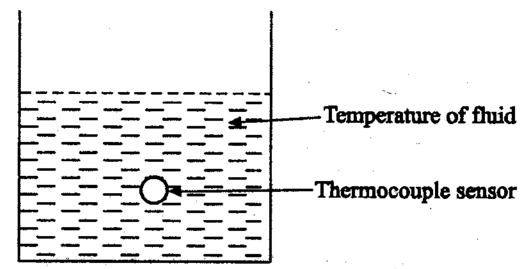


Fig. (2.10) Thermocouple (first-order transducer)

where,

Q - Overall heat-transfer coefficient

A - Heat transfer area

T_1 - Temperature indicated by the thermocouple

T_2 - Temperature of the fluid

M - Mass of the sensing portion of the thermocouple.

S - Specific heat of the sensing bead.

- The transfer function is given by

$$\frac{T_1(s)}{T_2(s)} = \frac{1}{\tau s + 1} \quad \text{... (2.20)}$$

where,

$$\tau = \frac{MS}{QA}$$

- The voltage output of a thermocouple is proportional to the difference in temperature of hot junction and cold junction.

$$V \propto T_1 \sim T_2$$

- As the cold junction is kept constant at 0°C, the voltage output is proportional to the temperature of the bead at the hot junction. (i.e)

$$V \propto T_1$$

$$V = KT_1$$

where,

V - Thermocouple output in volts.

K - proportionality constant.

- The overall transfer function of the thermocouple is given by

$$\frac{V(s)}{T_2(s)} = \frac{V(s)}{T_1(s)} \cdot \frac{T_1(s)}{T_2(s)} \quad \text{... (2.21)}$$

$$\boxed{\frac{V(s)}{T_2(s)} = \frac{K}{\tau s + 1}} \quad \text{... (2.22)}$$

- The equation (2.22) shows that the thermocouple is a first order transducer.
- When the hot junction of a thermocouple is kept inside a thermal wall in order to protect it from abrasive and corrosive effects of the surroundings, the transducer becomes a second order one.

2.3.2.1 *Responses of First - order transducer*

- First - order systems are characterised by a transfer function represented as

$$G(s) = \frac{Y(s)}{X(s)} = \frac{b_0}{a_1 s + a_0}$$

and rewritten as

$$G(s) = \frac{K}{1 + \tau s} \quad \text{... (2.23)}$$

where,

$$K = \frac{b_0}{a_0} \rightarrow \text{static sensitivity}$$

$$T = \frac{a_1}{a_0} \rightarrow \text{time constant of the system}$$

- Let us study the response of I-order transducer for standard input signals.

1. Response of I order transducer for step input

- If the I order transducer is excited by a unit step input function $X(s) = \frac{1}{s}$, then $Y(s)$ is given by

$$Y(s) = \frac{1}{s} \cdot \frac{K}{1 + \tau s}$$

So,

$$y(t) = k\left(1 - e^{-t/\tau}\right) \quad \text{... (2.24)}$$

- Equation (2.24) reveals the fact that $y(t)$ assumes a final value of k slowly with time.
- The speed of response is dependent on the value of τ.
- The smaller the value of τ, the higher the speed of response.

- Fig. (2.11) shows the response of a first-order transducer for a step-input function.

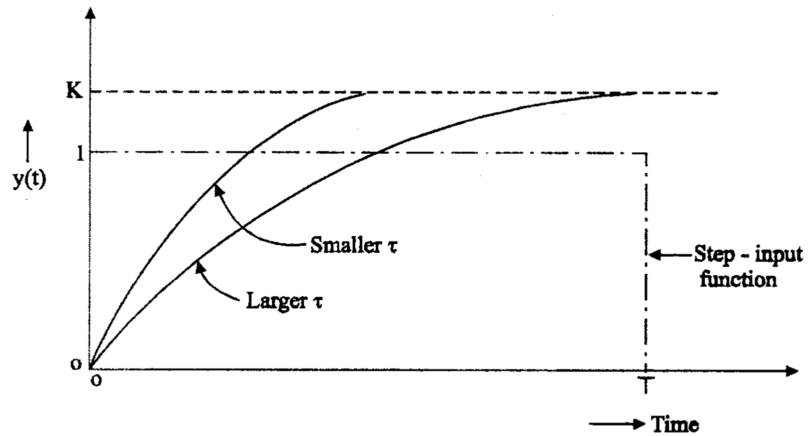


Fig. 2.11 step response of a first - order transducer

2. Response of I-order transducer for ramp input

- If the input function is of the unit-ramp type, then the input-output relationship of a I-order transducer is given by

$$Y(s) = \frac{1}{s^2} \cdot \frac{K}{1 + \tau s}$$

and when solved, $y(t)$ is given by

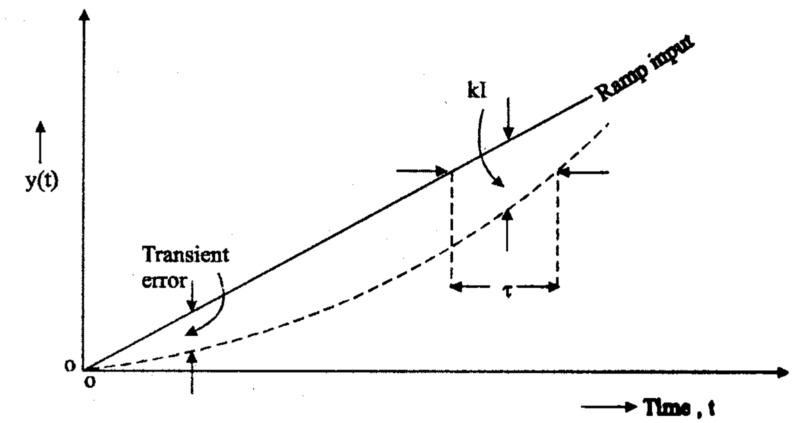


Fig. 2.12 Ramp of first-order transducer

$$y(t) = K[e^{-t/\tau} + t - \tau] \quad \text{... (2.25)}$$

- The ramp response of first-order transducer is shown in fig. (2.12)
- If the transducer is ideal, it should result in an output signal $y(t) = Kt$, but there is a deviation from this value due to its time constant.
- Hence the dynamic error is given by

$$\text{Dynamic error} = + K(\tau e^{-t/\tau}) - K\tau \quad \text{... (2.26)}$$

- The first term of the net dynamic error dies with time and hence it constitutes transient error, whereas the second term $K\tau$ becomes the steady state error.
- Under steady state conditions, the amplitude of output attains the true value after τ seconds only.

3. Response of I - order transducer for unit - impulse function

- The response for a unit - impulse function is represented by

$$Y(s) = \frac{K}{1 + \tau s}$$

$$y(t) = \frac{K}{\tau} e^{-t/\tau} \quad \text{... (2.27)}$$

- If the strength of the impulse is A units, the response becomes A times the one given by equ. (2.27).

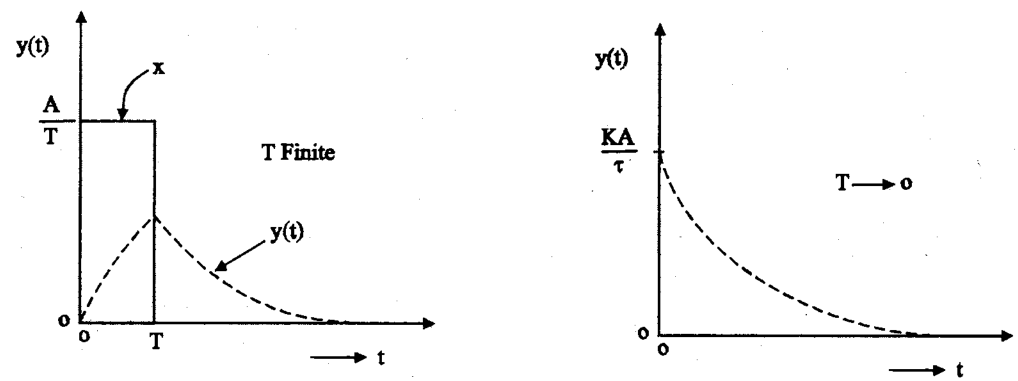


Fig. (2.13) Response of first-order system (a) for a prolonged impulse - input; (b) for an ideal impulse input.

- The impulse responses of I order transducer are shown in fig. (2.13) (a) & fig. (2.13) (b).

4. Frequency response of first - order transducer

- For sinusoidal input functions, the frequency response is determined from the relation

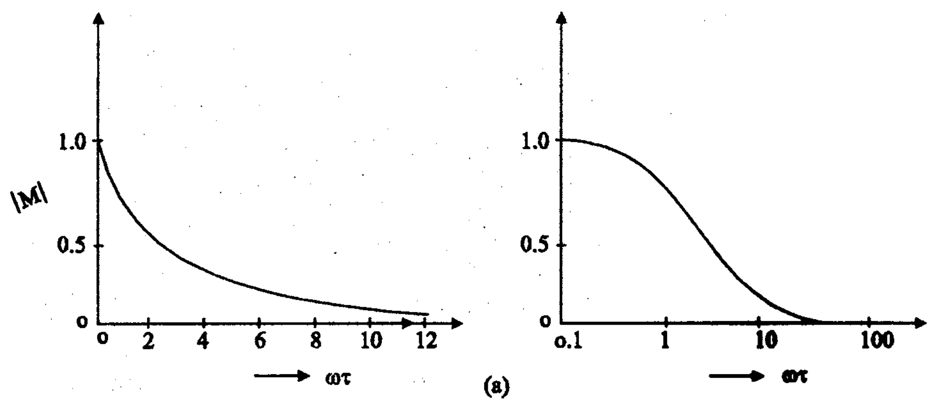
$$\frac{Y(j\omega)}{X(j\omega)} = \frac{K}{1 + j\omega\tau}$$

$$= \left| \frac{K}{1+\omega^2\tau^2} \right| \underline{|\tan^{-1} - \omega\tau}$$

$$= |M| \underline{|\phi} \qquad \text{... (2.28)}$$

- At zero frequency, i.e., under dc excitation, the value of $|M|$ becomes equal to K with $\phi = 0$.

- Treating the natural frequency of the system, ω_n, as given by $\frac{1}{\tau}$, the frequency response curves relating $|M|$ and $\underline{|\phi}$ with $\frac{\omega}{\omega_n}$ $(= \omega\tau)$ are shown in fig. (2.14).



(a)

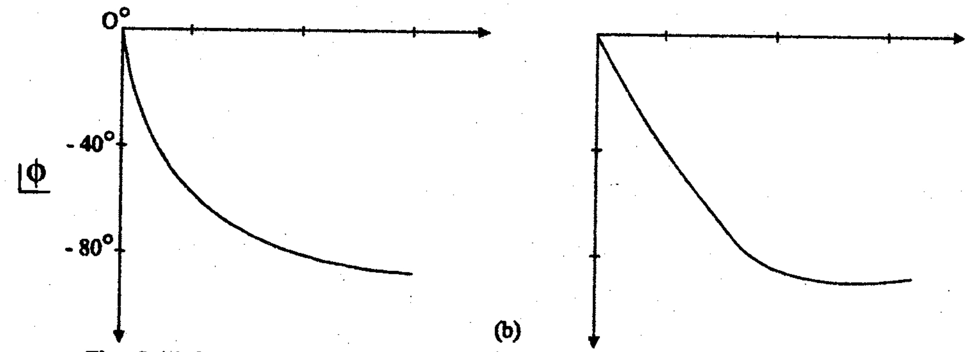


Fig. 2.14 frequency response characteristics of a first -order system: (a) for amplitude (b) for phase

2.3.3 Second - order transducer

- Second - order systems are characterized by a transfer function given by

$$G(s) = \frac{Y(s)}{X(s)} = \frac{b_0}{a_2 s^2 + a_1 s + a_0}$$

which can be rewritten as

$$G(s) = \frac{K}{s^2\left(\frac{a_2}{a_0}\right) + s\left(\frac{a_1}{a_0}\right) + 1} \qquad \text{... (2.28)}$$

where,

$$K \text{ - static sensitivity } \left(= \frac{b_0}{a_0}\right)$$

- The undamped natural frequency ω_n of the second - order system becomes $\sqrt{\left(\frac{a_0}{a_2}\right)}$

- The ratio $\left(\frac{a_1}{a_0}\right)$ signifies the damping conditions of the system.

- The damping factor (or damping ratio) is

$$\xi = \frac{a_1}{2\sqrt{a_0 a_2}} \quad \text{... (2.29)}$$

$$2\frac{\xi}{\omega_n} = \frac{a_1}{a_0} \quad \text{... (2.30)}$$

- The equation (2.28) can be rewritten as

$$\frac{Y(s)}{X(s)} = \frac{K}{\left[\frac{s^2}{\omega_n^2} + \frac{2\xi s}{\omega_n} + 1\right]} \quad \text{... (2.31)}$$

1. Response of II order transducer for step input

- When a second-order transducer is subjected to an unit step input,

$$X(s) = \frac{1}{s}$$

- The laplace transform of the output is given by

$$Y(s) = \frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \cdot \frac{1}{s} \quad \text{... (2.32)}$$

$$Y(s) = \frac{K}{s\left[\frac{s^2}{\omega_n^2} + \frac{2\xi s}{\omega_n} + 1\right]}$$

- $y(t)$ for different damping conditions is given by

$$\frac{y(t)}{K} = \left[1 - \frac{e^{-\omega_n \xi t}}{\sqrt{1-\xi^2}} \sin\{\omega_n \sqrt{(1-\xi^2)}\, t + \sin^{-1}\sqrt{1-\xi^2}\}\right] \text{ for } \xi < 1 \quad \text{... (2.33)}$$

$$\frac{y(t)}{K} = [1 - (1 + \omega_n t)\, e^{-\omega_n t}] \text{ for } \xi = 1$$

$$\frac{y(t)}{K} = \left[1 - \frac{e^{-\omega_n \xi t}}{\sqrt{\xi^2 - 1}} \sinh\{\omega_n \sqrt{(\xi^2 - 1)}\, t + \sinh^{-1}\sqrt{\xi^2 - 1}\}\right] \text{for } \xi > 1 \quad \text{... (2.34)}$$

- The step responses of second-order transducer for various values of damping ratios are shown in fig. (2.15).
- Whenever a second-order transducer is suddenly connected to an input, it is equivalent to the application of step input.
- To have a quick indication of the measured values, the time taken for the transducer response to reach the steady - state value should be minimum.
- As the second-order system subjected to step-input takes infinite time to reach the steady-state value, it is customary to define settling time for such systems.
- The settling time is the time taken for the output to reach and stay within a specified percentage of steady-state value.
- For example, 10% settling time means, the time taken for the system output to reach and stay within 90% to 110% of the steady-state value.

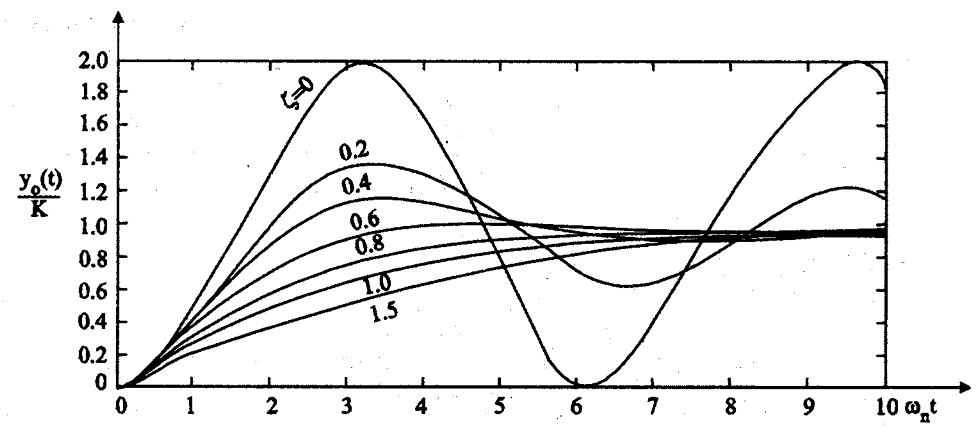


Fig. (2.15) step response of a II - order transducer for various value of ξ.

(2) Response of second - order transducer for ramp input

- Let us consider a second-order transducer subjected to ramp input given by

$$r(t) = At$$

$$R(s) = \frac{A}{s^2} \qquad \text{... (2.35)}$$

- The response of a second-order system for ramp input is given by

$$y(s) = \frac{K\omega_n^2}{s^2 + 2\,\xi\omega_n s + \omega_n^2} \cdot \frac{A}{s^2} \qquad \text{... (2.36)}$$

$$y(s) = \frac{KA}{s^2\left[\dfrac{s^2}{\omega_n^2} + \dfrac{2\xi s}{\omega_n} + 1\right]}$$

- By partial fraction,

$$y(s) = \frac{B_1}{s} + \frac{B_2}{s^2} + \frac{B_3 s + B_4}{s^2 + 2\xi\omega_n s + \omega_n^2} \qquad \text{... (2.37)}$$

- By comparing the coefficients of S, B_1, B_2, B_3, B_4 are determined

$$y(s) = KA\left[\frac{-2\dfrac{\xi}{\omega_n}}{s} + \frac{1}{s^2} + \frac{2\dfrac{\xi}{\omega_n s} + 4\xi^2 - 1}{s^2 + 2\xi\omega_n s + \omega_n^2}\right] \qquad \text{... (2.38)}$$

- The output $y(t) \cdot$ [for $\xi < 1$] is

$$y(t) = KAt - \frac{KA2\xi}{\omega_n}\left[1 - \frac{e^{-\xi\omega_n t}}{2\xi\sqrt{1-\xi^2}}\sin\left(\omega_n\sqrt{1-\xi^2}\,t + \phi\right)\right] \qquad \text{... (2.39)}$$

where,

$$\phi = \tan^{-1}\frac{2\xi\sqrt{1-\xi^2}}{2\,\xi^2 - 1} \qquad \text{... (2.40)}$$

- The ramp response of a II order system in shown in fig. (2.16).
- It is seen that there is a steady state error of $2\xi\dfrac{K}{\omega_n}$

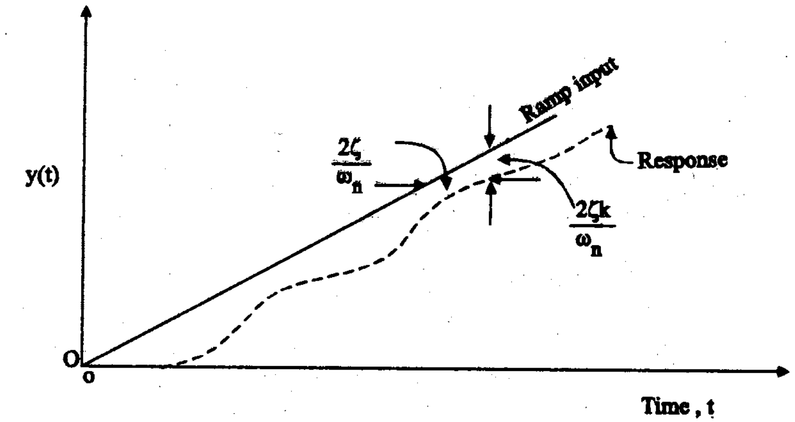


Fig. 2.16 Ramp response of a II order system

- The steady state error decreases as ω_n increases and is proportional to ξ.
- Under steady state conditions, there is a time lag of $\dfrac{2\xi}{\omega_n}$ in the indication of the true value.
- For a given ω_n if ξ is reduced, oscillations persist for a longer time, but the steady state time lag and steady error becomes less.
- The output for $\xi = 1$ is

$$y(t) = KAt - \frac{2KA}{\omega_n}\left[1 - e^{-\omega_n t}\left(1 + \frac{\omega_n t}{2}\right)\right]$$

- The output for $\xi > 1$ is

$$y(t) = KAt - \frac{2\xi KA}{\omega_n}\left[\begin{array}{l} 1 + \dfrac{2\xi^2 - 1 - 2\xi\sqrt{\xi^2-1}}{4\xi\sqrt{\xi^2-1}}\, e^{(-\xi + \sqrt{\xi^2-1})\,\omega_n t} \\ + \dfrac{-2\xi^2 + 1 - 2\xi\sqrt{\xi^2-1}}{4\xi\sqrt{\xi^2-1}}\, e^{(-\xi + \sqrt{\xi^2-1})\,\omega_n t)} \end{array}\right] \qquad \text{... (2.41)}$$

(3) Response of a II order transducer for terminated ramp input

- It is quite realistic to assume that electrical and electronic instruments are subjected to step and ramp-input excitations.

- But other physical instruments, designed for measurement of pressure or temperature are unlikely to experience step changes of input quantities.
- Hence the input is considered to change from the initial value in a ramp fashion until it becomes constant.

Such a change is treated as terminated ramp input function and is represented in fig. (2.17), assuming that

$$x(t) = \frac{t}{T} \quad \text{for} \quad 0 \le t \le T \qquad \text{... (2.42)}$$

$$= 1 \quad \text{for} \quad T \le t \le \infty$$

$$\frac{dy(t)}{dt} = 0 = y(t) \quad \text{at} \quad t = 0$$

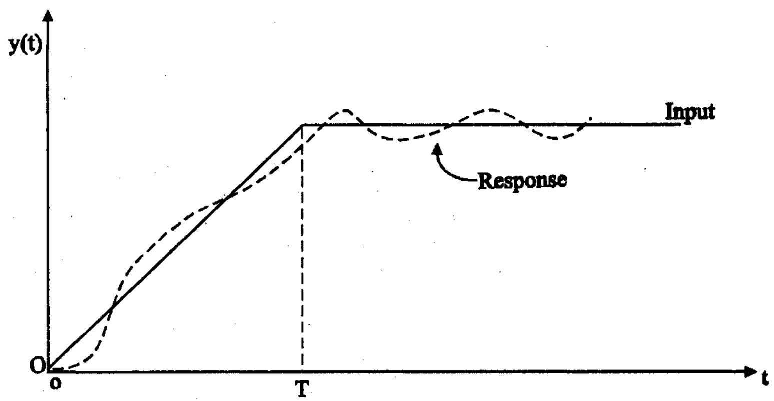


Fig. (2.17) Terminated ramp response of a second - order system

(4) Frequency response of a second-order transducer

- The frequency response of the II-order system is obtained from its transfer function and is given by

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{K}{\left[-\left(\frac{\omega}{\omega_n}\right)^2 + \frac{2j\omega\xi}{\omega_n} + 1\right]} \qquad \text{... (2.43)}$$

- Writing $\frac{\omega}{\omega_n} = \eta$, the ratio of the frequency of the forcing function to its natural frequency, the response is expressed as

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{K}{[(1-\eta^2)^2 + 4\xi^2\eta^2]^{1/2}} \angle\phi \qquad \text{... (2.44)}$$

where

$$\phi = \tan^{-1} \frac{2\xi\eta}{\sqrt{1-\eta^2}}$$

$$= |M|\ \phi$$

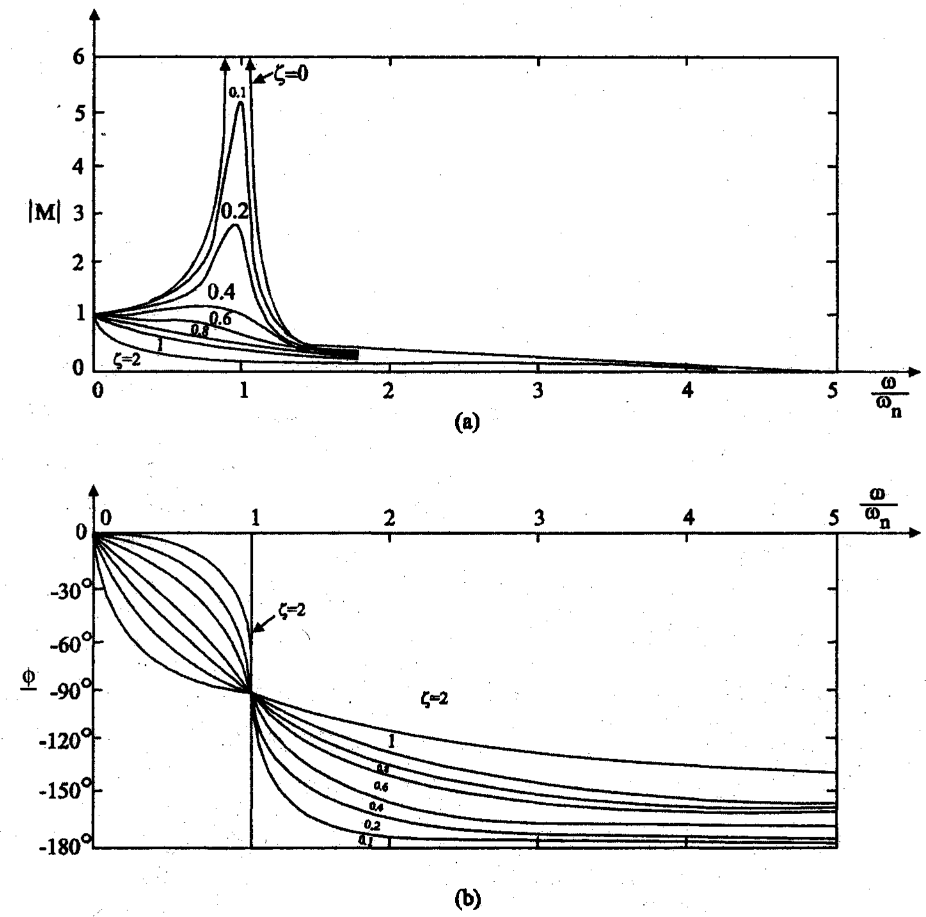


Fig. 2.18 Frequency response characteristics of a second order system for (a) amplitude (b) phase

- The frequency response characteristics of a second-order system for amplitude ($|M|$) and phase ($\angle\phi$) are shown in fig. (2.18).

2.3.4 Higher Order Transducers

- The system which can be described by higher order differential equations is higher order system.
- Many transducers have higher order dynamics which can be described by higher order differential equations.
- For analysis, they can be represented by either first-order or second-order differential equations with some assumptions.
- However, for accurate analysis, the higher order equations can be taken as it is and solved.
- The response of the higher order transducers would be similar to that of second-order transducers with a sluggish rise in the initial period.

Frequency response

- The response of a system to a frequency input is called frequency response of a system.
- The response of a transducer to a frequency input (frequency response of transducer) is an important characteristic since most of the signals can be considered to be a combination of signals of different frequencies.
- The sensitivity of a transducer should be same for all frequencies and phase shift should be either zero or it should increase linearly with frequency.
- That means, the amplitude plot of the frequency response should be flat for all frequencies.
- In general, this plot drops at higher frequencies.
- The term bandwidth is used to quantify the flat useful region of the amplitude plot of the frequency response.
- The bandwidth is defined as the frequency range in which the amplitude ratio is more than 0.707 of the final value.
- This is shown in fig. (2.19)

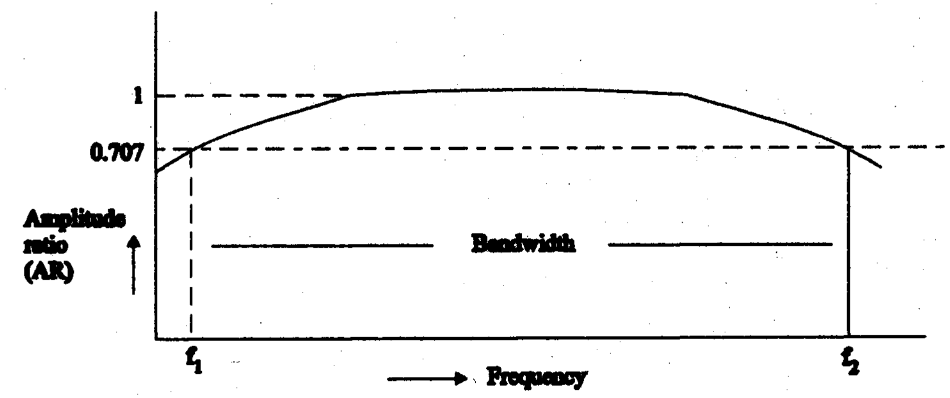


Fig. 2.19 Bandwidth of a transducer - frequency response

- If all the frequency components of the input lie within the bandwidth of the transducer, then the transducer will faithfully reproduce the input.
- If the frequency components of the input signal are outside the bandwidth of the transducer, then the output will be distorted.
- I' important information is in the frequencies outside the bandwith, then this information may be missed.

2.4 MATHEMATICAL MODEL OF TRANSDUCERS

- The mathematical models are the differential equations that describe the dynamics of transducers.
- These models can be derived from the knowledge of the components, their interconnection and the physical laws governing their functioning.
- A number of assumptions are needed to derive the equations representing the model.
- But practically, the components used, their values, their behaviour, their interconnections and the physical laws followed by them may not be precisely known.
- Therefore using conventional method, the model cannot be obtained.
- In such situations, the transducer can be assumed to be a black box, whose inputs and outputs are accessible for measurements.

- Number of methods are available to identify the transducer model by measuring the inputs and outputs of the transducer.
- If the order of the model is known already, then the method of identification becomes simple.

(1) Identification of transducer mathematical models

Identification from Impulse response

- Let the transfer function of the transducer be of the form $\frac{K}{(1+\tau s)}$ where K and τ are the only unknown parameters.
- When this transducer is excited with an input impulse, the output transform

$$Y(s) = \frac{K}{(1+\tau s)}$$

as $$R(s) = 1$$

- Therefore $y(t) = L^{-1}\, Y(s)$

$$= Ke^{-t/\tau} \quad \text{... (2.45)}$$

- The output of the transducer is shown in fig. (2.20)

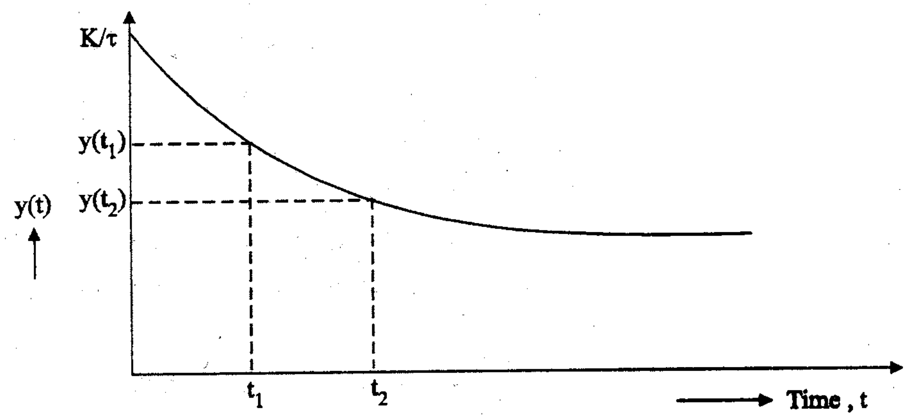


Fig. 2.20 First-order transducer response for impulse signal.

- From the experimentally obtained outputs of the transducer $y(t_1)$ and $y(t_2)$ at two different times t_1 and t_2, the two unknown parameters K and τ of the transducer can be estimated.

$$y(t_1) = \frac{K}{\tau} e^{-t_1/\tau} \quad \text{... (2.46)}$$

$$y(t_2) = \frac{K}{\tau} e^{-t_2/\tau} \quad \text{... (2.47)}$$

$$\frac{y(t_1)}{y(t_2)} = e^{(t_2-t_1)/\tau} \quad \text{... (2.48)}$$

$$T = \frac{(t_2 - t_1)}{\ln \frac{y(t_1)}{y(t_2)}}$$

- K can be calculated by substituting τ in one of the above equations.
- If the transfer function is of the form

$$\frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\xi}{\omega_n} s + 1}$$

- where ξ, ω_n and K are the unknown parameters.
- When such a system is subjected to an unit impulse, the response for the underemployed case will be as shown in fig. (2.21)

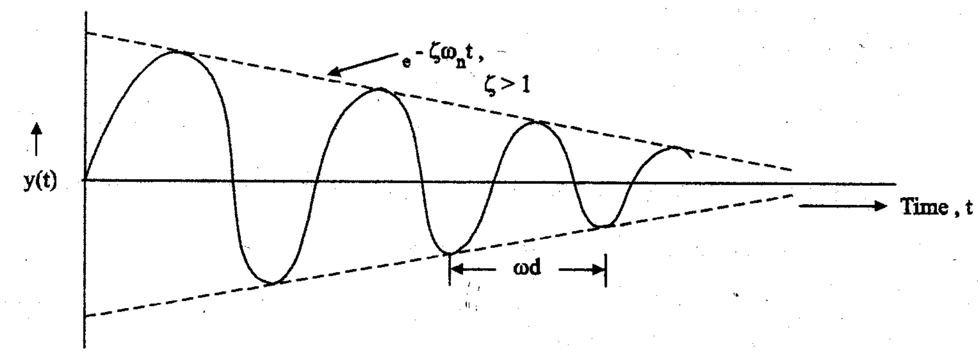


Fig. 2.21 Response of II - order transducer for impulse input signal.

$$y(t) = \frac{K\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin \omega_n [\sqrt{1-\xi^2}\, t] \qquad \ldots (2.49)$$

- From the experimental output curve, ξ, ω_n is calculated taking the envelope (dotted line) only.
- As the envelope is a decaying exponential curve, $\frac{1}{\xi\omega_n}$ is the time constant of the exponential curve.
- The time between two successive peaks T_d is determined which is equal to

$$\omega_d = \frac{2\pi}{T_d} = \omega_n \sqrt{1-\xi^2} \qquad \ldots (2.50)$$

- Now the values of ξ, and ω_n are determined from the values of $\frac{1}{\xi\omega_n}$ and the value of the envelope at $t = 0$ is

$$\frac{K\omega_n}{\sqrt{1-\xi^2}} \qquad \ldots (2.51)$$

which can be determined from the experimental response.

- As ξ and ω_n have already been evaluated, K can be calculated.

(2) Identification from step response

- When the transfer function of the transducer is of the form $\frac{K}{(1+\tau s)}$, the parameters K and τ have to be determined from the step response.
- The static sensitivity K is calculated as

$$K = \frac{\text{Steady state output charge}}{\text{Input change}} \qquad \ldots (2.52)$$

- For a second-order transducer, the parameters, K, ξ, and ω_n can be determined from the step response.

- The response of an under damped transducer for an unit step input is shown in fig. (2.22).
- The expression for the output $y(t)$ of the transducer is given by

$$y(t) = K\left[1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\right] \sin(\omega_n \sqrt{1-\xi^2}\, t + \phi) \qquad \ldots (2.53)$$

- The time instances at which the maximum and minimum values of the response curve occur can be found out by differentiating $y(t)$ with respect to time and equating to zero as shown below.

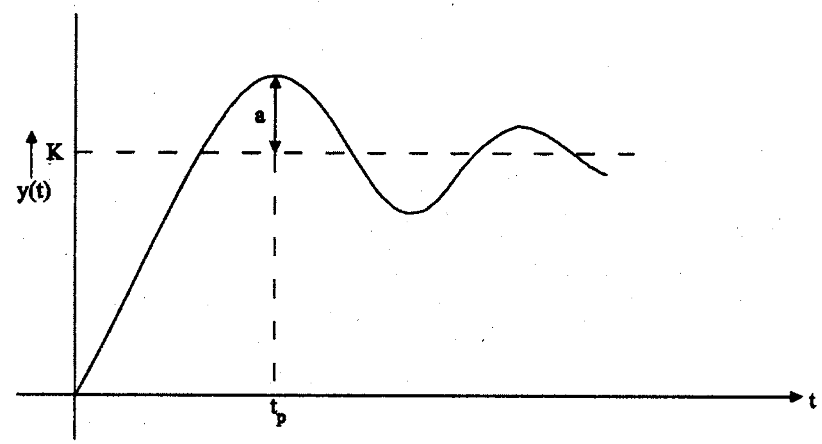


Fig. (2.22) second - order transducer response for step input.

$$\frac{dy(t)}{dt} = K\left\{ \xi\omega_n \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_n \sqrt{1-\xi^2}\, t + \phi) - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \cdot \cos(\omega_n \sqrt{1-\xi^2}\, t + \phi) \cdot \omega_n \sqrt{1-\xi^2} \right\} \qquad \ldots (2.54)$$

When this expression is equated to zero, one gets,

$$\tan(\omega_n \sqrt{1-\xi^2}\, t + \phi) = \frac{\sqrt{1-\xi^2}}{\xi}$$

- $\phi = \tan^{-1} \dfrac{\sqrt{1-\xi^2}}{\xi^2}$ (by definition)
- Therefore, $\tan(\omega_n \sqrt{1-\xi^2}\, t + \phi) = \tan \phi$ ... (2.55)
- This equation is true for all values of

$$t = 0, \frac{\pi}{\omega_n \sqrt{1-\xi^2}}, \frac{2\pi}{\omega_n \sqrt{1-\xi^2}}, \frac{3\pi}{\omega_n \sqrt{1-\xi^2}} \text{ etc.} \quad \text{... (2.56)}$$

- when $t = 0$, $y(t)$ is 0, minimum value

$$t = t_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} \quad \text{... (2.57)}$$

- $y(t)$ is the first maximum and t_p is the peak time.

$$t = t_v = \frac{2\pi}{\omega_n \sqrt{1-\xi^2}} \quad \text{... (2.58)}$$

- $y(t)$ - second minimum and t_v - valley time.
- As the oscillation is a damped one, the time at which the first maximum occurs will be the maximum overshoot.
- Therefore this overshoot shown as 'a' in fig. (2.22) can be obtained as

$a = y(t)|_{max} - y(t)|_{\text{steady state}}$

- $y(t)_{max}$ is obtained by substituting

$t_p = \dfrac{\pi}{\omega_n \sqrt{1-\xi^2}}$ in equation (2.59)

i.e.,

$$y(t)|_{max} = K\left\{\frac{1 - e^{-\xi\omega_n \frac{\pi}{\omega_n}\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} \sin\left(\omega_n \sqrt{1-\xi^2} \cdot \frac{\pi}{\omega_n \sqrt{1-\xi^2}} + \phi\right)\right\} \quad \text{... (2.60)}$$

$$= K\left\{1 + \frac{e^{-\pi\xi / \sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} \sin\phi\right\} \quad \text{... (2.61)}$$

$$= K\left[1 + e^{-\pi\xi/\sqrt{1-\xi^2}}\right] \text{ as } \sin\phi = \sqrt{1-\xi^2} \quad \text{... (2.62)}$$

$$y(t)|_{\textit{steady state}} = \underset{t\to\infty}{Lt}\; y(t)$$

$$= K(1-0)$$

$$= K$$

$$\therefore \text{ Overshoot, } a = K e^{-\pi\xi/\sqrt{1-\xi^2}} \quad \text{... (2.63)}$$

- From the step response plotted from experimental results, t_p, a and K can be obtained from equation (2.52). ξ can be calculated from equation (2.63) as a and K are already known.
- Substituting this value of ξ in equ. (2.58), ω_n can be determined.

TWO MARK QUESTIONS AND ANSWERS

1. Define transducer.

Transducer is a device which is used to convert non electrical quantities in to electrical quantities.

2. Classify transducer.

On the basis of transduction form used, transducer is classified as,

- As primary and secondary transducers
- As Active and passive transducers
- As analog and digital transducers
- As transducers and inverse transducers.

3. **Define static characteristics.**
Static characteristics of a measurement system are in general, those that must be considered when the system or instrument is used to measure a condition not varying with time.

4. **Mention different types of static characteristics.**
Static characteristics are,

(a) Accuracy

(b) Sensitivity

(c) Reproducibility

(d) Drift

(e) Static error

(f) Dead zone.

5. **What is dynamic characteristics?**
Many measurements are concerned with rapidly varying quantities and therefore, for such cases we must examine the dynamic relations which exist between the output and the input. This is normally done with the help of differential equations. Performance criteria based upon dynamic relations constitute the dynamic characteristics.

6. **Mention the applications of dynamic characteristics.**
The applications of dynamic characteristics are,

- Zero-order transducers
- First-order transducers
- Second-order transducers
- Higher-order transducers.

7. **What are the test inputs of the transducer?**
The test inputs of the transducer are,

- Impulse input
- Step input
- Ramp input

- Parabolic input
- Sinusoidal input.

8. **Define zero-order transducer.**
The input-output relationship of a zero-order transducer is given by,

$Y(t) = K\,r(t)$

where, $r(t)$ is the input, $Y(t)$ is the output, K is the static sensitivity of the transducer. Example for zero order transducer is potentiometer.

9. **What is mathematical model?**
Mathematical model is a mathematical representation of a physical model and is achieved from the later by utilizing the physical loss.

10. **What is frequency response of ZOT?**
Frequency response is thus defined as the steady state output of a transducer when it is excited with sinusoidal input. The frequency response is represented with the help of two plots namely amplitude ratio versus frequency and phase angle shift versus frequency.

11. **What is damping ratio?**
The damping ratio c is an important parameter which decides the nature of oscillation in the transducer output. When $c = 0$, the second order system is said to be un damped and the system behaves like an oscillator. When $c = 1$, the second order system is said to be critical damped on when $c > 1$, the second order system is said to be over damped.

12. **Define sensitivity.**
Sensitivity should be taken depending on the operating point. The sensitivity is expressed in output unit/input unit.

13. **Define linearity.**
Linearity is a measure of the maximum deviation of the plotted transducer response from a specified straight line.

14. **Compare accuracy and precision.**
Accuracy is the closeness to true value where as precision is the closeness amongst the readings. Precision is the degree of closeness with which a given value may be repeatedly measured.

15. What is threshold?

When the input to a transducer is increased from zero, there is a minimum value below which no output can be detected. This minimum value of the input is defined as the threshold of the transducer.

16. Define resolution.

When the input to a transducer is increased slowly from some non-zero arbitary value, the change in output is not detected at all until a certain input increment is exceeded. This increment is defined as the resolution.

17. Define hysteresis.

When the input to a transducer which is initially at rest is increased from zero to full-scale and then decreased back to zero, there may be two output values for the same input. Hysteresis effects can be minimized by taking readings corresponding to ascending and descending value of the input and then taking their arithmetic average.

18. What is range and span?

The range of the transducer is specified as from the lower value of input to higher value of input.

The span of the transducer is specified as, the difference between the higher and lower limits of recommended input values.

19. What is rise time?

Rise time is defined as the time required for the system to rise from 0 to 100 percent of its final value.

20. A thermometer has a time constant of 3.5 sec. It is quickly taken from a temperature 0°C to a water bath having temperature 100°C. What temperature will be indicated after 1.5 S?

$\theta = \theta_0\ [1 - \exp\ (1 - t/\tau)]$

$= 100\ [1 - \exp\ (1 - 1.5/3.5)] = 34.86°C$

21. A temperature-sensitive transducer is subjected to a sudden temperature change. It takes 10 sec for the transducer to reach equilibrium condition (5 time constant). How long will it take for the transducer to read half of the temperature difference?

Time to reach equilibrium conditions $= 5\,\tau = 10s.$

Time constant $\tau = 10/5$

$= 2$ sec

$\theta = \theta_0\ [1 - \exp\ (1 - t/\tau)]$

$0.5 = 1 - [\exp\ (-t/2)]$

$\therefore$ t = 1.39 sec

22. What is primary transducer?

Bourdon tube acting as a primary transducer, senses the pressure and convert the pressure into displacement. No output is given to the input of the a bourdon tube. So it is called primary transducer. Mechanical device can act as a primary transducer.

23. What is secondary transducer?

The output of the bourdon tube is given to the input of the LVDT. There are two stages of transduction, firstly the pressure is converted into a displacement by the bourdon tube then the displacement is converted into analog voltage by LVDT. Here LVDT is called secondary transducer. Electrical device can act as a secondary transducer.

24. What is passive transducer?

In the absence of external power, transducer cannot work and it is called a passive transducer. Example: Capacitive, inductive, resistance transducers.

25. What is active transducer?

In the absence of external power, transducer can work and it is called active transducer. Example: Velocity, temperature, light can be transduced with the help of active transducer.

26. What is analog transducer?

Analog transducers convert the input quantity into an analog output which is a continuous function of time. Thus a strain gauge, an LVDT, a thermocouple or a thermistors may be called analog transducer, as they give an output which is continuous function of time.

27. (a) At the input, an amplifier has a signal voltage level of 3 μ V and a noise voltage level of 1 μ V. What is the signal to noise ratio at the input?

(b) If the voltage gain of the amplifier is 20, what is the S/N ratio at the output?

(c) If the amplifier adds 5 μ V of noise, what is S/N ratio at the output? Calculate also the noise factor and the noise figure.

(a) S/N at the input is,

$$= \left(\frac{3 \times 10^{-6}}{1 \times 10^{-6}} \right)^2 = 9$$

(b) Voltage level of signal at the output $= 20 \times 3 = 60\,\mu\text{V}$

Voltage level of noise at the output $= 20 \times 1 = 20\,\mu\text{V}$

∴ Signal to noise ratio at the output

$$= \left(\frac{60 \times 10^{-6}}{20 \times 10^{-6}} \right)^2 = 9$$

(c) If the amplifier adds 5 . V to the noise, therefore the voltage level of noise at the output $= 20 + 5 = 25\,\mu\text{V}$

S/N ratio at the output

$$= \left(\frac{60 \times 10^{-6}}{25 \times 10^{-6}} \right)^2 = 5.76$$

Noise factor,

$$F = \frac{\text{S/N at input}}{\text{S/N at output}}$$

$$= \frac{9}{5.76}$$

$$= 1.56$$

Noise figure,

$$nf = 10 \log F$$

$$= 10 \log 1.56$$

$$= 1.93 \text{ dB}$$

28. The dead zone in certain pyrometer is 0.125% of span. The calibration is 400°C to 1000°C. What temperature change might occur before it is detected?

Span $= 1000 - 400 = 600°\text{C}$

Dead zone $= (0.125/100) \times 600$

∴ A change of 0.75°C must occur before it is detected.

29. A moving coil voltmeter has a uniform (scale with 100 divisions, the full scale reading) is 200 V and $\frac{1}{10}$ of a scale division can be estimated with a fail degree of certainty. Determine the resolution of the instrument in volt.

1 scale division $= 200/100 = 0.2\text{ V}$

Resolution $= \frac{1}{10}$ scale division

$$= \frac{1}{10} \times 2 = 0.2\text{ V}$$

30. A circuit was tuned for resonance by 8 different students and the value of resonant frequency in kHz was recorded as 532, 548, 543, 535, 546, 531, 543 and 536. Calculate, (a) Arithmetic mean, (b) Deviations from mean, (c) Average deviation, (d) Standard deviation, (e) Variance.

(a) The arithmetic mean of the readings is,

$$\overline{X} = \frac{\Sigma x}{n}$$

$$= \frac{532 + 548 + 543 + 535 + 546 + 531 + 543 + 536}{8}$$

$$= 539.25 \text{ kHz}$$

(b) The deviations are

$$d_1 = x_1 - \overline{X} = 532 - 539.25 = -7.25 \text{ kHz}$$

$$d_2 = x_2 - \overline{X} = 548 - 539.25 = 8.75 \text{ kHz}$$

$$d_3 = x_3 - \overline{X} = 543 - 539.25 = 3.75 \text{ kHz}$$

$$d_4 = x_4 - \overline{X} = 535 - 539.25 = -4.25 \text{ kHz}$$

$$d_5 = x_5 - \overline{X} = 546 - 539.25 = 6.75 \text{ kHz}$$

$$d_6 = x_6 - \overline{X} = 531 - 539.25 = -8.25 \text{ kHz}$$

$$d_7 = x_7 - \overline{X} = 543 - 539.25 = 3.75 \text{ kHz}$$

$$d_8 = x_8 - \overline{X} = 536 - 539.25 = -3.25 \text{ kHz}$$

(c) Average deviation is,

$$\overline{D} = \frac{\Sigma |d|}{n}$$

$$= \frac{7.25 + 8.75 + 3.75 + 4.25 + 6.75 + 8.25 + 3.75 + 3.25}{8}$$

$$= 5.75 \text{ kHz}$$

(d) $\because$ The number of readings is $8 < 20$, standard deviation

$$S = \sqrt{\frac{\Sigma d^2}{n-1}}$$

$$= \frac{\sqrt{(-7.25)^2 + (8.75)^2 + (3.75)^2 + (-4.25)^2 + (6.75)^2 + (-8.25)^2 + (3.75)^2 + (-3.25)^2}}{(8-1)}$$

$$= 6.54 \text{ kHz}$$

(e) Variance,

$$V = S^2 = 42.77 \text{ (kHz)}^2$$

31. A temperature sensing device can be modelled as a 1st order system with a time constant of 6 seconds. It is suddenly subjected to a step input of 25°C - 150°C. What temperature will be indicated in 10 seconds after the process has started.

Final steady state temperature, $\theta_0 = 150°C$

Initial temperature, $\theta_i = 25°C$

Time constant, $\tau = 6$ sec

$\therefore$ Temperature after 10 sec, $\theta = \theta_0 + (\theta_i - \theta_0)\,[\exp(-t/\tau)]$

$$= 150 + (25 - 150)\,[\exp(-10/60)]$$

$$= 126.4°C$$

32. A 6.25 mm long RTD with a steady state gain of 0.3925 Ω/°C and a time constant of 5.5 sec experiences a step change of 75°C in temperature. Before the temperature change, it has a stable 100 Ω resistance. Write the time domain equation for resistance and find its value after 15 sec of application of step input.

Gain of RTD is 0.3925 Ω/°C and a step input 75°C is applied to it. This is equivalent to the application of $0.3925 \times 75 = 29.44\ \Omega$ step input in terms of resistance.

$\therefore$ Change in value of resistance with time

$$= 29.44\ [1 - \exp(-t/5.5)]\ \Omega$$

Hence in order to obtain the time domain equation for resistance, the value of initial resistance must be added to it.

∴ Equation for resistance at any time 't' after the application of step input is,

$$R_t = 29.44\ [1 - \exp(-t/5.5)] + 100\ \Omega$$

The value of resistance at $t = 15$ sec is,

$$R_{15} = 29.44\ [1 - \exp(-15/5.5)] + 100$$

$$= 127.5\ \Omega$$

33. A Wheatstone bridge requires a change of 7 Ω in the unknown arm of the bridge to produce a change in deflection of 3 mm of the galvanometer. Determine the sensitivity. Also determine the deflection factor.

$$\text{Sensitivity} = \frac{\text{Magnitude of output response}}{\text{Magnitude of input}}$$

$$= \frac{3\ \text{mm}}{7\ \Omega}$$

$$= 0.429\ \text{mm}/\Omega$$

Inverse sensivity or scale factor

$$= \frac{\text{Magnitude of input}}{\text{Magnitude of output response}}$$

$$= \frac{7\ \Omega}{3\ \text{mm}}$$

$$= 2.33\ \Omega/\text{mm}$$

34. A 10,000 Ω variable resistance has a linearity of 0.1% and the movement of contact arm is 320° (a) Determine the maximum position deviation in degrees and the resistance deviation in ohm. (b) If this instrument is to be used as a potentiometer with a linear scale of 0 to 1.6 V, determine the maximum voltage error.

(a) Maximum displacement deviation

$$= \frac{\text{Percent linearity} \times \text{Full scale reading}}{100}$$

$$\frac{0.1 \times 320}{100} = 0.32°$$

Similarly, maximum resistance displacement

$$= \frac{0.1 \times 10{,}000}{100}$$

$$= 10\ \Omega$$

(b) A displacement 320° corresponds to 1.6 V and therefore 0.32° corresponds to a voltage of

$$(0.32/320) \times 1.6 = 1.6 \times 10^{-3}\ \text{V}$$

Maximum voltage error

$$= 1.6 \times 10^{-3}\ \text{V}$$

$$= 1.6\ \text{mV}$$

35. A multimeter having a sensitivity of 2000 Ω/V is used for the measurement of voltage across a circuit having an output resistance of 10 kΩ. The open circuit voltage of the circuit is 6 V. Find the reading of the multimeter when it is set to its 10 V scale. Find the percentage error.

Input resistance of voltmeter

$$Z_L = 2000 \times 10\ \Omega = 20\ \text{k}\Omega$$

Output resistance of circuit

$$Z_0 = 10\ \text{k}\Omega$$

Open circuit voltage of circuit under measurement

$$E_0 = 6\ \text{V}$$

Reading of voltmeter is

$$E_L = \frac{E_0}{1 + Z_0/Z_L}$$

$$= \frac{6}{1 + 10/20}$$

$$= 4\ \text{V}$$

∴ Percentage error in voltage reading

$$= \frac{(4-6)}{6 \times 100}$$

$$= -33\% \text{ or } 33\% \text{ low}$$

36. In a test, temperature is measured 100 times with variations in apparatus and procedures.

After applying the corrections, the results are,

Temperature °C	397	398	399	400	401	402	403	404	405
Frequency of occurrence	1	3	12	23	37	16	4	2	2

Calculate, (a) Arithmetic mean, (b) Mean deviation, (c) Standard deviation, (d) The probable error of one reading, (e) The standard deviation and the probable error of the mean, (f) The standard deviation of the standard deviation.

The computations are done in a tabular form as under,

Temperature T°C	Frequency of occurrence, f	TXf	Deviation d	$f \times d$	d^2	$f \times d^2$
397	1	397	− 3.78	− 3.78	14.288	14.288
398	3	1194	− 2.78	− 8.34	7.728	23.185
399	12	4788	− 1.78	− 21.36	3.168	38.020
400	23	9200	− 0.78	+ 17.94	0.608	13.993
401	37	14837	+ 0.22	+ 8.14	0.048	1.708
402	16	6432	+ 1.22	+ 19.52	1.488	23.814
403	4	1612	+ 2.22	+ 8.88	4.928	19.714
404	2	808	+ 3.22	+ 6.44	10.368	20.737
405	2	810	+ 4.22	+ 8.44	17.808	35.618
Total	100	40078		$\Sigma [f \times d]$ = 102.8		$\Sigma fd^2 = 191.08$

(a) Mean temperature $= \frac{40078}{100} = 400.78$°C

(b) Mean deviation, $\overline{D} = \frac{102.8}{100} = 1.208$ °C

(c) Standard deviation, $\sigma = \sqrt{\frac{191.08}{100}} = 1.380$°C

(d) Probable error of one reading

$$\gamma_1 = 0.6745$$

$$\sigma = 0.6745 \times 1.38$$

$$= 0.93°\text{C}$$

(e) Probable error of the mean

$$\gamma_m = \frac{0.93}{\sqrt{100}}$$

$$= 0.093°C$$

Standard deviation of the mean

$$\sigma_m = \frac{\sigma}{\sqrt{100}}$$

$$= \frac{1.38}{\sqrt{100}}$$

$$= 0.138°C$$

(f) Standard deviation of the standard deviation

$$\sigma_\sigma = \frac{\sigma_m}{\sqrt{2}}$$

$$= \frac{0.138}{\sqrt{2}}$$

$$= 0.0796°C$$

37. A value $R = 92.2 \pm 0.1\ \Omega$ (where $0.1\ \Omega$ is the standard deviation) is specified for a batch of 1000 resistors. How many would you estimate have values in the range $R = 92.2 \pm 0.15\ \Omega$? Assumes normal distribution consult probability tables.

Deviation, $x = \pm 0.15\ \Omega$

Standard deviation, $\sigma = \pm 0.1\ \Omega$

$\therefore$ Ratio, $t = \dfrac{x}{\sigma}$

$$= \frac{\pm 0.15}{\pm 0.1}$$

$$= 1.5$$

Corresponding to 1.5, the area under the Gaussian curve is 0.4332. Therefore the probable number of resistors having a value of

$$92.2 \pm 0.15\ \Omega = 2 \times 0.4332 \times 1000$$

$$= 866$$

38. The temperature of a furnace is increasing at a rate of 0.1°C/s. What is the maximum permissible time constant of a Ist order instrument that can be used, so the temperature is read with a maximum error of 5°C?

A ramp signal of 0.1°C/s is applied to the instrument and thus $A = 0.1$. Maximum steady state error for a ramp signal applied to a Ist order instrument is given by $e_{ss} = A\,\tau$.

Maximum allowable time constant

$$\tau = \frac{e_{ss}}{A}$$

$$= \frac{5}{0.1}$$

$$= 50 \text{ s}$$

Unit - III

Variable Resistance Transducer

3.1 INTRODUCTION

Electrical circuits consist of combinations of the three passive elements: resistor, inductor and capacitor. The primary parameters that describe them are respectively resistance, self or mutual inductance and capacitance. Any change in the parameter of the element can be recognized only when the element is made 'live' by electric energization or excitation, otherwise the element is in 'dead' state. Hence transducers that are based on the variation of the parameters due to application of any external stimulus are known as passive transducers. In this chapter, resistive, inductive and capacitive transducers are presented along with the several possibilities available for making use of them for measurement of physical and chemical variables. Wherever possible, sections are subdivided in such a way as to identify the element of the transducer and the measurand, such as strain-gauge flow transducer and capacitive strain transducer.

Basic characteristics of each transducer, its limitations and where necessary, relevant signal processing circuitry are presented. Additional insight is provided for transducers that are more powerful and popular, so as to acquaint the reader with the developments in transducer technology. Though the criteria for the design of transducers have been enumerated, details concerning actual designs are not given.

Basic Principle

It is generally seen that methods which involve the measurement of change in resistance are preferred to those employing other principles. This is because both alternating as well as direct currents and voltages are suitable for resistance measurements.

The resistance of a metal conductor is expressed by a simple equation that involves a few physical quantities. The relationship is

$$R = \frac{\rho L}{A}$$

where

R - Resistance; Ω

L - Length of conductor; m

A - Cross - sectional area of conductor; m^2 and

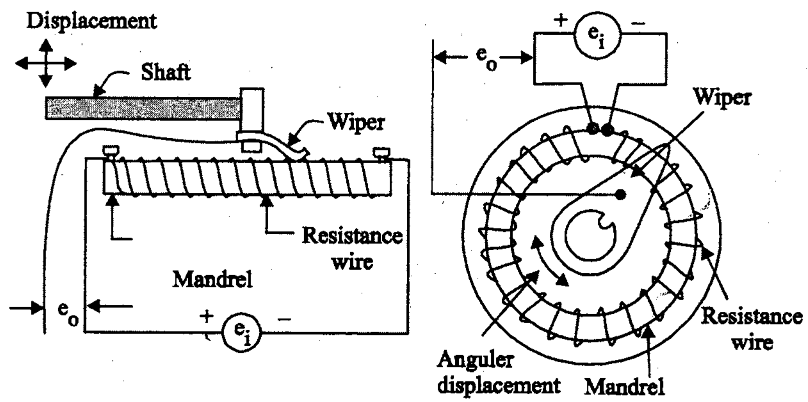
ρ - Resistivity of conductor material, Ωm

Any method of varying one of the quantities involved in the above relationship can be the design basis of an electrical resistive transducer. There are a number of ways in which resistance can be changed by a physical phenomenon. The translational and rotational potentiometers which work on the basis of change in the value of resistance which change in length of the conductor can be used for measurement of translational or rotary displacements. Strain gauges work on the principle that the resistance of a conductor or a semi conductor changes when strained. This property can be used for measurement of displacement, force and pressure. The resistivity of materials changes with change of temperature thus causing a change of resistance. This property may be used for measurement of temperature. Thus electrical resistance transducers have a wide field of application.

3.2 POTENTIOMETER

Basically a resistance potentiometer consists of a resistive element provided with a sliding contact. This sliding contact is called a wiper. The motion of the sliding contact may be translatory or rotational. A linear pot and a rotary pot are shown in figure 3.1 (a) and (b) respectively.

Some potentiometers use the combination of the two motions, ie translational as well as rotational. These potentiometers have their resistive element in the form of a helix and, therefore, they are called helipots.



(a) Linear (translational) POT (b) Rotary POT

Fig. 3.1 Resistive potentiometers (POTs)

The translational resistive elements are straight devices and have a stroke of 2 mm to 0.5 m. The rotational devices are circular in shape and used for measurement of angular displacement. They may have a full scale angular

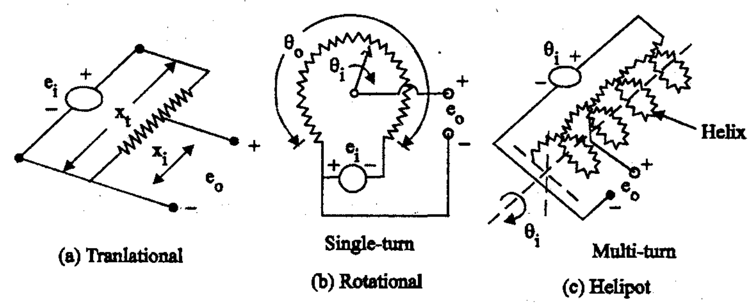


Fig. 3.2 Diagrams for translational, rotational and helipots.

displacement as small as 10°. A full single turn potentiometer may provide accurate measurements upto 357°. Multiturn potentiometers may measure upto 3500° of rotation through use of helipots.

Fig 3.2 shows the diagrams for translational, single turn rotational, and multiturn helix potentiometers.

Let e_i and e_o - input and output voltages respectively; V,

x_t - total length of translational pot; m,

x_i - displacement of wiper from its zero position; m,

R_p - total resistance of the potentiometer; Ω

If the distribution of the resistance with respect to translational movement is linear, the resistance per unit length is $\frac{R_p}{x_t}$

The output voltage under ideal condition is:

$$e_o = \left(\frac{\text{resistance at the output terminals}}{\text{resistance at the input terminals}}\right) \times \text{input voltage}$$

$$= \left(\frac{R_p\,(x_i/x_t)}{R_p}\right) e_i = \frac{x_i}{x_t} \times e_i$$

Under the ideal circumstances, the output voltage varies linearly with displacement as shown in figure 3.3

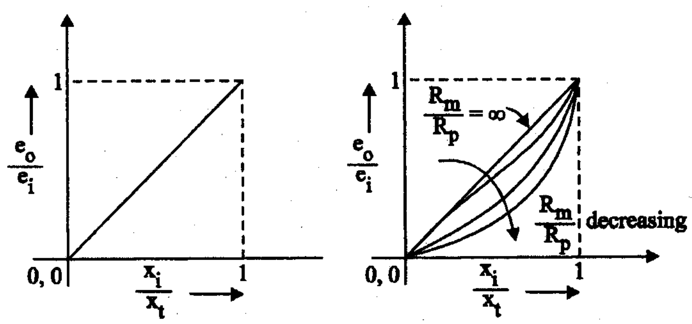


Fig. 3.3 Characteristics of potentiometers

$$\text{Sensitivity } S = \frac{\text{Output}}{\text{Input}} = \frac{e_o}{x_i} = \frac{e_i}{x_t}$$

Thus under ideal conditions the sensitivity is constant, and the output is faithfully reproduced and has a linear relationship with input. The same is true of rotational motion.

Let θ_i = input angular displacement in degrees,

and θ_t = total travel of the wiper in degrees

$$\therefore \text{ output voltage } e_o = e_i\left(\frac{\theta_i}{\theta_t}\right)$$

3.3 STRAIN GAUGES

If a metal conductor is stretched or compressed, its resistance changes on account of the fact that both length and diameter of conductor change. Also there is a change in the value of resistivity of the conductor when it is strained and this property is called piezoresistive effect. Therefore, resistance strain gauges are also known as piezoresistive gauges. The strain gauges are used for measurement of strain and associated stress in experimental stress analysis. Secondly, many other detectors and transducers, notably the load cells, torque meters, diaphragm type pressure gauges, temperature sensors, accelerometers and flow meters employ strain gauges as secondary transducers.

3.3.1 Theory of Strain Gauges

The change in the value of resistance by straining the gauge may be partly explained by the normal dimensional behaviour of elastic material. If a strip of elastic material is subjected to tension, as shown in figure 3.4 or in other words positively strained, its longitudinal dimension will increase while there will be a reduction in the lateral dimension. So when a gauge is subjected to a positive strain, its length increases while its areas of cross-section decreases as shown in Figure 3.4.

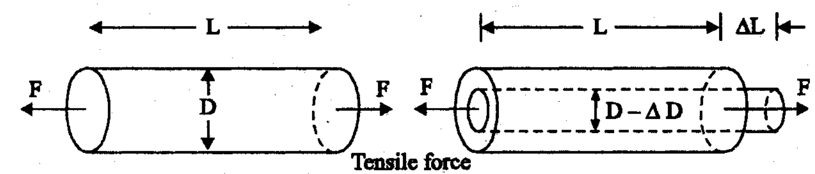


Fig. 3.4 Change in dimensions of a strain gauge element when subjected to a tensile force

Since the resistance of a conductor is proportional to its length and inversely proportional to its area of cross section, the resistance of the gauge increases with positive strain. The change in the value of resistance of strained conductor is more than what can be accounted for an increase in resistance due to dimensional changes. The extra change in the value of resistance is attributed to the change in the value of resistivity of a conductor when strained.

Let us consider a strain gauge made of circular wire. The wire has the dimensions: Length $= L$, area $= A$, diameter $= D$ before being strained. The material of the wire has a resistivity ρ.

Resistance of unstrained gauge $R = \frac{\rho L}{A}$

Let a tensile stress s be applied to the wire. This produces a positive stain causing the length to increase and to decrease as shown in figure 3.4.

Thus when the wire is strained there are changes in its dimensions. Let $\Delta L =$ change in length,

$\Delta A =$ change in area, $\Delta D =$ change in diameter and

$\Delta R =$ change in resistance

In order to find how ΔR depends upon the material physical quantities, the expression for R is differentiated with respect to stress s. Thus we get:

$$\frac{dR}{ds} = \frac{\rho}{A}\frac{\partial L}{\partial s} - \frac{\rho L}{A^2}\frac{\partial A}{\partial s} + \frac{L}{A}\frac{\partial \rho}{\partial s} \qquad (3.1)$$

Dividing Eqn (3.1) throughout by resistance $R = \frac{\rho L}{A}$, we have

$$\frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} - \frac{1}{A}\frac{\partial A}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \qquad (3.2)$$

It is evident from Eqn, (3.2), that the per unit change in resistance is due to:

(i) per unit change in length $= \frac{\Delta L}{L}$,

(ii) per unit change in area $= \frac{\Delta A}{A}$, and

(iii) per unit change in resistivity $= \frac{\Delta \rho}{\rho}$

$$\text{Area, } A = \frac{\pi}{4}D^2 \therefore \frac{\partial A}{\partial s} = 2\cdot\frac{\pi}{4}D\cdot\frac{\partial D}{\partial s} \qquad (3.3)$$

$$\frac{1}{A}\frac{\partial A}{\partial s} = \frac{(2\pi/4)\,D}{(\pi/4)\,D^2}\cdot\frac{\partial D}{\partial s}$$

$$= \frac{2}{D}\frac{\partial D}{\partial s} \qquad (3.4)$$

$\therefore$ Eqn, 3.2 can be written as:

$$\frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} - \frac{2}{D}\times\frac{\partial D}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \qquad (3.5)$$

Now,Poisson's ratio

$$V = \frac{\text{lateral strain}}{\text{longitudinal strain}} = -\frac{\partial D/D}{\partial L/L} \qquad (3.6)$$

or $$\frac{\partial D}{D} = -V\times\frac{\partial L}{L}$$

$$\therefore \frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} + V\frac{2}{L}\frac{\partial L}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \qquad (3.7)$$

or small variations, the above relationship can be written

as: $$\frac{\Delta R}{R} = \frac{\Delta L}{L} + 2V\frac{\Delta L}{L} + \frac{\Delta \rho}{\rho} \qquad (3.8)$$

The gauge factor is defined as the ratio of per unit changes in resistance to per unit change in length.

$$\text{Gauge factor } G_f = \frac{\Delta R/R}{\Delta L/L} \qquad (3.9)$$

(or) $$\frac{\Delta R}{R} = G_f\frac{\Delta L}{L} = G_f \times \epsilon \qquad (3.10)$$

where ϵ = strain $= \frac{\Delta L}{L}$

The gauge factor can be written as:

$$= 1 + 2V + \frac{\Delta\rho/\rho}{\epsilon} \qquad (3.11)$$

$$= 1 \quad + \quad 2V \quad + \quad \frac{\Delta \rho / \rho}{E}$$

Resistance change due to change of length	Resistance change due to change in area	Resistance change due to piezoresistive effect

$$G_f = \frac{\Delta R/R}{\Delta L/L} = 1 + 2V + \frac{\Delta \rho/\rho}{\Delta L/L}$$

The strain is usually expressed in terms of microstrain.

$$1 \text{ microstrain } 1 \frac{\mu \text{ m}}{\text{m}}$$

If the change in the value of resistivity of a material when strained is neglected, the gauge factor is:

$$G_f = 1 + 2V \tag{3.12}$$

Eqn 3.12 is valid only when Piezoresistive Effect (i.e) change in resistivity due to strain is almost negligible.

The Poisson's ratio for all metals is between 0 and 0.5. This gives a gauge factor of approximately 2. The common value for Poisson's ratio for wires is 0.3. This gives a value of 1.6 for wire wound strain gauges.

Types of Strain Gauges

The following are the major types of strain gauges:

1. Unbonded metal strain gauges
2. Bonded metal wire strain gauges
3. Bonded metal foil strain gauges
4. Vacuum deposited thin metal film strain gauges
5. Sputter deposited thin metal strain gauges
6. Bonded semiconductor strain gauges
7. Diffused metal strain gauges

Strain gauges are broadly used for two major types of application and they are:

(i) experimental stress analysis of machines and structures, and

(ii) construction of force, torque, pressure, flow and acceleration transducers

3.3.2 Unbonded metal Strain Gauges

An unbonded metal strain gauge consists of a wire stretched between two points in an insulating medium such as air. It made of various copper nickel chrome nickel or nickel iron alloys. They are about 0.025 mm diameter are fixed with some initial tension between two frames which can move relative to each other. This initial tension or preload is necessary to avoid buckling under compression or negative displacement and this preloading should be greater than any expected compression or negative displacement. A simplified figure is shown in figure 3.5.

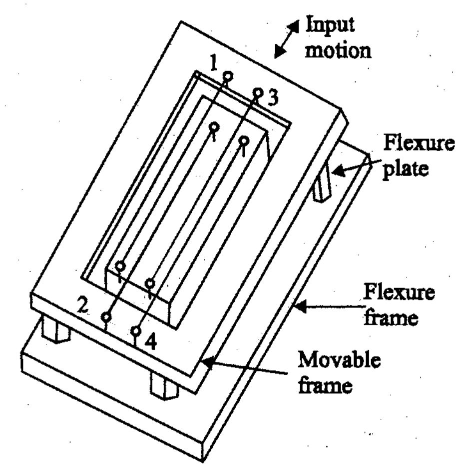


Fig. 3.5 (a) Unbounded type strain gage

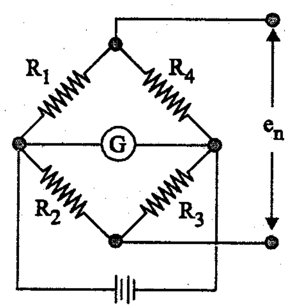


Fig. 3.5 (b) Circuit Connection

Unbonded type strain gauge for rotational motion is shown in figure 3.6

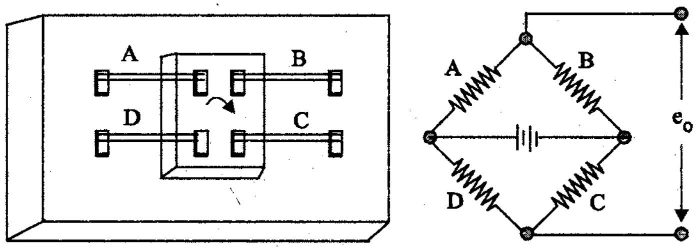


Fig. 3.6 Unbonded type strain gage for rotational stress

The angular motion gives to the inner member which is pivoted to the outer stationary member, increases the tension on the wires and reduces the preload on the other two wires. For example, clockwise twist given to the centre beam increases the tension on wires A and C and reduces the reloaded tension on wires B and D. If they are connected in a bridge as shown then the output voltage available is four times the voltage that would have been obtained due to a single wire. This arrangement is useful for measurement of Torsional Strain and angular displacement. This type of gauges can be used to measure only very small displacements of the order of 0.004 cm full scale. Normally these gauges are used as sensors for force, pressure and acceleration. In these cases the strain wires serve as the necessary spring elements to transduce force to displacement and this displacement is sensed as a resistance variation. The range of force and deflection values are decided by the size, length of wires and the number of wires used.

The sensitivity for a bridge excitation of 5 volts is 40 mv full scale output for 0.006 cm full scale displacement. The nominal value of resistance of the bridge arms is 350 ohms. The thermal sensitivity shift is 0.02% per degree celsius between – 18°C and 120°C.

3.3.3 Bonded Wire Strain Gauges

Construction

A resistance wire strain gauge consists of a grid of fine resistance wire of about 0.025 mm in diameter or less. The grid is cemented to carrier (base) which may be a thin sheet of bakelite or a sheet of teflon. The wire is covered on top with a thin sheet of material so as to prevent it from any mechanical damage. The spreading of wire permits a uniform distribution of stress over the grid. The carrier is bonded with an adhesive material to the specimen under study.

This permits a good transfer of strain from carrier to grid of wires. The wires cannot buckle as they are embedded in a matrix of cement and hence faithfully follow both the tensile and compressive strains of the specimen. Since, the materials and the wire sizes used for bonded wire strain gauges are the same as used for unbonded wire strain gauges, the gauge factors and resistances for both are comparable. The most commonly used forms of strain gauges are shown in figure 3.7.

The nominal values of resistance for these gauges range from 40 to 2000 ohms, but 120, 350 and 1000 are common values.

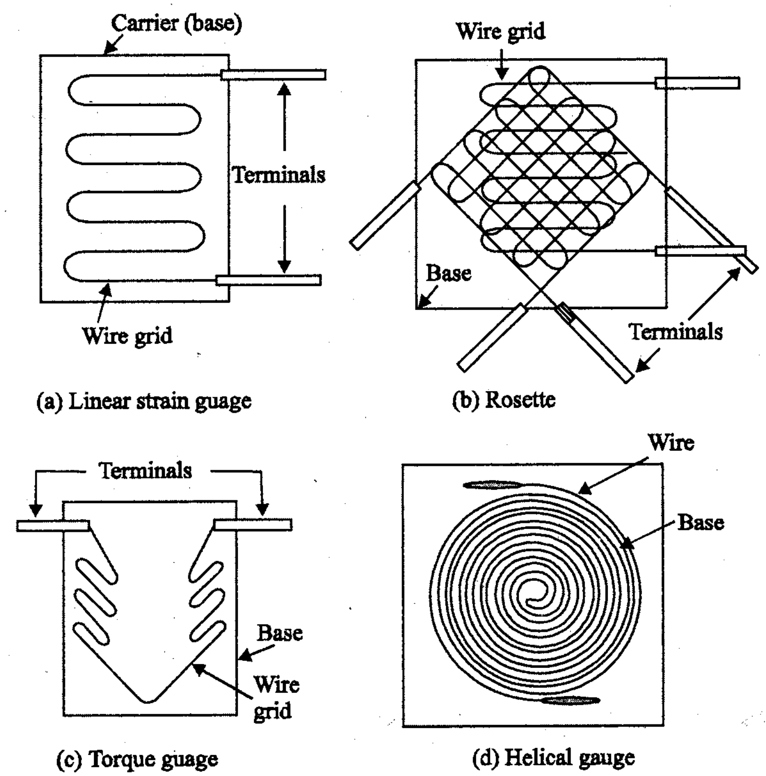


Fig. 3.7 Resistance wire strain gauge

Base (Carrier) Material

1. Epoxy - 200°C to 150°C
2. Bakelite cellulose or fiber glass materials - 200°C to 300°C

The carrier material should have the following properties.

- High dielectric strength
- Minimum temperature restrictions
- Minimum Thickness consistent with other factors
- High mechanical strength
- Good adherence to cements used

Adhesives

Ethylcellulose cement, nitrocellulose cement, bakelite cement and epoxy cement are some of the commonly used adhesive materials. The temperature range upto which they can be used is usually below 175°C.

Leads

The leads should be of such materials which have low and stable resistivity and also a low resistance temperature coefficient.

The recommended lead wire insulation material of the temperature range is:

Nylon	–	75°C
Vinyl	–	65°C to 75°C
Polyethylene	–	75°C to 95°C
Teflon	–	75°C to 260°C

3.3.4 Bonded Metal foil Strain Gauges

Construction

This class of strain gauges is only an extension of the bonded metal wire strain gauges. The bonded metal wire strain **gauges** have been completely superseded by bonded metal foil strain gauges.

The sensing elements of foil gauges are **formed from** sheets less than 0.005 mm thick by photo-etching processes, which allow **greater** flexibility with regard to shape.

In figure 3.8, for example, the three linear grid gauges are designed with fat end turns. This local increase in area reduces the transverse sensitivity which is a spurious input since the gauge is designed to measure the strain component along the length of grid elements.

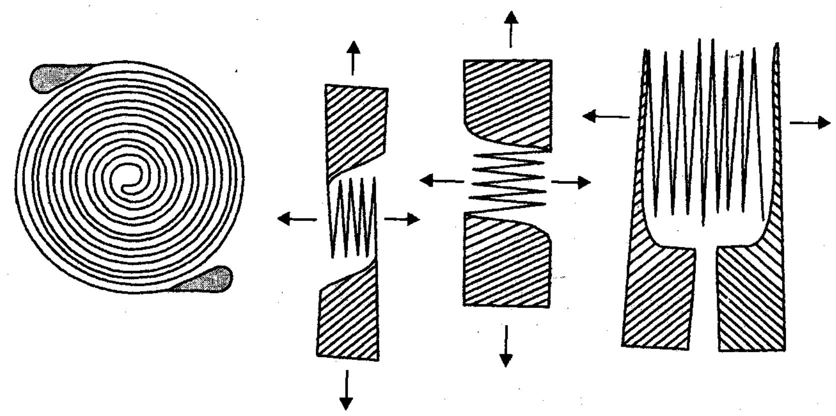


Fig. 3.8 Metal foil strain gauges

For foil type strain gauges, the manufacturing process also easily provides convenient soldering tabs, which are integral to the sensing grid, on all four gauges as shown in Figure 3.8.

Foil type of gauges are employed for both stress analysis as well as for construction of transducers. Foil type of gauges are mounted on a flexible insulating carrier film about 0.025 mm thick which is made of polymide, glass phenolic etc. Typical gauge resistances are 120, 350 and 1000 Ω with the allowable gauge current of 5 to 40 mA which is determined by the heat dissipation capabilities of the gauge. The gauge factors typically range from 2 to 4.

Material for foil type Strain Gauge

Material		Gauge factor
Nichrome	–	2.5
Constantan	–	2.1
Isoelastic	–	3.6
Nickel	–	– 12
Platinum	–	4.8

3.3.5 Evaporation Deposited Thin Metal Strain Gauges

Evaporation deposited thin film metal strain gauges are mostly used for the fabrication of transducers. They are of sputter deposited variety. Both processes begin with a suitable elastic metal element. The elastic metal element converts the physical quantity into a strain. To cite an example of a pressure transducer, a thin, circular metal diaphragm is formed. Both the evaporation and sputtering processes form all the strain gauge elements directly on the strain surface, they are not separately attached as in the case of bonded strain gauges.

In the evaporation process, the diaphragm is placed in a vacuum chamber with some insulating material. Heat is applied until the insulating material vapourises and then condenses, forming a thin dielectric film on the diaphragm. Suitably shaped templates are placed over the diaphragm, and the evaporation and condensation processes are repeated with the metallic gauge material, forming the desired strain gauge pattern on top of the insulating substrate.

In the sputtering process, a thin dielectric layer is deposited in vacuum over the entire diaphragm surface. The detailed mechanism of deposition is, however, entirely different from the evaporation method. The complete layer of metallic gauge is sputtered on the top of the dielectric material without using any substrate. The diaphragms are now removed from the vacuum chamber, and microimaging techniques using photo masking materials are used to form the gauge pattern. The diaphragms are then returned to the vacuum chamber. Sputter etching techniques are used to remove all unmasked metal layer, leaving behind the desired gauge pattern.

Resistance and gauge factors of film gauges are identical to those of foil gauges. Since no organic cementing materials are used, thin film gauges exhibit a better time and temperature stability.

3.3.6 Semiconductor strain gauges

Semiconductor strain gauges are used where a very high gauge factor and a small envelope are required. The resistance of the semi conductors changes with change in applied strain. Unlike in the case of metallic gauges where the change in resistance is mainly due to change in dimensions when strained, the semi conductor strain gauge depend for their action upon piezo-resistive effect.

Semi conducting materials such as silicon and germanium are used as resistive materials for semi conductor strain gauges.

A typical strain gauge consists of a strain sensitive crystal material and leads are sandwiched in a protective matrix. The production of these gauge employs conventional semi conductor technology using semi conducting wafer (or) filaments which have a thickness of 0.05 mm and bonding them on a suitable insulating substrates, such as teflon. Gold leads are generally employed for making the contacts. Some of the typical semi conductor strain gauges are shown in fig 3.9. These strain gauges can be fabricated along with integrated circuit (IC) operational amplifiers which can act as a pressure sensitive transducers.

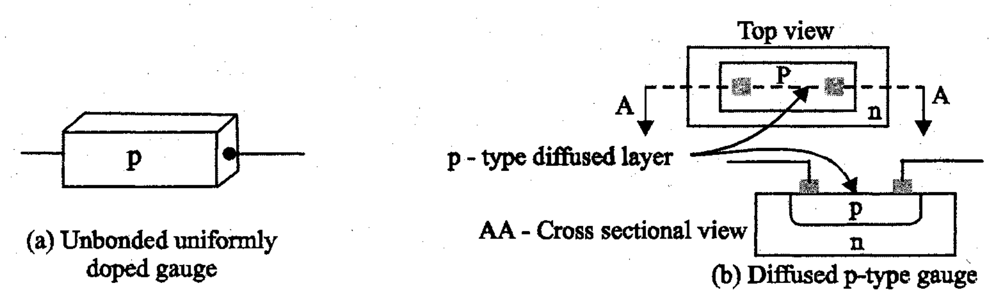


Fig. 3.9 Semi-conductor strain gauge

Advantages

1. High gauge factor.
2. Hysteresis characteristics are excellent.
3. High fatigue life.
4. Very small in size.

Disadvantages

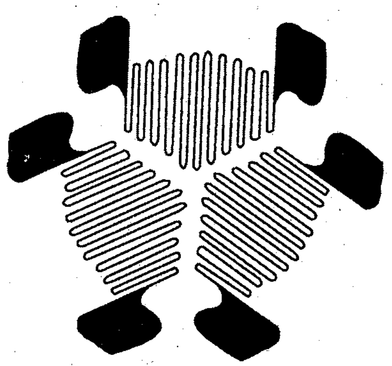
1. Very sensitive to changes in temperature.
2. Linearity is poor.

3.3.7 Diffused strain gauges

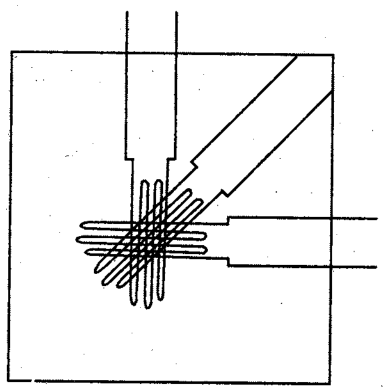
The Diffusion process used in IC manufacture is employed. In pressure transducer, for example, the diaphragm would be of silicon rather than metal and the strain gauge effect would be realized by depositing impurities in the diaphragm to form an intrinsic strain gauge. This type of construction may allow lower manufacturing costs in some designs, as a large number of diaphragms can be made on a single silicon wafer.

3.3.8 Rosettes

In addition to single element strain gauge, a combination of strain gauge called "Rosettes" are available in many combinations for specific stress analysis (or) transducer application.

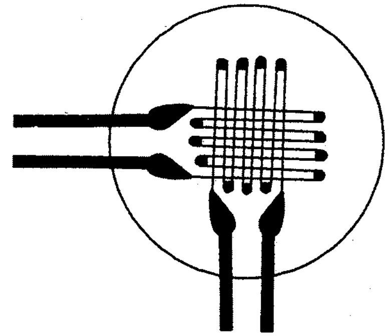


3-Element Rosette
60° Planar
(foil)

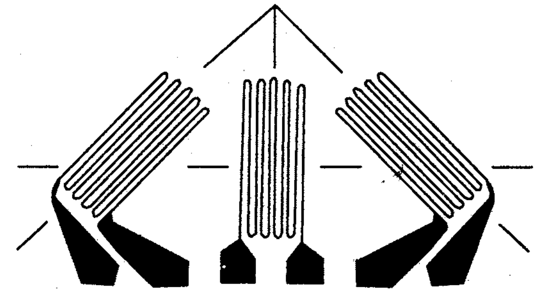


3-Element Rosette
45° Stacked
(wire)

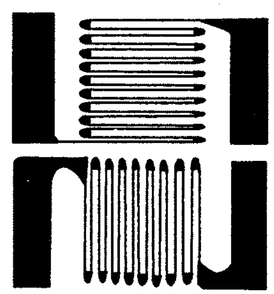
Fig. 3.10 Some forms of Rosettes



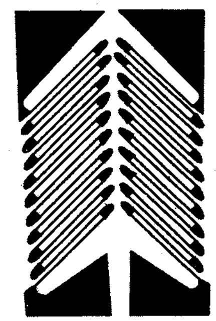
FABX-50-12SX
2-Element Rosette
90° Stacked
(foil)



3-Element Rosette
45° Stacked
(foil)



2-Element Rosette
90° Planar
(foil)



2-Element Rosette
45° Planar
(foil)

Fig. 3.10 Some forms of Rosettes

3.4 RESISTANCE THERMOMETERS OR RESISTANCE TEMPERATURE DETECTOR (RTD)

3.4.1 Introduction

Resistance thermometers are primary electrical transducers enabling measurement of temperature changes in terms of resistance changes. The resistive element is usually made of a solid material, a metal, metallic alloy or a semiconductor compound. The resistivity of metals increases with temperature, while that of semi conductors and insulators generally decreases.

Wire wound elements employ considerable length of wire, and if free to expand, the length also increases with increase in temperature. Hence as temperature changes, the change in resistance will be due to changes in both length and resistivity. Materials used for resistance thermometers have temperature coefficient of resistivity much larger than the coefficient of thermal expansion.

3.4.2 Resistance thermometers

Resistance thermometers use conductive elements like nickel and copper or tungsten and nickel/iron alloys. The variation of resistance R with temperature T for most metallic materials can be represented by an equation of the form

$$R_T = R_o\,(1 + a_1 T + a_2 T^2 + \dots a_n T^n) \tag{3.13}$$

where R_o is the resistance at T = 0°C

The changes in resistance for different metals are given in the form of graph in figure 3.11.

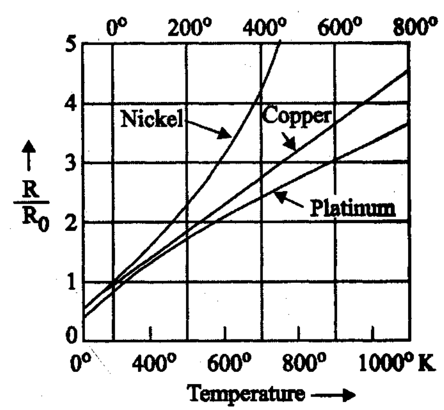


Fig. 3.11 Characteristics of materials used for resistance thermometers.

For engineering purposes and also if range of variation of temperature is narrow then

$$R_t = R_o\,(1 + \alpha\overline{t - t_o} \tag{3.14}$$

$$R_t = R_o\,(1 + \alpha\,\Delta\,t) \tag{3.15}$$

where α is the temperature coefficient as t_o and, R_o is the resistance at t_o

Construction

Resistance elements are generally long, spring like wires enclosed in a metal sheath. The construction of practical resistance thermometer is shown in figure

3.12. The resistance element is surrounded by a porcelain insulator which prevents short circuit between wire and the metal sheath.

Two leads are attached to each side of the platinum wire. When this instrument is placed in a liquid or a gas medium whose temperature is to be

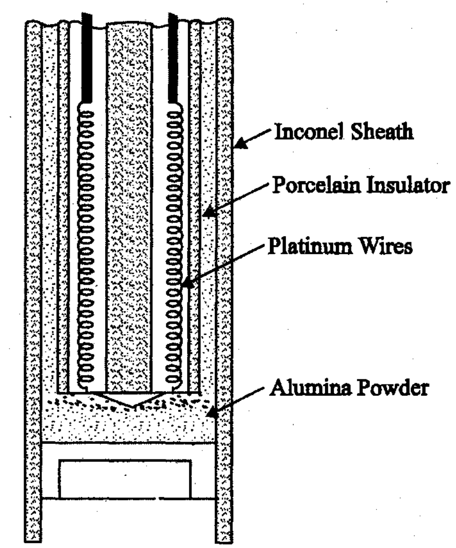


Fig. 3.12 A Resistance Thermometer

measured, the sheath quickly reaches the temperature of the medium. This changes in temperature causes the platinum wire inside the sheath to heat or cool, resulting in a proportional change in the wires resistance. This change in resistance can be directly calibrated to indicate the temperature.

Metals used for Resistance Thermometers

Metal	Temperature Range °C	
	Min	Max
Platinum	– 260	110
Copper	0	180
Nickel	– 220	300
Tungsten	– 200	1000

RTD Circuits (Resistance Thermometer)

The variation in resistance is measured and converted into a voltage signal with the help of a bridge circuit - Bridge circuits employ either deflection mode of operation or the null mode. (manually or automatically balance). Figure 3.13 is a bridge for null method of measurement.

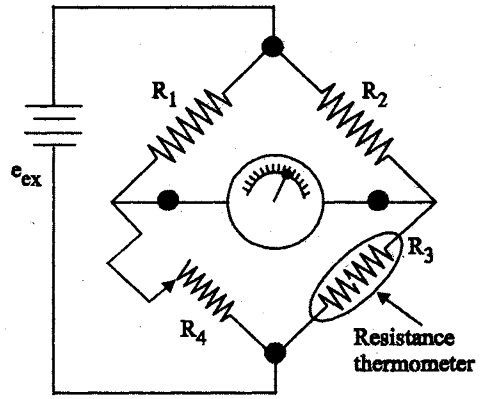


Fig. 3.13 Null balance bridge circuit of resistance thermometer

R_4 is varied until balance is achieved. When better accuracy is required the arrangement shown in figure 3.14 is preferred.

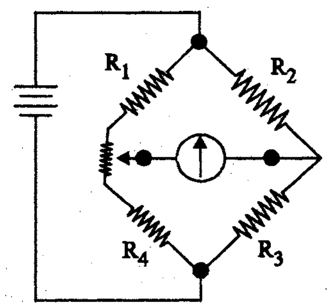


Fig. 3.14 Bridge balance circuit for better accuracy

In this circuit the contact resistance in the adjustable resistor has no influence on the resistance of the bridge legs.

If long lead wires subjected to temperature variations are unavoidable, then three wire resistance thermometer is used with the circuit configuration as shown in figure 3.15.

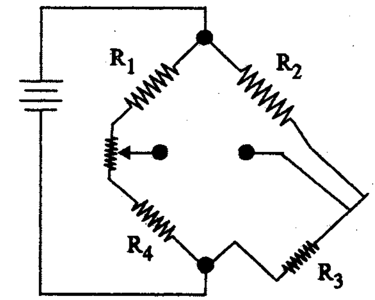


Fig. 3.15 Three wire resistance thermometer circuit

To get a fairly linear relationship between the output voltage and the temperature, the values of R_1 and R_2 of the above circuits are made atleast 10 times greater than that of the thermometer.

Advantages

- Good Reproducibility
- Fast in response
- Small in size
- High Accuracy
- Wide temperature range
- Temperature compensation is not required

Disadvantages

- Cost is high
- Excitation needed
- Large bulb size than thermocouple
- Produce mechanical vibration

3.5 THERMISTORS

3.5.1 Introduction

Thermistors are thermal resistors with a high negative temperature coefficient of resistance.

They are made of manganese, nickel, copper, iron, uranium and cobalt oxides which were milled, mixed in proper proportions with binders pressed into the desired shape and sintered.

Construction

Thermistors are composed of sintered mixture of metallic oxides such as manganese, nickel, cobalt, copper, iron and uranium. They are available in variety of sizes and shapes. The thermistors may be in the form of beads, rods and discs. Some of the commercial forms are shown in figure 3.16.

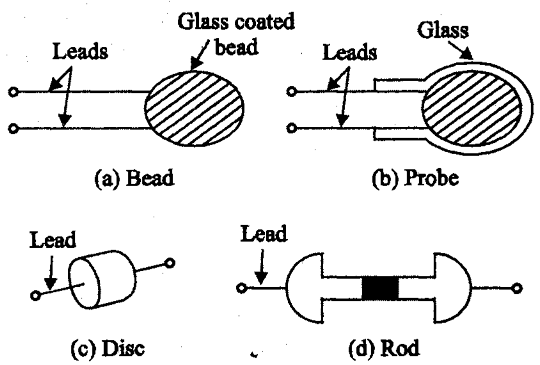


Fig. 3.16 Different terms of construction of thermistors

A thermistor in the form of a bead is smaller in size and the bead may have a diameter of 0.015 mm to 1.25 mm. Beads may be sealed in the tips of solid glass rods to form probes which may be easier to mount than the beads. Glass probes have a diameter of about 2.5 mm and a length which varies from 6 mm to 50 mm. Discs are made by pressing material under high pressure into cylindrical flat shapes with diameters ranging from 2.5 mm to 25 mm.

Thermistors

Thermistor is a contraction of a term "thermal resistor". Thermistors are generally composed of semi-conductor materials. Although positive temperature co-efficient of units (which exhibit an increase in the value of resistance with increase in temperature) are available, most thermistors have a negative coefficient of temperature resistance ie. their resistance decreases with increase of temperature. The negative temperature coefficient of resistance can be as largé as several percent per degree celsius. This allows the thermistor circuits

to detect very small changes in temperature which could not be observed with a RTD or a thermocouple.

In some cases the resistance of thermistor at room temperature may decrease as much as 5 percent for each 1°C rise in temperature. This high sensitivity to temperature changes makes thermistors extremely useful for precision temperature measurements control and compensation.

Thermistors are widely used in applications which involve measurements in the range of $-60°C$ to $15°C$. The resistance of thermistors ranges from $0.5\ \Omega$ to $0.75\ M\,\Omega$. Thermistor is a highly sensitive device. The price to be paid off for the high sensitivity is in terms of linearity. The thermistor exhibits a highly non-linear characteristic of resistance versus temperature.

Characteristics of Thermistor

Three important characteristics of thermistor make them extremely useful in measurement and control applications. These are:

(i) the resistance - temperature characteristics

(ii) the voltage current characteristics

(iii) the current-time characteristics

Thermistors have a large negative temperature coefficient and it is highly nonlinear. The resistance at different temperatures can be found out using the following equation,

$$R_T = R_o\, e^{\beta\,(1/T - 1/G_o)} \qquad (3.16)$$

where

R_T - resistance at temperature T

R_o - resistance at temperature T_o

β - constant characteristic of material

e - base of natural log

and $T_1\ T_o$ - absolute temperature K

The value of β for the semi conductor made of the above material is 4000.

The temperature coefficient α for thermistor is expressed as

$$\alpha = \left. \frac{1}{R_T} \frac{d R_T}{dT} \right|_{T = T_o} \tag{3.17}$$

$$R_T = R_o \, e^{\beta (1/7 - 1/T_o)}$$

$$\frac{dR_T}{dT} = R_o \, \beta \, e^{\beta (1/T - 1/T_o)} \cdot \frac{1}{T^2}$$

Due to self heating the resistance decreases and the current increases. As the current is more the heating is also more and hence resistance will decrease. Some kind of chain action takes place here. This process will continue until the thermistor reaches the maximum temperature possible for the amount of power available at which time a steady state will exist.

Figure 3.19 show typical current time characteristic curves for a semiconductor material. The thermal dissipation constant for typical thermistor ranges from 0.1 m W/°C for glass covered beads to 7 m W/°C for relatively large discs. All are measured in still air. Other semiconductor temperature sensors include carbon resistors, silicon and germanium devices.

Carbon resistors are merely the commercial carbon-composition elements commonly used as resistance elements in electronic circuitry. The normal power rating is from 0.1 to 1 watt and the resistance value varies from 2 to 150 ohm. They are also used for cryogenic temperature measurements in the range 1 to 20 K. From about 20 K downward these elements exhibit a large increase in resistance with decrease in temperature given by the relation

$$\log_{10} R = \frac{K}{\log_{10} T} = A + \frac{B}{T} \tag{3.19}$$

R is the resistance, T is the temperature **in Kelvin** and A, B and K are constants determined by calibration

$$\left. \frac{1}{R_T} \frac{dR_T}{dT} \right|_{T = T_o} = - \frac{\beta}{T_o^2} \tag{3.18}$$

when $T_o = 25°C = 298$ K

$$\alpha = -(4000/298^2) = -0.045$$

The resistivity versus temperature graphs are shown in figure 3.17.

The voltage to current characteristics of thermistors is as shown in figure 3.18.

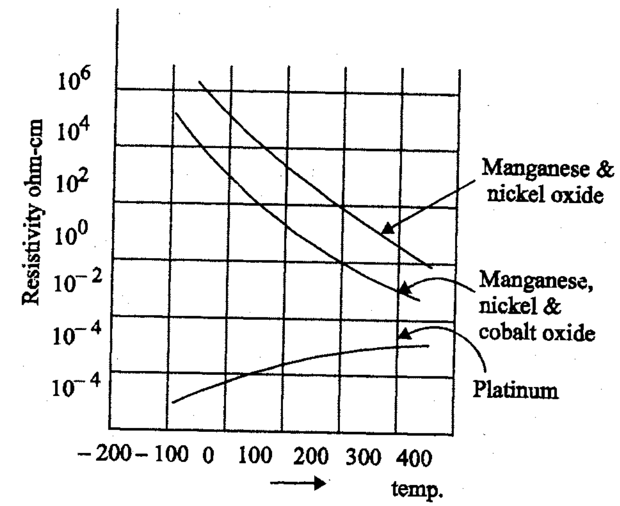


Fig. 3.17 Resistive curves for thermistors

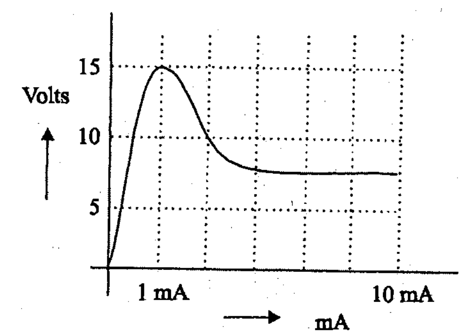


Fig. 3.18 V-I characteristics of thermistors

The current through the semiconductor element is time dependent for a constant voltage as the resistance varies due to self heating as shown in figure 3.19 of the individual resistors. Reproducitibility of the order of 0.2% is obtained in the range of 0 to 20°C.

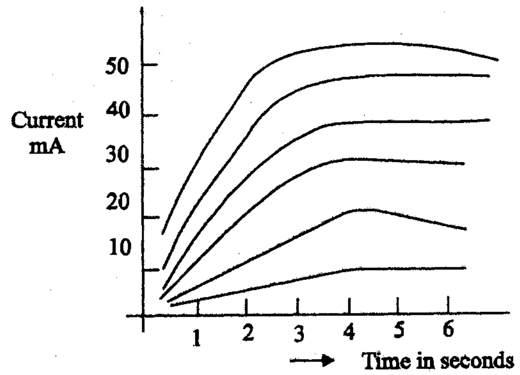


Fig. 3.19 Current variation due to self heating in thermistor

Silicon with boron impurities can be designed to have either a positive or negative temperature coefficient over a particular temperature range. A typical element shows from the normal value at 25°C a change of 80% at − 150°C to + 180% at 200°C.

Germanium doped with arsenic and gallium is used for cryogenic temperatures where it exhibits a large decrease in resistance with increase in temperature.

Applications

1. Measurement of power at high frequencies.
2. Vacuum measurement.
3. Measurement of thermal conductivity.
4. Measurement of level, flow and pressure of liquids.
5. Measurement of composition of gases.

Advantages

- Very high sensitivity.
- It can be manufactured in any size or shape.
- Good stability.
- Fast in Response. (In the order of ms)

Disadvantages

- Highly non-linear.
- In low temperature, sensitivity also low.
- Its upper limit is set by instability.

3.5.2 Temperature Compensation

Because Thermistors have a negative temperature coefficient of resistant - opposite to the positive coefficient of most electrical conductors and semiconductors they are widely used to compensate for the effects of temperature on both component and circuit performance.

Disk type thermistors are used for this purpose where the maximum temperature does not exceed ± 125°C. A properly selected thermistor, mounted against or near a circuit element, such as a copper meter coil, and experiencing the same ambient temperature changes, can be connected in such a way that the total circuit resistance is constant over a wide range of temperatures. This is shown in the curves of figure 3.20 which illustrates the effect of a compensation network.

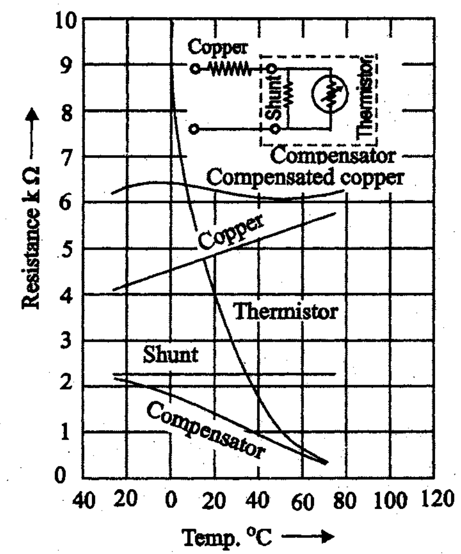


Fig. 3.20 Temperature compensation of a copper conductor by of a thermistor network

The compensator consists of a thermistor, shunted by a resistor. The negative temperature coefficient of this combination equals the positive coefficient of the copper coil. The coil resistance of 5000 Ω at 25°C, varies from approximately 4500 Ω at 0°C to 5700 Ω at 60°C, representing a change of about ± 12 percent. With a single thermistor compensation network, this variation is reduced to about ± 15 Ω or ± 1/4 percent. With double or triple compensation networks, variations can be reduced even further.

3.6 HOT WIRE ANEMOMETER

3.6.1 Introduction

Hot wire anemometers are hot wire resistance transducer which are used for measurement of flow rates of fluids. In hot wire anemometers resistive wire is used as a basic sensor, which is heated initially by passing an electric current. This heated resistive wire mounted on a probe is exposed to air flow or wind, which is cooled because of fanning effect. The amount of cooling depends on the velocity of air flow.

The resistance of the probe when it is hot is different from that when it is cooled. This difference in resistance, or this variation in resistance is converted into a voltage variation. Broadly hot wire anemometers are commonly used in two different modes.

1. Constant current type

2. Constant temperature type

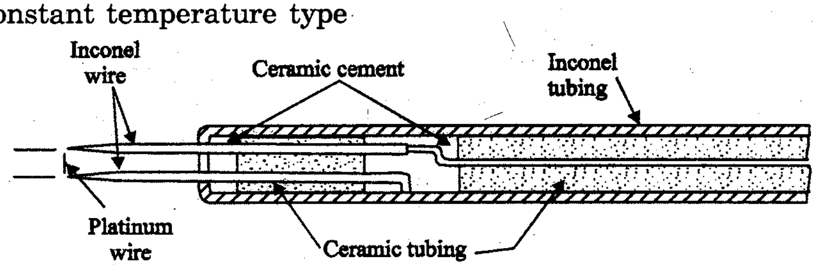


Fig. 3.21 Hot wire anemometer probe

3.6.2 Basic principle

The two types of anemometers use the same basic principle but in different ways.

In the constant current mode, the fine resistance wire carrying a fixed current is exposed to the flow velocity. The flow of current through the wire generates heat on account of $t^2 R$ loss. This heat is dissipated from the surface of the wire by convection to the surroundings. (The loss of heat due to conduction and radiation is negligible). The wire attains equilibrium temperature when the heat generated due to $i^2 R$ loss is equal to the heat dissipated due to convective loss.

The circuit is so designed that $i^2 R$ heat is essentially constant and therefore the wire temperature must adjust itself to change the convective loss until equilibrium is reached. The resistance of the wire depends upon the temperature and the temperature depends the rate of flow. Therefore, the resistance of wire becomes a measure of the flow rate.

In the constant temperature mode, the current required to maintain the resistance and hence temperature constant, becomes a measure of flow velocity.

$$\text{Heat generated } = I^2 R_w \qquad (3.20)$$

where

I - current through the wire; A,

R_W - resistance of wire; Ω,

Heat dissipated due to convection $= hA\,(\theta_W - \theta_f)$

where

h - coefficient of heat transfer, $W/m^2 - °C$,

A - heat transfer area; m^2,

θ_W - temperature of wire; °C,

and θ_f - temperature of flowing fluid, °C, For equilibrium conditions, we can write the energy balance for the hot wire as,

$$I^2 R_w = hA\,(\theta_w - \theta_f) \tag{3.21}$$

Now from h is mainly, a function of flow velocity for a given fluid density. From King's Law, for a range of velocities, this function can be written as,

$$h = C_0 + C_1 \sqrt{V} \tag{3.22}$$

where C_0 and C_1 are constants and V is the flow velocity of fluid in m/s.

Hence Eqn, 3.21, can be written as:

$$I^2 R_w = (C_0 + C \sqrt{V})\,A\,(\theta_w - \theta_f) \tag{3.23}$$

3.6.3 Constant Temperature Anemometer

Now, Eqn (3.23) can be written as:

$$I^2 = (C_0 + C_1 \sqrt{V})\,\frac{A\,(\theta_w - \theta_f)}{R_w} \tag{3.24}$$

For constant temperature θ_w of wire, its resistance R_w is constant. A and θ_f are already constant and therefore Eqn, 3.24 can be written as:

$$I^2 = K_1 + K_2 \sqrt{V} \tag{3.25}$$

where K_1 and K_2 are constants.

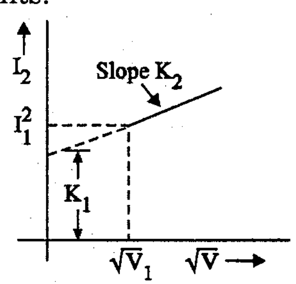


Fig. 3.22 Relationship between I^2 and $\sqrt{V}$

Hence, a straight line relationship exists between I^2 and $\sqrt{V}$ as shown in figure 3.22.

For the purpose of measurement, the hot wire anemometer which is in the form of an insulated probe is connected in a whetstone bridge as shown in fig 3.23.

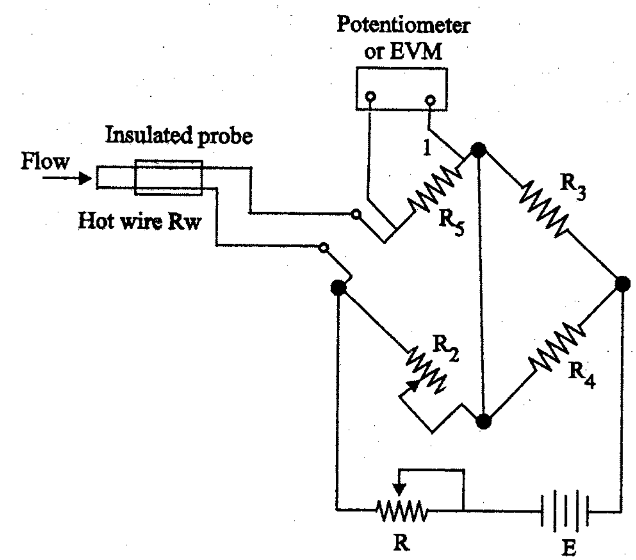


Fig. 3.23 Bridge circuit used for constant temperature Hot wire anemometer

A standard resistor R_s is connected in series with the hot wire anemometer. A galvanometer is used to detect the balance conditions. The current through the hot wire is determined by measuring voltage drop across the standard resistor R_s with the help of a d.c potentiometer or an Electronic voltmeter (EVM). R_4 is very large as compared to R_2 so that most of the current flows through R_4.

The measuring circuit is first calibrated by exposing the hot wire to known velocities and using the same fluid for which it is ultimately used. The pressure and temperature of the fluid should be maintained at the same values during calibration and usage later. The velocities of fluid are measured accurately by some other method like static Pitot tube. The output is recorded over a range of velocity.

In calibration V is set at some known value V_1. Then R_4 is adjusted to set the hot wire current I at a value low enough to prevent wire burn out but high enough to give adequate sensitivity to velocity. The resistance R_w will come to a definite temperature and resistance. Then the resistor R_2 is adjusted to balance the bridge. This adjustment is essentially a measurement of wire temperature, which is held fixed at all velocities.

The first on the calibration curve is thus plotted as $I_1^2 \sqrt{V_1}$. Now V is changed to a new value, causing wire temperature and hence R_w to change there by unbalancing the bridge. Then R_w, and thus wire temperature is restored to its original value by changing I (by changing R) till balance is restored. The value of R_2 is not changed as this assures the R_w has remained constant and so has the temperature. The new point is plotted on the calibration curve, and this procedure is repeated for other velocities.

A plot of I^2 V/s $\sqrt{V}$ show in figure 3.22 is used as the calibration curve for the specified medium of flow. Once calibrated, the probe can be used to measure unknown velocities by balancing the bridge and finding the value of I. The corresponding value of V can be found from the calibration curve.

The method described above can be used for measurement of average (steady) velocities as it is manual in nature. This mode of operation can be extended to measure both average and fluctuation components of velocity by making the bridge balancing operation automatic, rather than manual, through feedback arrangements.

3.6.4 Constant Current Anemometer

In the constant-current mode of operation, the current through the hot wire is kept at a suitable value. The hot wire anemometer is connected in a bridge circuit as shown in figure 3.24. The bridge is calibrated first.

The value of current I through the anemometer is selected and set at a proper value taking precautions so that the burn out of hot wire does not occur. The hot wire is subjected to different known values of velocities V of the fluid under test. This changes the value of R_w and therefore unbalances the bridge thereby producing an out of balanced voltage e_0 which is measured by a high

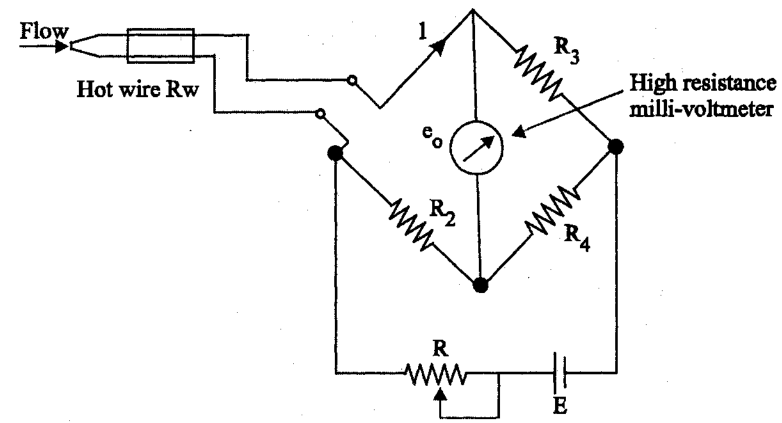


Fig. 3.24 Bridge circuit used for constant current Hot wire anemometer

resistance milli voltmeter. A calibration curve showing a plot of out of balance voltage e_o V/s flow velocity V is shown in figure 3.25.

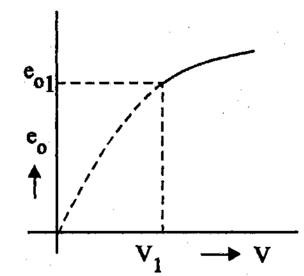


Fig. 3.25 Relationship between out of balance voltage e_o and flow velocity V (calibration curve)

The value of any unknown value of flow velocity can be found from the calibration curve corresponding to the out of balance voltage e_o. Suppose while measuring the velocity of a fluid, an out of balance voltage e_{01} is obtained, the velocity corresponding to this is V_1 as found from the calibration curve. The range of velocities for which constant current type anemometer can be used is necessarily low because of the possibility of the wire burn out when the flow stops. This means that choice of lower value of I for the upper limit of velocity or a lower value of velocity for an upper limit with a satisfactory value of I.

The measuring circuit of the constant current anemometer can be used for the measurement of steady velocities as well as the rapidly fluctuating components such as the turbulent components superimposed on an average velocity.

3.7 HUMIDITY MEASUREMENT USING RESISTIVE TRANSDUCERS

Humidity

Humidity is the measure of water vapour present in a gas. It is usually measured as absolute humidity, relative humidity or dew point temperature.

Absolute humidity or Specific humidity

It is the mass of water vapour present per unit volume.

$$W = \frac{m_v}{m_a}$$

Relative Humidity

It is the ratio of water vapour pressure actually present to water vapour pressure required for saturation at a given temperature. The ratio is expressed in percent. Relative humidity (*RH*) is always dependent upon temperature.

$$\phi = \frac{m_v}{m_{sat}} = \frac{P_v}{P_g}$$

P_v - actual partial pressure

P_g - saturation pressure of vapour

Construction

A typical resistive hygrometer is shown in figure 3.26. It shows a mixture of lithium chloride and carbon which acts as conducting film. This is put on an insulating substrate between metal electrodes. A mixture of lithium chloride and carbon exhibits a change in resistivity with humidity. This material with a binder may be coated on a wire or an electrodes.

Resulting resistance changes over a wide range, e.g. 10^4 to $10^9\ \Omega$ as the humidity changes from 100 to 0 percent. This makes it impractical to design a single element to operate from 1 to 100 percent relative humidity.

Instead several elements are used, each in a narrow range, with provision for switching elements. Resistance is measured either with a whetstone bridge or by a combination of current and voltage measurements. Most of these must not be exposed to conditions of 100 percent humidity as the resulting

condensation may damage the device. Either they must be operated in a constant temperature environment or temperature corrections must be made. These are accurate to within ± 2.5 percent or ± 1.5 percent in some cases. Response times are typically of the order of few seconds. These are currently the most common electronic hygrometers.

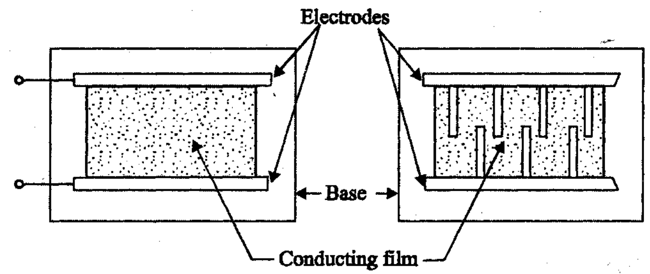


Fig. 3.26 Resistive hygrometer

Working Principle

The resistance of the element changes when it is exposed to variations in humidity. The higher the relative humidity, the more moisture the lithium chloride will absorb, and the lower will be its resistance.

The resistance of the sensing unit is a measure of the relative humidity. Resistance should be measured by applying a.c to the whetstone bridge. D.C voltage is not applied because it tends to breakdown the lithium chloride to its lithium and chloride atoms. The current flow is a measure of the resistance and hence of the relative humidity.

Thus hygrometer is called Dunmore type of hygrometer. The resistance/relative humidity relationship is quite non-linear, and generally a single transducer can cover only a small range of the order of 10 percent humidity. Where large ranges, as great as 5 to 99 percent relative humidity, are needed, seven or eight of transducers, each designed for a specific part of the total range, are combined in a single package.

These transducers are widely used for continuous recording and/or control or relative humidity. Another electrical type of transducer, the sulfonated polystyrene ion-exchange device called the pope cell exhibits a non-linear change of resistance from a few $M\ \Omega$ at 0 percent to about $1000\ \Omega$ at 100 percent relative humidity and a single transducer can cover the entire range. Accuracy is comparable to that of the Dunmore transducer.

TWO MARK QUESTIONS AND ANSWERS

1. What is potentiometer?

Basically a resistance potentiometer, or simply a POT, (a resistive potentiometer used for the purposes of voltage division is called a POT) consists of a resistive element provided with a sliding contact. The POT is a passive transducer.

2. List the materials used for potentiometer.

Materials used for potentiometer are

(a) Wire wound potentiometer

1. Platinum
2. Nickel chromium
3. Nicker copper
4. Some other precious resistive element

(b) Non wire wound potentiometer

(i) Cermet

(ii) Hot moulded carbon

(iii) Carbon film

(iv) Thin metal film

3. Classify potentiometers.

Potentiometers are classified,

(a) Based on operation

(i) Linear potentiometer

(ii) Rotary potentiometer

(iii) Helipot

(iv) Non-linear potentiometer

(b) Based on material used

(i) Wire wound potentiometer

(ii) Non-wire wound potentiometer

4. What are the advantages and disadvantages of potentiometer?

The advantages of potentiometer are,

(a) Inexpensive.

(b) Useful for measurement of large amplitudes.

(c) Efficiency is very high.

(d) Frequency response of wire wound potentiometers is limited.

The disadvantage of potentiometer is,

(a) Require a large force to move.

5. Define resistive transducer. Give example.

The resistance of the metal conductor is expressed by a simple expression, $R = eL/A$ which involves a few physical quantities.

where,

R	–	Resistance in Ω
L	–	Length of conductor in m
A	–	Cross sectional area in m^2
e	–	Resistivity of conductor material in Ωm

The device in which any one of the above properties is changed for measurement purpose is called a resistive transducer.

Example: Strain gauge, potentiometer, resistance thermometer.

6. List the factors influencing the choice of transducers.

Factors influencing the choice of a transducer are,

(a) Operating principle

(b) Sensitivity

(c) Operating range

(d) Accuracy

(e) Cross sensitivity

(f) Loading effect

(g) Environmental compatibility

(h) Insensitivity to unwanted signals

(i) Usage and ruggedness

(j) Stability and reliability

(k) Static characteristics

7. What is gauge factor?

The gauge factor is unit resistance change per unit strain.

8. What are the different types of strain gauge?

The various types of strain gauge are,

(a) Unbonded metal strain gauges

(b) Bonded metal wire strain gauges

(c) Bonded metal foil stain gauges

(d) Vacuum deposited thin metal film stain gauges

(e) Sputter deposited thin metal strain gauges

(f) Bonded semiconductor strain gauges

(g) Diffused metal strain gauges.

9. What are the factors to be considered for bonded strain gauge?

The following factors are considered for bonded strain gauge.

(a) Filament construction

(b) Material of the filament wire

(c) Base carrier material or backing material

(d) Cement used to bond the filament to the carrier

(e) Lead wire connections.

10. What is strain?

Strain is a ratio of changing length to original length.

11. What is Young's modulus?

Young's modulus is a ratio of stress and strain, $\frac{dR/R}{dl/l}$

12. What is resistance thermometer?

A resistance thermometer consists of a resistive element which is exposed to the temperature to be measured. If the conductors or metals are used to measure the temperature, they are known as resistance thermometers and if semiconductors are used then they are known as thermistors.

13. What are the different approximation methods of resistance thermometer?

The approximation methods of resistance thermometer are,

- Linear approximation
- Quadratic approximation

14. What is self heating error of thermometer?

Resistance thermometer bridges may be excited with either DC or AC. The direct or rms alternating current through the thermometer is usually in the range of 2 to 20 mA. This current causes an $I^2 R$ heating which raises the temperature of the thermometer above its surrounding, causing the so called self heating error.

15. What are the advantages and disadvantages of resistance thermometers?

The advantages of resistance thermometer are,

(a) They are suitable for measuring large temperature differences and high temperatures.

(b) They are very accurate which make them suitable for small temperature measurement.

(c) Well designed resistance thermometers have excellent stability.

(d) Unlike thermocouples, they do not need a reference junction and this favors them in many aerospace and industrial applications.

The disadvantages of resistance thermometer are,

(a) Their relatively large volume compared to thermocouples results in monitoring an average temperature over the length of the resistor rather than a point temperature.

(b) They need auxiliary apparatus and power supply.

(c) The resistance element is usually more expensive than a thermocouple.

(d) There are errors due to self heating and thermoelectric effect of the resistive element and connecting leads (dissimilar metal junctions).

16. What is the principle of hot wire anemometer?

Another resistance variation type transducers is hot wire anemometer. In general, anemometers are devices used for measurement of velocity of flow.

17. Why dynamic compensation is required for hotwire anemometer?

To avoid the fluctuation, we need dynamic compensation circuits for the hot wire anemometer.

18. What are the applications of thermistors?

The applications of thermistors are,

- Measurement of power at high frequencies.
- Measurement of thermal conductivity.
- Measurement of level, flow and pressure of liquids.
- Measurement of composition of gases.
- Vacuum measurements.
- Providing time delay.

19. Mention the features of thermistors.

The features of thermistors are,

- Compact, rugged and inexpensive.
- Good stability.
- The response time of thermistors can vary from a fraction of a second to minute.

- Self heating of thermistors is avoided.
- Thermistors can be installed at a distance from their associated measuring circuits.

20. Mention the materials used for thermistors.

Mixture of metallic oxides such as manganese, nickel, cobalt, copper, iron and uranium are use for thermistors.

21. Give the principle of stain gauge.

If a metal conductor is stretched or compressed, its resistance changes on the fact that both length and diameter of conductor change. There is a change in the value of resistivity of the conductor, when it is strained. This property is called piezo-resistive effect. The strain gauges are resistive transducers used for measurement of strain and associated stress in experimental stress analysis.

22. Mention the applications of strain gauge.

The applications of strain gauge are, it is

- Used to measure pressure
- Used to measure torque
- Used to measure acceleration
- Used to measure force

23. List the strain gauge materials with its gauge factor.

Sl.No.	Material	Gauge factor
(a)	Nickel	– 12.1
(b)	Manganin	+ 0.47
(c)	Nichrome	+ 2.0
(d)	Constantan	+ 2.1
(e)	Soft iron	+ 4.2
(f)	Platinum	+ 4.8
(g)	Carbon	+ 20
(h)	Doped crystal	100 – 5000

24. Define Poisson's ratio.

Poisson's ratio is defined as the ratio of lateral strain to longitudinal strain.

$$\text{Poisson's ratio, } r = \frac{OD/D}{OL/L}$$

25. Define stress and strain.

Stress is defined as the deforming force per unit area.

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} \text{ N/m}$$

Strain is defined as the ratio of change in dimension to original dimension.

$$\text{Strain} = \frac{\text{Change in dimension}}{\text{Original dimension}} \text{ (dimensionless)}$$

26. Write a note on semiconductor strain gauge.

Semiconductor strain gauges are used where a very high gauge factor and a small envelope are required. The resistance of the semiconductor changes with change in applied strain. They depend on piezo-resistive effect. Semiconducting materials like silicon and germanium are used as resistive material.

27. Write a note on gauge sensitivity of full bridge and half bridge circuit.

Gauge sensitivity of a full bridge circuit for strain measurement is

$$s_g = 4k\, R_g\, G_f$$

Gauge sensitivity of a half bridge circuit is

$$s_g = 2k\, R_g\, G_f$$

where,

k – Scaling factor
R_g – Resistance of gauge material
G_f – Gauge factor

28. Explain how linearity and sensitivity of a linear potentiometer conflicting with each other when loaded with o/p devices.

For high sensitivity, the i/p voltage should be large and in turn resistance R_p should be high. On the other hand, for higher linearity, the resistance of the potentiometer R_p should be made as small as possible. If R_p is low power dissipation goes up which requires low input voltage and hence lower sensitivity. Thus linearity and sensitivity are two conflicting requirements.

29. What is meant by Poisson's arrangement in construction of strain gauge. List its features.

Poisson's arrangement in construction of strain gauge is a method of temperature compensation that utilizes two active gauges R_{g1} and R_{g3} which are bonded at right angles to the structural membrane.

(a) Temperature compensation is obtained.

(b) Bridge sensitivity is increased by a factor $(1 + r)$ where r is the Poisson's ratio at the material used.

30. How is the resolution of a linear resistive potentiometer determined?

The resolution of a potentiometer is the smallest change in displacement that can be measured. If the excitation is fixed then it is the smallest change in resistance that can be obtained by slider movement. To get high resolution a single slide wire can be used as the resistance element of the potentiometer.

31. Mention two advantages of thermistors over resistance thermometer.

The advantages of thermistors over resistance thermometer are,

- Thermistor gives high output and it is fast acting.
- Relatively small in size, low thermal capacity and it offers high value of temperature coefficient.

32. What is loading effect? Explain with example.

The incapability of the system to faithfully measure, record or control the input signal in undistorted form is called the loading effect.

Example: The output of a potentiometer is normally connected to a meter which has a definite input impedance and hence a current will be drawn by this meter. Due to the presence of meter resistance R_m, there exists a non-linear relationship between $/V_0]$ and displacement X_1. Thus in order to keep linearity, the resistance of the potentiometer R_p should be as small as possible.

33. Why is dynamic compensation network used with hot wire instruments?

The time constant T cannot be reduced much below 0.001 sec in actual practice, which would limit the flat frequency response to less than 160 Hz. This is quite inadequate for turbulence studies since frequencies of 50 kHz and more are of interest. This limitation is overcome by the use of electrical dynamic compensation network.

34. What is piezoresistive effect?

If a metal conductor is stretched or compressed, its resistance changes on the fact that both length and diameter of conductor change. There is a change in the value of resistivity of the conductor, when it is strained. This property is called piezo resistive effect.

35. What is RTD? List the general requirements of RTD.

RTD is also known as resistance thermometer. Resistance of material changes with temperature changes. This property is used in temperature measurement.

Requirements for RTD material are,

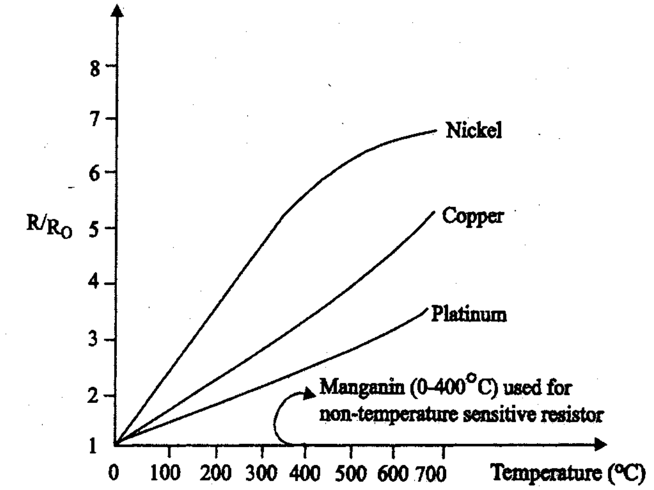
(a) The change in resistance of a material per unit change in temperature should be as large as possible.

(b) The resistivity of material should be high, so that minimum volume of material is used for the construction.

(c) The resistance should have a continuous and stable relationship with temperature.

(d) The material should have positive temperature resistance coefficient.

36. Draw the characteristics of various RTD material.



Characteristics of various RTD material

37. Define thermistors.

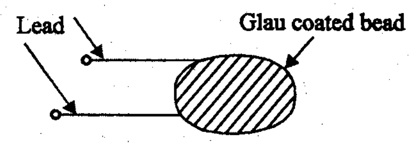
- Thermistors are also known as 'thermal resistors' or semiconducting resistance temperature transducers.
- Thermistors are thermal resistors with a high negative temperature coefficient of resistance.
- It is highly sensitive and it exhibits highly non linear characteristics.

38. What are the different forms of thermistors?

Thermistors are composed of 'sintered mixture of metallic oxides' such as manganese, nickel, cobalt, copper, iron, uranium.

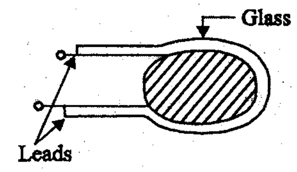
They are classified into four forms

(a) Bead form



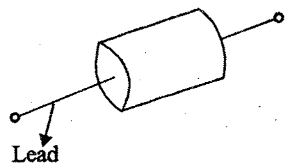
It has diameter of 0.15 mm to 1.25 mm

(b) Probe form



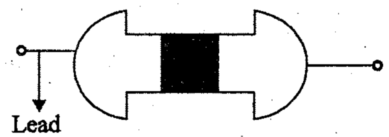
It has diameter of 2.5 mm and length of 6 mm to 50 mm

(c) Disc form



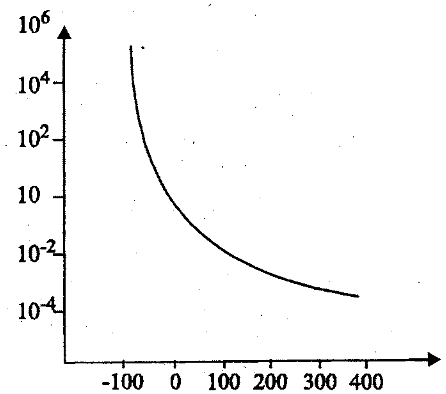
Disc are made by pressing material under high pressure into cylindrical flat shape with dia ranging from 2.5 mm to 25 mm.

(d) Rod form

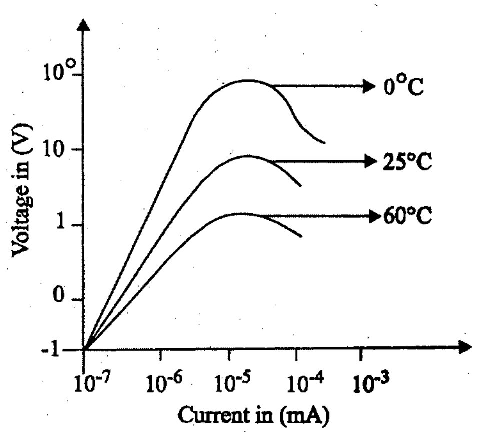


39. Illustrate the performance characteristics of thermistor.

Between $-100°C$ and $400°C$, the thermistor changes its resistivity from 10^5 and 10^{-2} Ωm, a factor of 10^7



Resistive curve for thermistor



V-I characteristics

40. What is hot wire anemometer? Mention its applications?

Hot wire anemometer is used to study varying flow conditions.

When a fluid flows over a heated surface, heat is transferred from the surface and therefore its temperature reduces. The rate of reduction of temperature is related to flow rate.

41. Compare RTD and thermistor.

Sl.No.	RTD	Thermistor
(a)	When temperature increases, the resistance of materials increases. It has positive temperature coefficient.	When temperature decreases, the resistance of material decreases. It has negative temperature coefficient.
(b)	Nickel, copper, platinum are used.	Sintered mixture of metallic oxides are used.
(c)	To approximate the curve linear and quadratic equations are used.	To approximate the curve, Steinhart equation is used
(d)	It is used to measure large range of temperature.	Small change in temperature can be detected.

42. Define humidity, relative humidity and absolute humidity.

Humidity is a measure of water vapour present in gas.

Absolute humidity is the mass of water vapour present per unit volume.

Relative humidity is the ratio of water vapour pressure actually present to water vapour pressure required for saturation at a given temperature. The ratio is expressed in percent. Relative humidity (RM) depends upon temperature.

43. Classify hygrometers.

Hygrometer is also known as 'humidity sensors'.

It is classified as,

(a) Resistive hygrometer.

(b) Capacitive hygrometer.

(c) Aluminium oxide hygrometer.

(d) Crystal hygrometer.

44. A strain gauge having gauge factor of 4 is used for testing machine. If the gauge resistance is 100 Ω and the strain is 20×10^{-6}, how much will be the resistance of strain gauge change?

$GP = 4; R = 100\ \Omega; \varepsilon = 20 \times 10^{-6}\ \Delta R = ?$

$$GP = \frac{(\Delta R/R)}{\varepsilon}$$

$$\Delta = 4 \times 20 \times 10^{-6} \times 100$$

$$= 8 \times 10^{-3}$$

45. A semiconductor gauge having a resistance of 1000 Ω and gauge factor – 133 is subjected to a compressive strain of 500 micro strain. Calculate the new value of resistance of strain gauge change.

$R = 1000\ \Omega; GP = -133; \varepsilon = 500 \times 10^{-6}; \Delta R = ?$

$$GP = \frac{(\Delta R/R)}{\varepsilon}$$

$$\Delta R = -133 \times 500 \times 10^{-6} \times 1000$$

$$= -66.5\ \Omega$$

46. A strain gauge has a resistance of 120 Ω unstrained and gauge factor is – 12. What is the resistance value if the strain is 1%?

$GP = -12; R = 120\ \Omega; \Delta R = ?\ \varepsilon = 1/100 = 0.01$

$$GP = \frac{(\Delta R/R)}{\varepsilon}$$

$$\Delta R = -12 \times 120 \times 0.01$$

$$= -144.72\ \Omega$$

47. A strain gauge with a gauge factor of 4 has a resistance of 500 Ω. It is to be used in a test in which the strain to be measured may be as low as 50×10^{-6}. What will be the change in resistance of the gauge?

$R = 500\ \Omega; GP = 4; \varepsilon = 5 \times 10^{-6}; \Delta R = ?$

$$GP = \frac{(\Delta R/R)}{\varepsilon}$$

$$\Delta R = 4 \times 500 \times 5 \times 10^{-6}$$

$$= 0.01\ \Omega$$

UNIT IV

Variable Inductance and Variable Capacitance Transducers

4.1 VARIABLE INDUCTANCE TRANSDUCER

The variable inductance transducers work, generally, upon one of the following three principles

(i) Change of self inductance

(ii) Change of mutual inductance

and (iii) Production of eddy currents

4.1.1 Transducers working on principle of change of Self-Inductance

The self inductance of a coil $L = \frac{N^2}{R}$

where N - number of turns, and

R - reluctance of the magnetic circuit

The reluctance of the magnetic circuit $R = \frac{l}{\mu A}$

$\therefore$ Inductance, $L = N^2 \mu (A/l)$

$$= N^2 \mu G \qquad \text{... (4.1)}$$

where μ - effective permeability of the medium in and around the coil; H/m.

$G = A/l$ - geometric form factor

A - area of cross-section of coil: m^2, and

l - length of coil, m

It is clear from Eqn, (4.1) that the variation in inductance may be caused by:

(i) change in number of turns, N,

(ii) change in geometric configurations, G,

and (iii) change in permeability, μ

Inductive transducers are mainly used for measurement of displacement. The displacement to be measured is arranged to cause variation of any three variables in Eqn (4.1) and thus alter the self-inductance L by ΔL.

The different types of inductive transducers for measurement of translational and rotary displacements are shown in figure 4.1.

4.1.2 Differential output of Inductive Transducers

Normally the change in self-inductance ΔL is adequate for detection for subsequent stages of instrumentation system. However, if the succeeding instrumentation responds to ΔL, rather than to $L + \Delta L$ the sensitivity and accuracy will be much higher. The transducer can be designed to provide two outputs one of which is an increase of self-inductance and the other is a decrease in self-inductance. The succeeding stages of instrumentation system measure the difference between the outputs, i.e $2\Delta L$. This is known as the differential output. The advantages of differential outputs are

(i) The sensitivity and accuracy are increased.

(ii) The output is less affected by external magnetic fields.

(iii) The effective variations due to temperature changes are reduced.

(iv) The effects of changes in supply voltage and frequency are reduced.

The differential arrangement consists of a coil which is divided into two parts. In response to a physical signal, which is normally a displacement, the inductance of one part increases from L to $L + \Delta L$ while that of the other part decreases from L to $L - \Delta L$. The change is measured as the difference of the two resulting in an output of $2\Delta L$ instead ΔL when only a single winding is used. The differential arrangements are shown in figure 4.1.

4.1.3 Transducers working on principle of change of Mutual Inductance

An inductance transducer working on the principle variation of mutual inductance uses multiple coils. The mutual inductance between two coils is

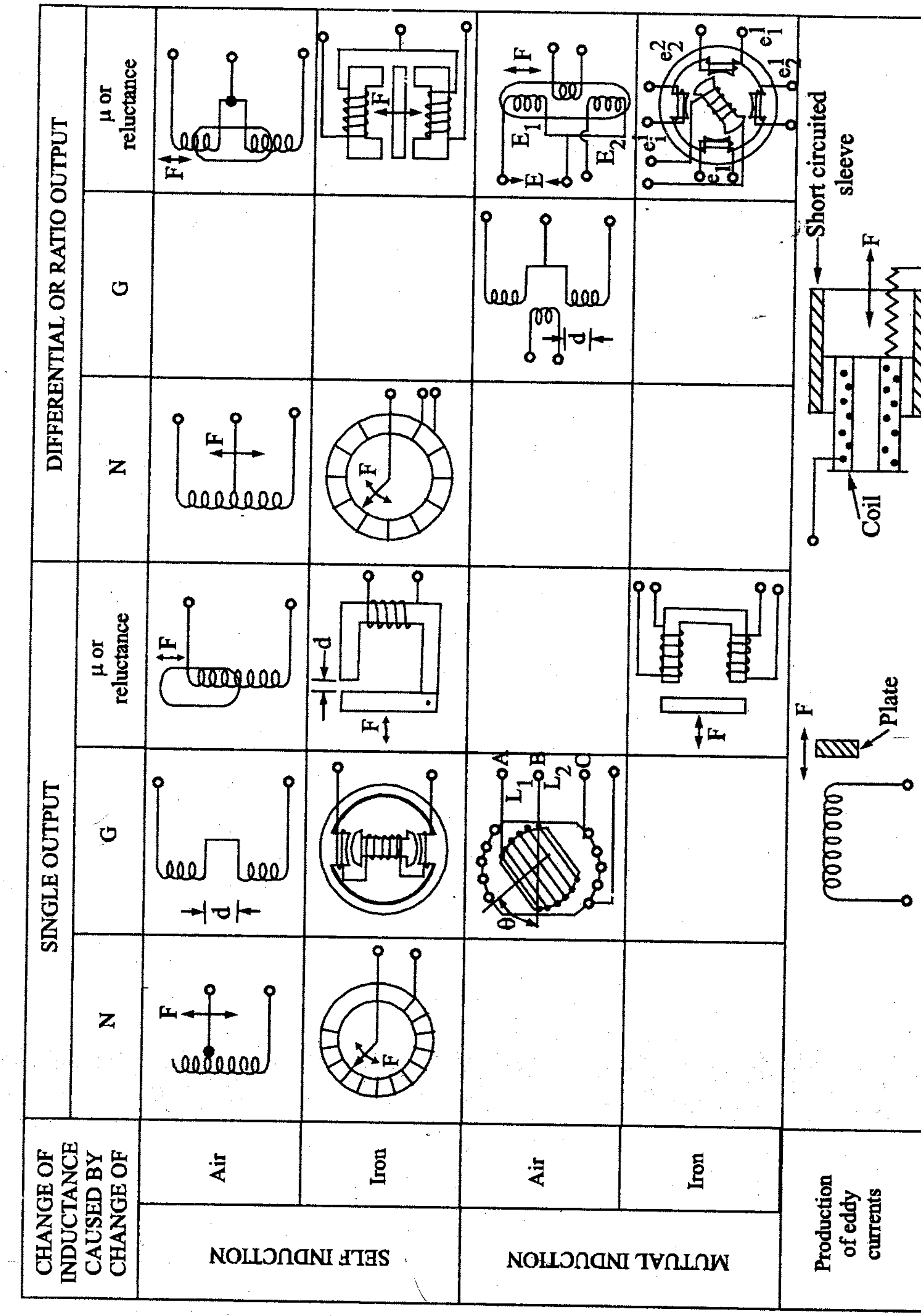


Fig. 4.1 Variable Inductance Transducers

$$M = K\sqrt{L_1 L_2} \quad \text{... (4.2)}$$

where L_1 and L_2 -self inductance of two coils and K - coefficient of coupling

Thus mutual inductance between the coils can be varied by variation of self-inductances or the coefficient of coupling. However, the mutual inductance can be converted into a self-inductance by connecting the coils in series. The self-inductance of such an arrangement varies between $L_1 + L_2 - 2M$ to $L_1 + L_2 + 2M$ with one of the coils being stationary while the other is movable. The self-inductance of each coil is constant but the mutual inductance changes depending upon the displacement of the movable coil.

The different arrangements of measurement of translational and rotary displacements are shown in figure 4.1.

In the differential arrangement, the fixed coil is divided into two parts. The movement of the movable coil increases the mutual inductance of one part by ΔM and decreases that of the other by ΔM.

4.1.4 Types of Inductive Transducers

Inductive transducers can be classified as air cored or iron cored.

Air or iron cored coils can be used for inductive transducers. Both have their own advantages and disadvantages.

Air cored coils

Air cored coil transducers can be operated at a higher carrier frequency because of absence of eddy current losses in air cores. The inductance of air cored coils is independent of the current carried by the coil as the permeability of air is constant and does not depend upon the current carried by the coil. Hence air cored coil transducers can be used for measurement of displacement variations occurring at fairly high frequencies.

Iron cored coils

The greatest disadvantage of iron cored coils transducers is that their inductance is not constant but depends upon the value of the current carried by the coil. Also at high frequencies, the eddy current loss tends to be high and therefore iron cored coil transducers cannot be used beyond a particular

frequency. The frequency of supply voltage should not exceed 20 kHz for iron core transducers to keep the core losses to acceptable values. Hence for accurate measurements the frequency of the input displacement should not exceed 2 kHz.

The advantages of iron cored coil transducers are:

(i) Their size is much smaller then that air cored transducers on account of high permeability of iron cores.

(ii) Iron cored transducers are less likely to cause external magnetic fields because their magnetic field is confined to the iron core of the transducer on account of high permeability and are less affected by stray magnetic fields on account of the high magnetic field produced by them.

Most iron cored transducers are of the variable reluctance type where the length of air gap in the magnetic circuit is varied. In most applications the reluctance of magnetic circuit is primarily that of air gap.

4.2 TRANSDUCERS WORKING ON PRINCIPLE OF PRODUCTION OF EDDY CURRENTS

These inductive transducers work on the principle that if a conducting plate is placed near a coil carrying alternating current, eddy currents are produced in the conducting plate. The conducting plate acts as a short-circuited secondary winding of a transformer. The eddy currents flowing in the plate produce a magnetic field of their own which acts against the magnetic field produced by the coil. This results in reduction of flux and thus the inductance of the coil is reduced. The nearer is the plate to the coil, the higher are the eddy currents and thus higher is the reduction in the inductance of the coil. Thus the inductance of the coil alters with variation of distance between the plate and the coil.

A number of arrangements are possible and two arrangements are shown in figure 4.1. The plate may be at right angle to the axis of the coil. The displacement of the plate causes a change in the inductance of the coil. In the other arrangement a conducting sleeve runs in parallel and coaxially over a coil. If the short-circuited sleeve is away from the coil, the inductance of the coil is high while if the sleeve is covering the coil, its inductance is low. The change in inductance is a measure of displacement.

4.3 INDUCTION POTENTIOMETERS

Two coils coupled to each other, such that the orientation of one of them with respect to the other determines the induced emf in one of them, may be used for measurement of angular deflections over a range of $\pm 90°$. The two coils shown in figure 4.2 constitute an equivalent of a transformer with variable coupling between primary and secondary. The mutual inductance M is maximum when the coils are coaxial, and zero when they are in quadrature. If θ_i is the angle between the coil axes, the mutual inductance and the induced emf in the secondary coils are given by

$$M = M_{max} \cos \theta_i$$

$$e_o = (K E_m \sin w_{ex} t) \cos\theta_i \qquad \ldots (4.3)$$

where $K = a$ constant

$E_m \sin w_{ex} t =$ excitation voltage of frequency w_{ex}

Although the above system can be considered to function as a variable self-inductance potentiometer, with the effective self-inductance given by

$$L_{eff} = L_1 + L_2 \pm 2M_m \cos\theta_i$$

Provision of a closed magnetic circuit with iron core yields some of the advantages.

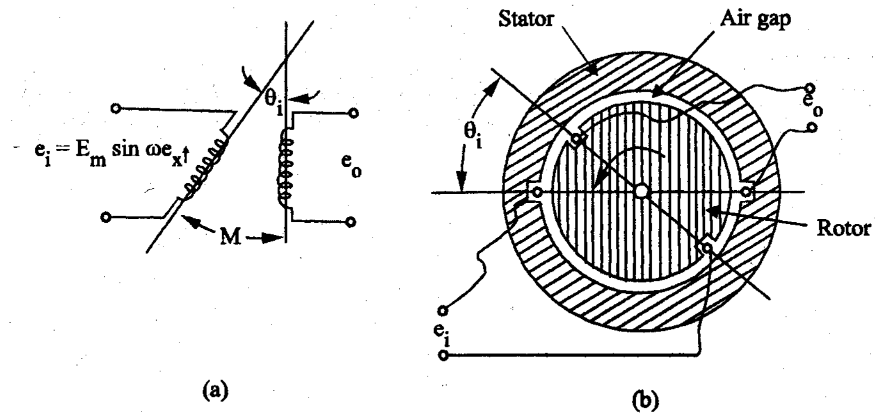


Fig. 4.2 (a) Coupled-coils for angular displacement; (b) rotary induction potentiometer

Figure 4.2 (a) shows such an arrangement, with the two coils mounted, one on the stator and other on the rotor. The rotor is usually dumbbell shaped or of any other suitable shape, which, as far as possible, provides uniform gap over the entire periphery. The coils may be concentrated or distributed over the periphery. The concentrated coil system gives an output voltage which is proportional to θ_i over a very small range around the null point as seen from Eq 4.2 (b), where as provision of distributed windings results in the extension of the linear range to $\pm 90°$. The devices of this kind belong to the class of induction potentiometers, under the patent names of linvar, indpot, etc. They are normally designed for use at excitation frequencies of 50 Hz or 400 Hz, providing sensitivities of the order of 1 volt/degree of rotation. The devices are available in different sizes ranging from 10 mm to 75 mm in diameter. The need for provision of a pair of slip rings and brushes to deliver the output signal makes the induction potentiometer less popular as compared to microsyn, for which the range of measurement is limited to $\pm 5°$.

4.4 LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT)

The most widely used inductive transducer to translate the linear motion into electrical signals is the linear variable differential transformer (LVDT). The basic construction of LVDT is shown in figure 4.3. The transformer consists of a single primary winding P and two secondary windings S_1 and S_2 would on a cylindrical former. The secondary windings have equal number of turns and are identically placed on either side of the primary winding. The primary winding

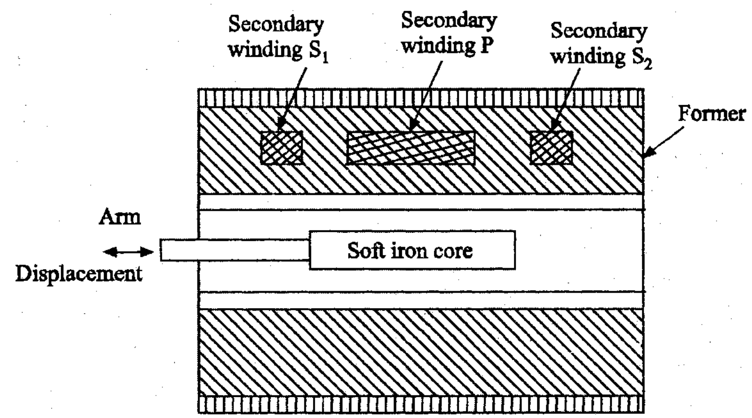


Fig. 4.3 Linear variable differential transformer (L.V.D.T.)

is connected to an alternating current source. A movable soft iron core is placed inside the former.

The displacement to be measured is applied to the arm attached to the soft iron core. In practice the core is made of high permeability, nickel iron which is hydrogen annealed. This gives low harmonics, low null voltage and a high sensitivity. This is slotted longitudinally to reduce eddy current losses. The assembly is placed in a stainless steel housing and the end lids provide electrostatic and electromagnetic shielding. The frequency of a.c. applied to primary windings may be between 50 Hz to 20 kHz.

Since the primary winding is excited by an alternating current source, it produces an alternating magnetic field which in turn induces alternating current voltages in the two secondary windings.

The output voltage of secondary, S_1 is E_{s1} and that of secondary, S_2 is E_{s2}. In order to convert the outputs from S_1 and S_2 into a single voltage signal, the two secondaries S_1 and S_2 are connected in series opposition as shown in fig. 4.4 (b). Thus the output voltage of the transducer is the difference of the two voltages. Differential output voltage,

$$E_o = E_{s1} - E_{s2} \qquad \text{... (4.4)}$$

When the core is at its normal (NULL) position, the flux linking with both the secondary windings is equal and hence equal emfs are induced in them. Thus at null position: $E_{s1} = E_{s2}$. Since the output voltage of the transducer is the difference of the two voltages, the output voltage E_o is zero at null position.

Now if the core is moved to the left of the NULL position, move flux links with winding S_1 and less with winding S_2. Accordingly output voltage E_{s1}, of the secondary winding S_1, is greater than E_{s2}, the output voltage of secondary winding S_2. The magnitude of output voltage is, thus, $E_o = E_{s1} - E_{s2}$ and the output voltage is in phase with the primary voltage. Similarly, if the core is moved to the right of the null position, the flux linking with winding S_2 becomes larger than that linking with winding S_1. This results in E_{s2} becoming larger than E_{s1}. The output voltage in this case is $E_o = E_{s2} - E_{s1}$ and 180° out of phase

with the primary voltage. Therefore, the two differential voltages are 180° out of phase with each other.

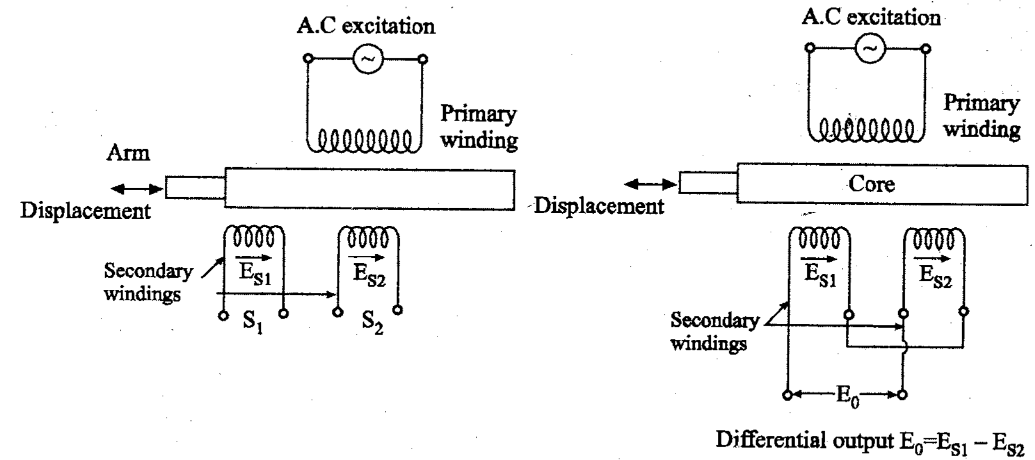


Fig. 4.4 Circuits of an LVDT

The amount of voltage change in either secondary winding is proportional to the amount of movement of the core. Hence, we have an indication of amount of linear motion. By noting which output voltage is increasing or decreasing, we can determine the direction of motion. In other words any physical displacement of the core causes the voltage of one secondary winding to increase while simultaneously reducing the voltage in the other secondary winding. The difference of two voltages appears across the output terminals of the transducer and gives a measure of the physical position of core and hence the displacement.

As the core is moved in one direction from the null position, the differential voltage i.e. the difference of the two secondary voltages, will increase while maintaining an in-phase relationship with the voltage from the input source. In the other direction from the null position, the differential voltage will also increase but will be 180° out of phase with the voltage from the source. By comparing the magnitude and phase of the output (differential) voltage with that of the source, the amount and direction of the movement of the core and hence of displacement may be determined.

The amount of output voltage may be measured to determine the displacement. The output signal may also be applied to a recorder or to a controller that can restore the moving system to its normal position.

The output voltage of an LVDT is a linear function of core displacement within a limited range of motion, about 5 mm from the null position. Figure 4.5 shows the variation of output voltage against displacement for various positions of the core. The curve is practically linear for small displacements (up to about 5 mm). Beyond this range of displacement, the curve starts to deviate from a straight line.

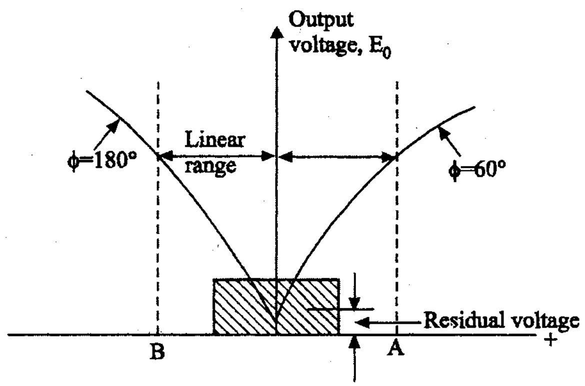


Fig. 4.5 Variation of output voltage with linear displacement for an LVDT

Figure 4.5 shows the variation of output voltage versus displacement for various positions of core. The current is practically linear for a limited range of displacement from the null position. Beyond this range of displacement the curve starts to deviate from a straight line.

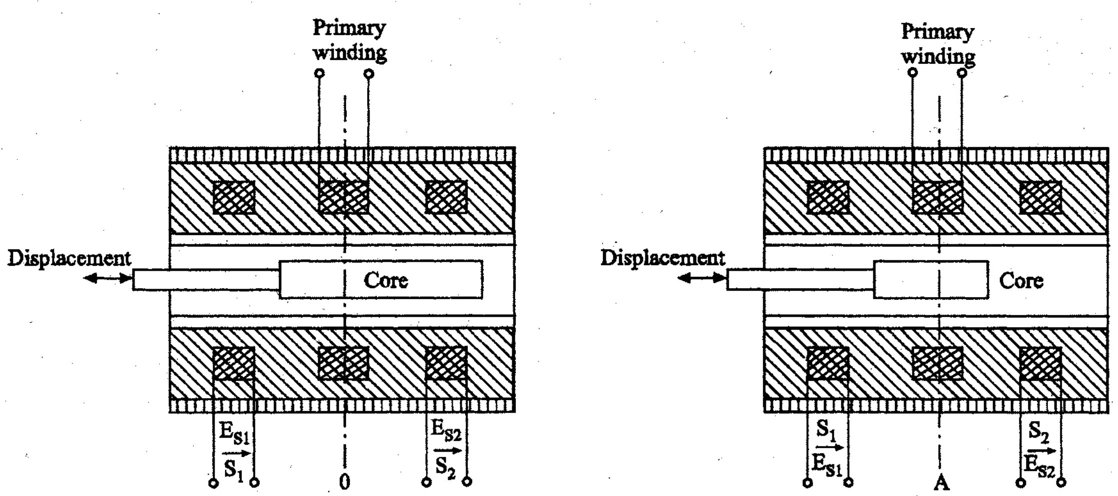


Fig. 4.6 Core of LVDT at different positions

Figure 4.6 shows the core of an LVDT at three different positions. In fig 4.6 (b) the core is at null position, it is symmetrical with respect to both the secondary windings. This is called the null position. At this position $E_{s1} = E_{s2}$ and hence the output voltage $E_o = 0$. When the core is moved to the left as in fig 4.6 (a) and is at A, E_{s1} is greater than E_{s2} and therefore phase angle $\phi = 0$. When the core is moved to the right towards B shown in fig 4.6 (c) E_{s2} is greater than E_{s1} and hence the output voltage is negative or a phase angle of 180°.

The characteristics are linear up to $O - A$ and $O - B$ but after that they become non-linear as shown in fig 4.6. Ideally the output voltage at the null

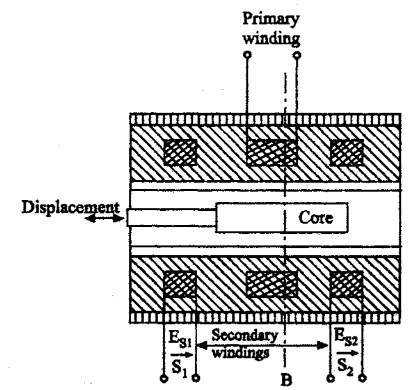


Fig. 4.6 (c) Core of LVDT at different positions

position should be equal to zero. However, in actual practice there exists a small voltage at the null position. This may be on account of presence of harmonics in the input supply voltage and also due to harmonics produced in the output voltage on account of use iron core. There may be either an incomplete magnetic or electrical unbalance or both which result in a finite output voltage at the null position. This finite residual voltage is generally less than 1% of the maximum output voltage in the linear range. Other causes of residual voltage are stray magnetic fields and temperature effects. The residual voltage is shown in fig 4.7. However, with improved technological methods and with the use of better a.c sources, the residual voltage can be reduced to almost a negligible value.

4.5 ROTARY VARIABLE DIFFERENTIAL TRANSFORMER (RVDT)

A variation of linear variable differential transformer (LVDT) may be used to sense angular displacement. This is the Rotary Variable Differential Transformer (RVDT). The circuit of a RVDT is shown in fig 4.8. It is similar to the LVDT except that its core is cam shaped and may be rotated between the windings by means of a shaft.

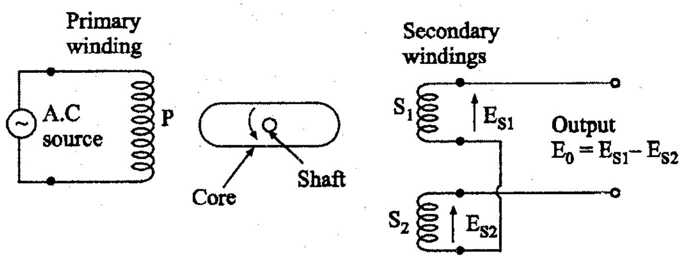


Fig. 4.8 Rotary variable differential transformer (RVDT)

The operation of a RVDT is similar to that of an LVDT. At the null position of the core, the output voltages of secondary windings S_1 and S_2 are equal and in opposition. Therefore, the net output is zero. Any angular displacement from the null position will result in a differential voltage output. The greater this angular displacement, the greater will be the differential output. Hence the response of the transducer is linear.

Clockwise rotation produces an increasing voltage of a secondary winding of one phase while counter clock-wise rotation produces an increasing voltage of opposite phase. Hence, the amount of angular displacement and its direction may be ascertained from the magnitude and phase of the output voltage of the transducer.

4.6 VARIABLE RELUCTANCE PRESSURE TRANSDUCER

Reluctance in a magnetic circuit is equivalent to resistance in an electrical circuit. Whenever the spacing (or coupling) between the two magnetic devices (or coils) changes, the reluctance between them also changes. Thus a pressure sensor can be used to change the spacing between two coils by moving one part of the magnetic circuit. This motion changes the reluctance between the coils, which in turn changes the voltage induced by one coil in the other. The change in the induced voltage can then be interpreted as a change in pressure.

(a) Linear Variable Differential Transformer

It is the most widely used inductive transducer to translate linear motion in to an electrical signal. Figure 4.9 shows an LVDT for the measurement of pressure.

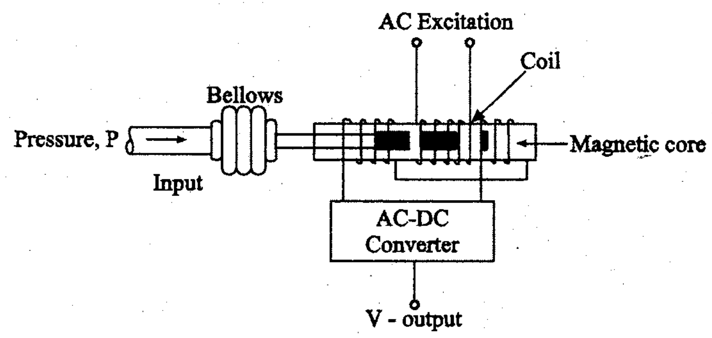


Fig. 4.9 Linear Variable Differential Transformer (LVDT)

Construction and Working

It consists of a primary winding (or coil) and two secondary windings (or coils). The windings are arranged concentrically next to each other. They are wound over a hollow bobbin which is usually of a non-magnetic and insulating materials. A ferromagnetic core (armature) is attached to the transducer sensing shaft (such as bellows). The core is generally made of a high permeability ferromagnetic alloy and has the shape of a rod or cylinder.

A.C excitation is applied across the primary winding and the movable core varies the coupling between it and the two secondary windings. When the core is in the centre position, the coupling to the secondary coils is equal. As the core moves away from the centre position, the coupling to one secondary, and hence its output voltage, increases while the coupling and the output voltage of the other secondary decreases.

Any change in pressure makes the bellows expand or contract. This motion moves the magnetic core inside the hollow portion of the bobbin. It causes the voltage of one secondary winding to increase, while simultaneously reducing the voltage in the other secondary winding. The difference of the two voltage appears across the output terminals of the transducers and gives a measure of the physical position of the core and hence the pressure.

Advantages

- It possesses a high sensitivity.

- It has infinite resolution.
- It is very rugged in construction and can usually tolerate a high degree of shock & vibration without any adverse effects.
- The output voltage of this transducer is practically linear for displacements of about 5 mm.
- It shows a low hysteresis, hence repeatability is excellent under all conditions.
- It is stable and easy to align and maintain due to simplicity of construction, small size and light body.

Disadvantages

- Temperature affects the performance of the transducer.
- Relatively large core displacements are required for appreciable amount of differential output.
- They are sensitive to stray magnetic fields but shielding is possible.

(b) Servo Pressure Transducer

Working principle

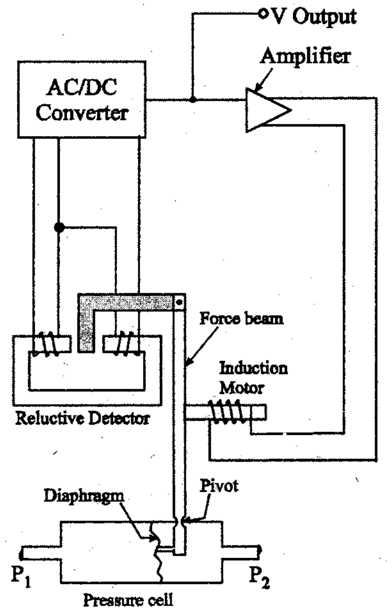


Fig. 4.10 A Servo Pressure Transducer

An increase in pressure P_1 over P_2 (fig 4.10) flexes the diaphragm and moves the short end of the force beam. The force beam pivots, and the long end moves a magnetic material in the reluctive detector. The signal from the reductive detector is converted from a.c power to d.c power, and sent to an amplifier. The amplifier responds by activating an inductive motor that moves the force beam back towards its original position. Very little flexing ever occurs in the diaphragm, even over the entire range of the instrument. As a result, the diaphragm lasts a long time.

Servo pressure transducers are available in a multitude of pressure ranges. The devices are generally used for measurement of pressure below 500 psi.

They do not respond to high frequency pressure oscillations. Other servo pressure instruments use capacitive detectors, and some use a Bourdon tube as the sensing element.

4.7 INDUCTIVE THICKNESS TRANSDUCER

In industry, the measurement of the thickness of rolled sheets or mass-produced objects is a common requirement. The material of the test sheet or object may be magnetic (iron or steel) nonmagnetic and conducting

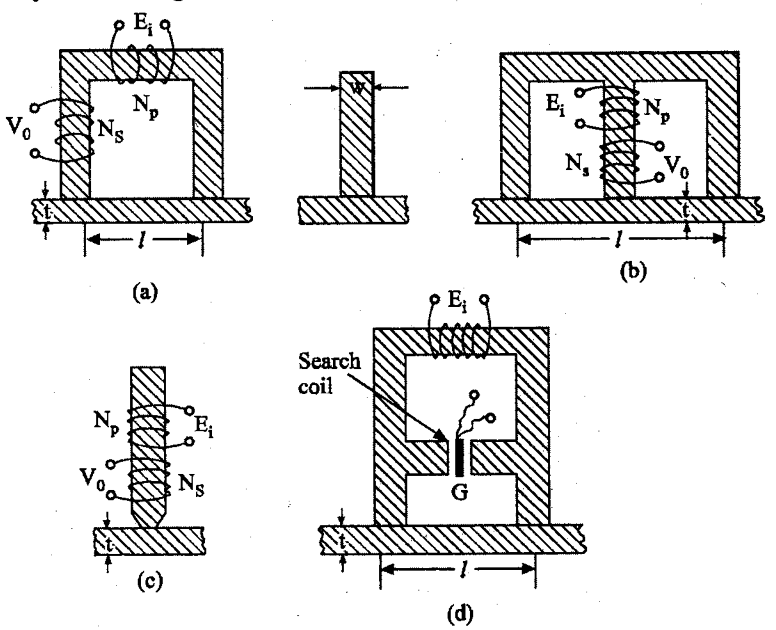


Fig. 4.11 Different arrangements for measurement of thickness of metallic and magnetic sheets

(Aluminium or Copper) or nonmagnetic and nonconducting (bakelite or paint). Inductive transducers meant for such purposes are known as inductive thickness gauges. As the thickness is of primary interest, it is important that the properties of the materials, such as permeability and resistivity, should remain constant. Each gauge is suitably designed for use with the test object and calibrated by making use of reference sheets or slabs of known thickness but of the same material of the test object.

Variable reluctance type inductance transducers prove handy for most of the applications. An E -, U -, I - shaped yoke of high permeability material is provided with one coil for the self-inductance type and a pair of coils for the mutual inductance type. The magnetic path is completed through the test piece of magnetic material, as shown in figure 4.11. The yoke is usually laminated to limit the eddy currents produced when the coil is excited by alternating current. The attraction force of the yoke on the armature and weight of the yoke may help in reducing the air gap between the yoke and the test piece. However, the surfaces of the test piece and the yoke are kept smooth for a closer contact.

If the reluctance of the yoke is made negligible as compared to that of the test piece, the self-inductance L of the coil is proportional to that of the test piece, the self-inductance L of the coil is proportional to the thickness t of the test piece and is given by

$$L = \frac{N^2 \mu_0 \mu_r bt}{l}$$

where b and l are the width and length respectively of the test piece, and μ_r is the relative permeability of the material.

The thickness of sheets of magnetic material as well as insulating material may be obtained by any of the arrangements as shown in figure 4.11. In the case of insulating material, the sheet is kept between the yoke, and a magnetic material backing of known thickness. The reluctance of the path is almost governed by the thickness of insulating sheet.

Measurement of thickness of test pieces ranging from 25 μ m to 2.5 mm is possible by the above methods with an accuracy of 2 - 5%.

The primary coil of the system shown in fig 4.11 is excited from a relatively high frequency source as the reluctance variation with the thickness of the sample will be very small. However, it is possible to measure variations in the thickness of conducting material sheets. The induced emf of the secondary coil may be used for direct indication and calibration.

An alternative is shown in fig 4.11 where the test object of magnetic material forms a low reluctance shunt path for the magnetic flux across the gap G. The induced emfs of the search coil serve as the output signals of the transducer. The primary coil is excited from a constant voltage source of suitable frequency.

4.8 CAPACITIVE TRANSDUCER

The principle of operation of capacitive transducers is based upon the familiar equation for capacitance of a parallel plate capacitor.

Capacitance,

$$C = \varepsilon A / d = \Sigma_r \Sigma_0 A / d \qquad \text{... (4.5)}$$

where A - overlapping area of plates; m^2

d - distance between two plate; m

$\in = \Sigma_0 \Sigma_r =$ permittivity of medium, f/m

$\in_r$ - relative permittivity

$\in_0$ - permittivity of free space; 8.85×10^{-12} f/m

A parallel plate capacitor is shown in figure 4.12

The capacitive transducer works on the principle of change of capacitance which may be caused by:

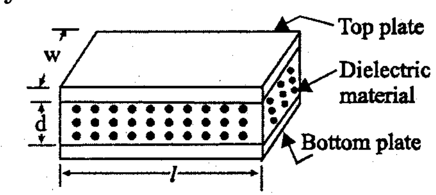


Fig. 4.12 Schematic diagram of a parallel plate capacitive transducer

(i) change in overlapping area A,

(ii) change in the distance d between the plates, and

(iii) change in dielectric constant

These changes are caused by physical variables like displacement, force and pressure in most of the cases. The change in capacitance may be caused by change in dielectric constant as in the case in measurement of liquid or gas levels.

The capacitance may be measured with bridge circuits. The output impedance of a capacitive transducer is: $X_c = 1/2\pi fc$,

where c - capacitance

f - frequency of excitation in Hz

In general, the output impedance of a capacitive transducer is high. This fact calls for a careful design of the output circuitry.

The capacitive transducers are commonly used for measurement of linear displacement. These transducers use the following effects:

(i) change in capacitance due to change in overlapping area of plates and

(ii) change in capacitance due to change in distance between the two plates.

4.8.1 Transducer using change in Area of plates

The capacitance is directly proportional to the area, A of the plates. Thus the capacitance changes linearly with change in area of plates. Hence this type of capacitive transducer is useful for measurement of moderate to large displacements say from 1 mm to several cm. The elementary diagrams of two types of capacitive transducers are shown in figure 4.13 (a) & 4.13 (b). The area changes linearly with displacement and also the capacitance. Figure 4.13 shows the variation of capacitance.

For a parallel plate capacitor, The capacitance is

$$C = \frac{\in A}{d} = \frac{\in xw}{d} f \qquad \text{... (4.6)}$$

where x - length of overlapping part of plates, m

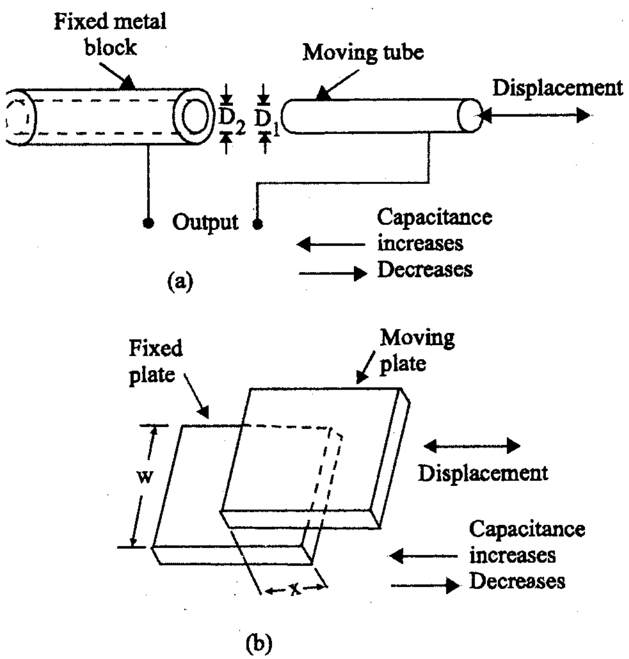


Fig. 4.13 Capacitive transducers working on the principle of change of capacitance with change of area

and w - width of overlapping part of plates, m

Sensitivity,

$$s = \frac{\partial c}{\partial x} = \in \frac{w}{d} f/m \qquad \text{... (4.7)}$$

The sensitivity is constant and therefore there is linear relationship between capacitance and displacement.

Sensitivity for a fractional change in

Capacitance

$$S = \frac{\partial C}{C\,\partial x} = \frac{1}{x} \qquad \text{... (4.8)}$$

This type of a capacitive transducer is suitable for measurement of linear displacements ranging from 1 mm to 10 mm. The accuracy is as high as 0.005%.

For a cylindrical capacitor the capacitance is:

$$e = \frac{2\pi \in x}{\log_e (D_2/D_1)} f \quad \text{... (4.9)}$$

where x - length of overlapping part of cylinders; m,

D_2 - inner diameter of outer cylindrical electrode; m,

and D_1 - outer diameter of inner cylindrical electrode; m

Sensitivity,

$$S = \frac{\partial C}{\partial x} = \frac{2\pi \in}{\log_e (D_2/D_1)} f/m \quad \text{... (4.10)}$$

Therefore, the sensitivity is constant and the relationship between capacitance and displacement is linear as shown in figure 4.14.

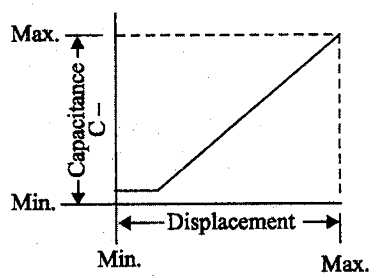


Fig. 4.14 Capacitance displacement curve of capacitive transducer (working on principle of change of plate area caused by change in displacement)

The principle of change of capacitance with change in area can be employed for measurement of angular displacement. Fig 4.15 (a) shows a two-plate capacitor. One plate is fixed and the other is movable. The angular displacement

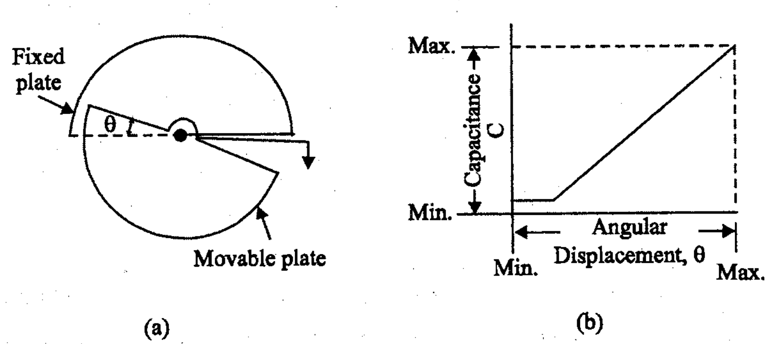


Fig. 4.15 Capacitive transducer for measurement of angular displacement

to be measured is applied to movable plate. The angular displacement changes the effective area between the plates and thus changes the capacitance. The capacitance is maximum when the two plates completely overlap each other i.e when $\theta = 180°$.

$\therefore$ Maximum value of capacitance

$$\in_{max} = \frac{\in A}{d} = \frac{\pi \in r^2}{2d} \quad \text{... (4.11)}$$

Capacitance at angle θ is, $C = \dfrac{\in \theta r^2}{2d}$ ... (4.12)

where θ - angular displacement in radian

$$S = \frac{\partial C}{\partial \theta} = \frac{\in r^2}{2d} \quad \text{... (4.13)}$$

Therefore, the variation of capacitance with angular displacement is linear. This is shown in figure 4.15 (b). It should be understood that the above mentioned capacitive transducer can be used for a maximum angular displacement of 180°.

4.8.2 Transducer using change in Distance between plates

Capacitive transducer utilizing the effect of change of capacitance with change in distance between the two plates. One is a fixed plate and the displacement to be measured is applied to other plate which is movable. Since, the capacitance, C, varies inversely as the distance x, between the plates the

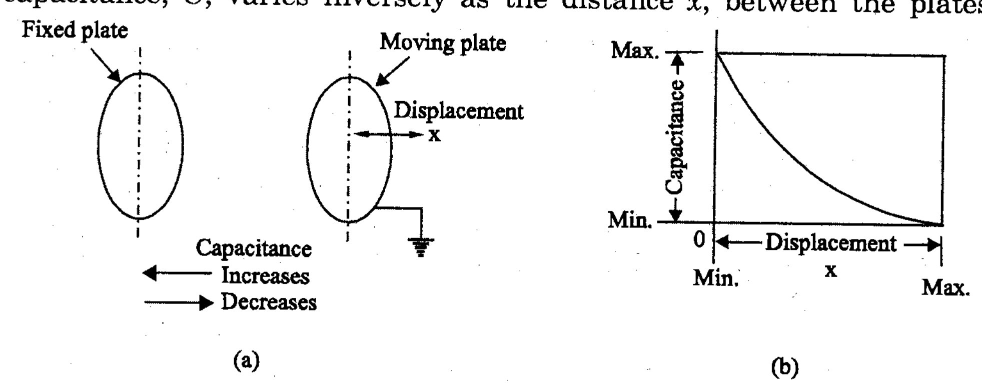


Fig. 4.16 Capacitive transducer using the principle of change of capacitance with change of distance between plates

response of this transducer is not linear and as shown in figure 4.16 (b). Thus this transducer is useful only for measurement of extremely small displacements.

Sensitivity

$$S = \frac{\partial C}{\partial x} = -\frac{\in A}{x^2} \quad \text{... (4.14)}$$

From this equation it is clear that the sensitivity of this type of transducer is not constant but varies over the range of the transducer. Thus, as explained earlier this transducer exhibits non-linear characteristics.

The relationship between variation of capacitance C with variation of distance between plates, x, is hyperbolic and is only approximately linear over a small range of displacement. The linearity can be closely approximated by use of a piece of dielectric material like mica having a high dielectric constant. In this type of transducer, a thin piece of mica thinner than the minimum gap distance is inserted between the plates.

Rotational displacement can be measured with an arrangement shown in figure 4.17. As the rotor plates of the capacitor are displaced in the counter clockwise direction the capacitance increases.

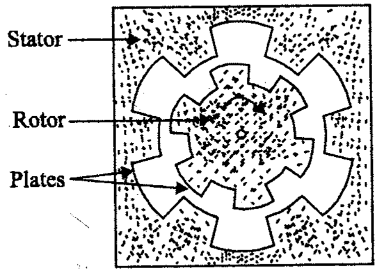


Fig. 4.17 Capacitive transducer

The change in the capacitance is a measure of the angular displacement. This capacitive transducer can be effectively used for measurement of torque.

4.8.3 Different measurements of Capacitive Transducers

- Capacitive Level Transducer
- Capacitive Displacement Transducer
- Capacitive Thickness Transducer

- Capacitive Strain Transducer
- Capacitive Pressure Transducer
- Capacitive Proximity Transducer
- Capacitive Moisture Transducer
- Capacitive Hygrometer
- Capacitive Microphone

4.8.3.1 Capacitive Level Transducer (Variation of Dielectric constant)

Capacitive Transducers using the principle of change of capacitance with change of dielectric are normally used for measurement of liquid levels. Figure 4.18 shows a capacitive transducer used for measurement of level of non-conducting liquid.

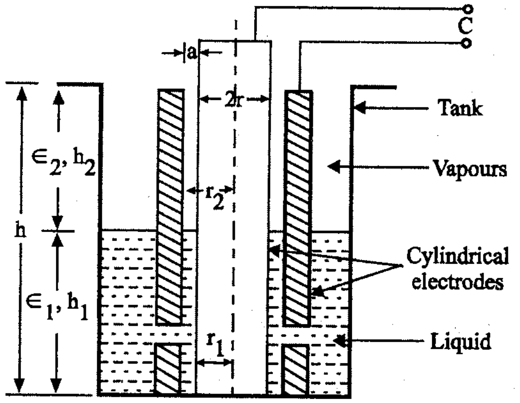


Fig. 4.18 Capacitive transducer for measurement of level of a non-conducting liquid

The electrodes are two concentric cylinders and the non-conducting liquid acts as the dielectric. At the lower end of the outer cylinder there are holes which allow passage of liquid. In case these holes are small, they provide mechanical damping of the surface variation.

The value of capacitance for the capacitor is

$$C = 2\pi \in_0 \frac{\in_1 h_1 + \in_2 h_2}{\log_e (r_2/r_1)} \quad \text{... (4.15)}$$

where h_1 - height of liquid; m,

h_2 - height of cylinder above liquid; m,

ϵ_1 - relative permittivity of liquid,

ϵ_2 - relative permittivity of vapour above liquid,

r_2 - inside radius of outer cylinder; m,

r_1 - outside radius of inner cylinder; m,

ϵ_0 - permittivity of free space; f/m

Relationship (4.15) is based upon the assumption

$n >> r_2$ and $r_2 >> r_2 - r_1 >> a$

Now, $r_2 = r + a$ and $r_1 = r$

$$\therefore C = 2\pi \epsilon_0 \frac{\epsilon_1 h_1 + \epsilon_2 h_2}{\log_e\left(1 + \frac{a}{r}\right)} \quad \text{... (4.15)}$$

4.8.3.2 *Capacitive Displacement Transducer*

The most popular form of variable capacitor used in displacement measurement is parallel plate capacitor with a variable air gap.

The simplest form of displacement transducer is a parallel plate capacitor with plate movable as shown in figure 4.19.

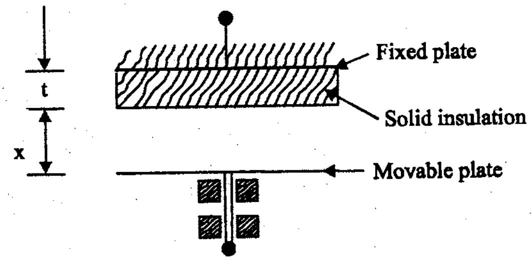


Fig. 4.19 Simple capacitive Displacement Transducer

The capacitance is given by,

$$C_x = \frac{A}{\frac{x}{\epsilon_0} + \frac{t}{\epsilon_1 \epsilon_0}} = \frac{A \epsilon_0}{x + \frac{t}{\epsilon_r}} \quad \text{... (4.16)}$$

where A is the common area between the plates

't' is the thickness of the solid dielectric medium

ϵ_r is the relative permittivity of the solid portion

ϵ_0 is the permittivity of air

If the air gap is increased by x then the capacitance will get reduced to

$$C_x - \Delta C = \frac{A \epsilon_0}{x + \Delta x + \frac{t}{\epsilon_r}} \quad \text{... (4.17)}$$

The sensitivity is,

$$\frac{\Delta C}{C_x} = \frac{C_x - (C_x - \Delta C)}{C_x}$$

$$= 1 - \frac{x + \frac{t}{\epsilon_r}}{x + \Delta x + \frac{t}{\epsilon_r}} = \frac{\Delta x}{x + \Delta x + \frac{t}{\epsilon_r}} \quad \text{... (4.18)}$$

$$\frac{\Delta C}{C_x} = \frac{\Delta x}{\left(x + \frac{t}{\epsilon_r}\right) + \Delta x} \quad \text{... (4.19)}$$

If Δx is very small compared to $x + \frac{t}{\epsilon_r}$ it can be deleted, then

$$\frac{\Delta C}{C_x} = \frac{\Delta x}{x + \frac{t}{\epsilon_r}} \quad \text{... (4.20)}$$

The per unit variation of capacitance is proportional to Δx. Thus it is linear over a small range of Δx. The range of linearity can be increased by having another fixed electrode as shown in figure 4.20 (a). The circuit connection is shown in figure 4.20 (b), which is a unity ratio arm wheatstone bridge.

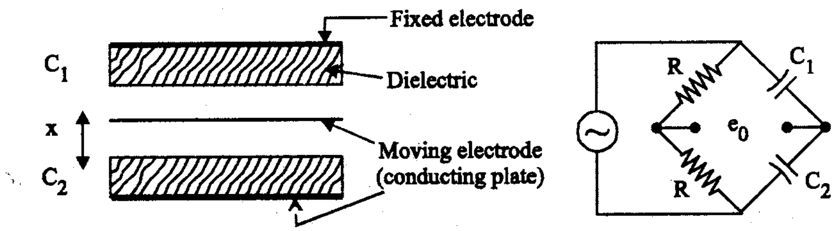


Fig. 4.20 (a) and 4.20 (b) Two fixed plate capacitive transducer and its circuitry

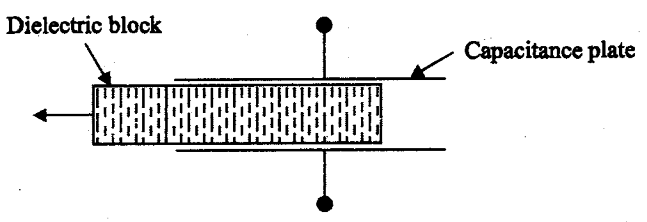


Fig. 4.21 Capacitive transducer for large displacement

For large linear displacements, capacitive transducers where the plates are fixed and the dielectric medium is moved as shown in figure 4.21 can be used.

4.8.3.3 Capacitive Thickness Transducer

If the material is being tested is an insulator, capacitive method using an arrangement shown in figure 4.22 may be used.

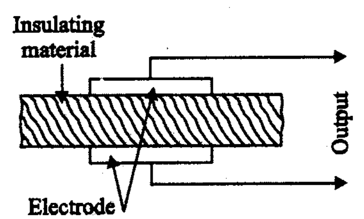


Fig. 4.22 Electrode of thickness of insulating materials

Two metal electrodes are placed on the two sides of the insulating material being tested. This arrangement forms a parallel plate capacitor, the two electrodes acting as the two plates with the insulating material acting as the dielectric. The capacitance naturally depends upon the thickness of the insulating material under the test. Thus by measuring the capacitance of this arrangement, the thickness of the insulating material may be determined.

4.8.3.4 Capacitive Strain Transducer

A strain gauge based on the principle of capacitance variation with plate separation is developed making use of two arched metal strips to support the electrodes of the capacitor, as shown in figure 4.23 (a). When the structure is strained, there is a change in the differential height of the arches as well as the gap between the electrodes. The height variation of each arch strip is calculated from

$$x^2 - x_0^2 = -\frac{15}{64} \in L\,(\in L + 2\,W_0) \qquad \ldots (4.21)$$

where $\in$ - strain

x - height of arch under strain

x_0 - initial height of arch under no strain

w_0 - unstrained width of arch

L - gauge length

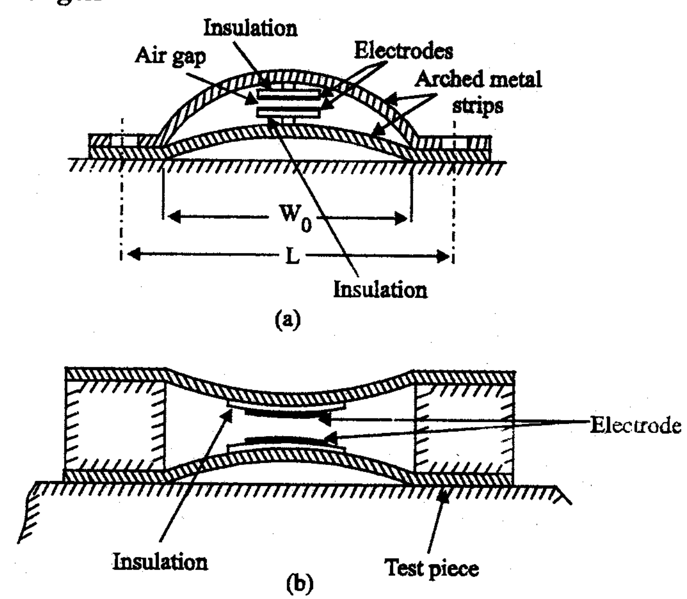


Fig. 4.23 Capacitive strain transducers using (a) plate separation (b) gap changing by arching

The gauge factor $\left(= \frac{\Delta C / C_0}{\epsilon} \right)$ is about 100 and the gauge is used for measurements of strain up to $\pm 5000° \mu$ at temperatures as high as 600°C.

An alternative arrangement is shown in figure 4.23 (b) in which the bowing of the arched metallic parts due to strain changes the gap between the electrodes. The flexible insulating strips and electrodes are cemented to the arched parts. The capacitance between the two live electrodes gives a measure of the strain.

4.8.3.5 Capacitive Pressure Transducer

Differential pressure can be transduced by a three terminal capacitor as shown in figure 4.24.

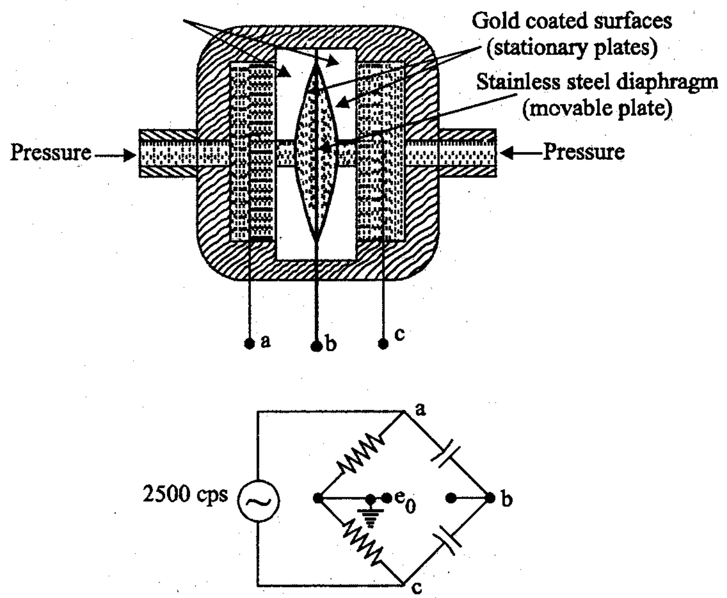


Fig. 4.24 Differential capacitor pressure pick up

Spherical cavity of a depth of about 0.025 mm is ground in to the glass disk. These depressions are gold coated to form fixed plates of a differential capacitor. A thin stainless steel diaphragm is clamped between the disks which serves as the movable plate. With equal pressures applied to both parts, the diaphragm is in a neutral position and the bridge is balanced and $e_o = 0$.

If one pressure is greater than the other the diaphragm deflects to the low pressure side, giving an output e_o in proportion to the differential pressure. For the opposite pressure difference e_o exhibits a 180° phase change. The high impedance level requires a cathode follower amplifier at e_o. A direction sensitive d.c output can be obtained by conventional phase sensitive demodulation and filtering.

4.8.3.6 Capacitive Proximity Transducer

In certain applications, the proximity of an object with respect to the fixed plate of the transducer is desired. Electrical circuits that develop output voltages proportional to the separation between the plates are available. The circuit shown in figure 4.25 uses an operational amplifier of high gain, giving output signal e_o proportional to x_o.

The moving object is provided with a plane conducting surface, if it does not behave like one. The object is earthed and the fixed plate is so designed as to have much smaller area than the movable surface and is provided with a guard ring as shown in figure 4.25. The output signal e_o is given by,

$$e_o = -\frac{C_f}{C_x} e_i = -\frac{C_f x_o}{\varepsilon_o A} E_m \sin \omega_{ex} t \qquad \text{... (4.22)}$$

where C_f = capacitance of the standard capacitor

$E_m \sin \omega_{ex} t$ = sinusoidal applied voltage

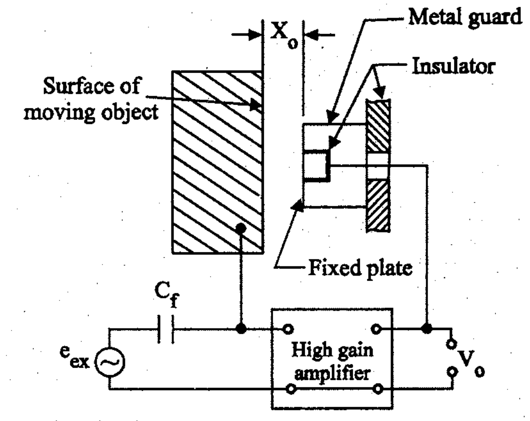


Fig. 4.25 A proximity transducer system along with signal processing circuit

4.8.3.7 *Capacitance Moisture Transducer*

The dielectric constant of pure water is about 80 and that of most insulating materials, solids or liquids is less than 10, and so it is possible to measure the moisture content of these materials by measuring the dielectric constant of the moist solid or solution of the substance in water. The technique can be extended for application to other combinations, if the variation in the dielectric constant is due to variation of the proportion of one substance in the mixture. The equivalent series on shunt resistance of the capacitor, representing the dielectric losses of the sample, may also be used to indicate the moisture content.

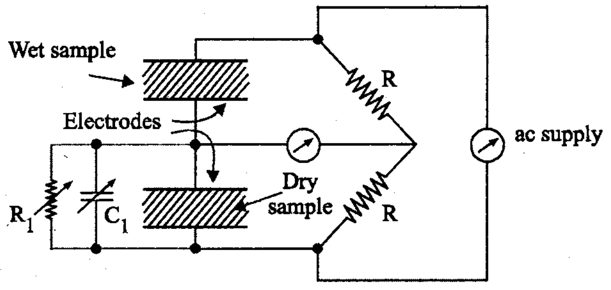


Fig. 4.26 A capacitive moisture transducer

Two identical capacitors, one holding the test sample and the other the dry sample, may be used in an ac bridge circuit, and the equivalent loss resistance as well as the capacitance may be measured by balancing the bridge. As the capacitance value increase with moisture and equivalent shunt resistance falls, the arm with dry sample may be shunted by a variable capacitor and resistor as shown in figure 4.26, and their values may be calibrated against the moisture content. Otherwise, the unbalance voltage may be directly used for calibration. One particular advantage of solids is that no additional means are necessary for them to compact the test material between the electrodes for good contact as is the case with resistive moisture transducers.

4.8.3.8 *Capacitance Hygrometer*

A more practical form of hygrometer employs the arrangement shown in figure 4.27 (a). The central part of the transducer is an aluminium rod acting as one electrode. The rod is oxidized over part of its length over which is provided a thin layer of graphite or of an evaporated metal. Moisture is absorbed through this thin porous layer, by the aluminium oxide, and the equivalent capacitance

between the outer metallic layer and aluminium rod undergoes variation because of the amount of moisture absorbed. When equilibrium is reached with the moist atmosphere the resistance and capacitance of the capacitance are measured.

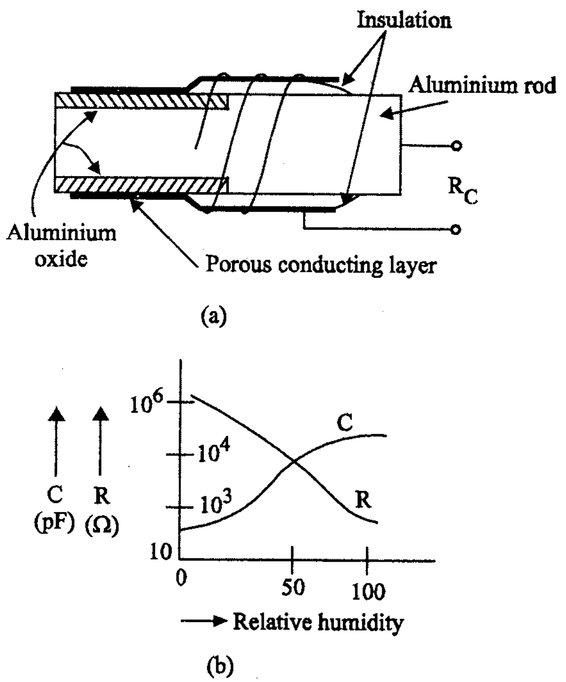


Fig. 4.27 (a) A capacitive hygrometer; (b) characteristic curves showing the effect of humidity on *R* and *C*

The variation of both components is shown in figure 4.27 (b) and can be used as a measure of the relative humidity. To some extent, the resistance variation is linear, but capacitance variation is non-linear.

4.8.3.9 *Capacitive Microphone*

Figure (4.28) is a simplified version of a typical capacitor microphone. The pressure response is found by assuming a uniform pressure p_i to exist all around the microphone at any instant of time. This is actually the case of sufficiently low sound frequencies but reflection and diffraction effects distort this uniform field at higher frequencies. The diaphragm is generally a very thin metal membrane which is stretched by suitable clamping arrangement. Diaphragm thickness ranges from about 0.0025 to 0.050 mm. The diaphragm is deflected

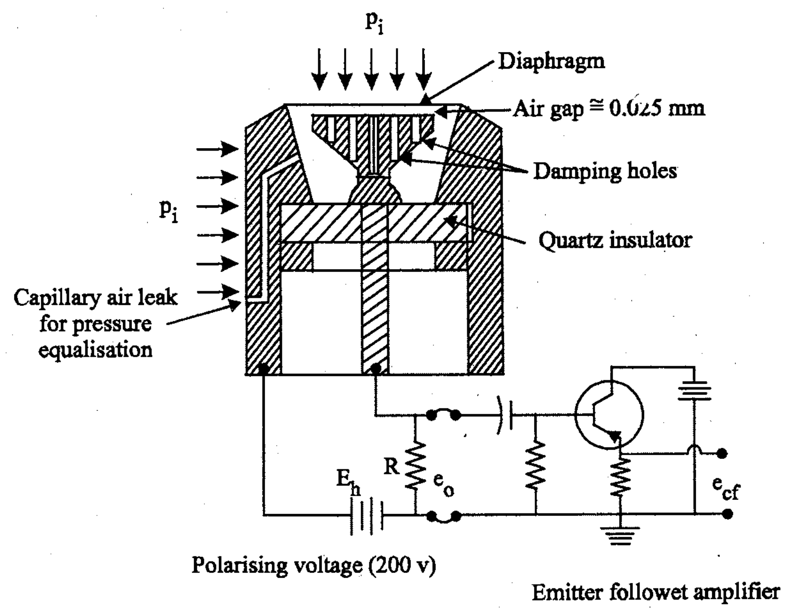


Fig. 4.28 Condenser microphone

by the sound pressure and acts as a moving plate of a capacitance displacement transducer. The other plate of the capacitor is stationary and may contain properly designed damping holes. The damping effect is used to control the resonant peak of the diaphragm response. A capillary air leak is provided to give equalisation of steady pressure on both sides of the diaphragm to prevent diaphragm busting.

The variable capacitor is connected into a simple series circuit with a high resistance R and polarised with a dc voltage of about 200 volts. This polarising voltage acts as a circuit excitation and also determines the neutral diaphragm position.

TWO MARK QUESTIONS AND ANSWERS

1. What is inductance transducer?

Transducers based on the variation of inductance are another group of important devices used in many applications. In these transducers, self inductance or the mutual of a couple of coils is changed when the quantity to be measured is varied.

2. Mention three principles of inductance transducer.

The three principles of inductance transducer are,

- Change of self inductance.
- Change of mutual inductance.
- Production of eddy currents.

3. What is LVDT?

The Linear Variable Differential Transformer (LVDT) is the most common mutual inductance element. This can be considered to be a versatile transducer element for most of the electromechanical measuring systems with regards to resolution, hysteresis, dynamic response, temperature characteristics, linearity and life.

4. What are the advantages and disadvantages of LVDT?

The advantages of LVDT are,

(a) High range.

(b) Friction and electrical isolation.

(c) Immunity from external effects.

(d) High input and high sensitivity.

(e) Ruggedness.

(f) Low hysteresis.

(g) Low power consumption.

The disadvantages of LVDT are,

(a) Relatively large displacements are required for appreciable differential output.

(b) They are sensitive to stray magnetic fields but shielding is possible.

(c) Many a times, the transducer performance is affected by vibrations.

(d) The receiving instrument must be selected to operate on AC.

(e) The dynamic response is limited.

(f) Temperature affects the performance of the transducer.

5. What are the applications of LVDT?

The applications of LVDT are,

- Displacement measurement and LVDT gauge heads.
- LVDT pneumatic servo follower.
- LVDT load cells.
- LVDT pressure transducer.

6. What is null voltage?

Ideally, the output voltage at the null position should be equal to zero. However, in actual practice there exists a small voltage at the null position.

7. Explain the principle of induction potentiometer.

The primary is excited with alternating current. This induces a voltage into the secondary. The amplitude of this output voltage varies with the mutual inductance between the two coils and this varies with the angle of rotation.

8. Explain the principle of variable reluctance accelerometer.

Variable reluctance accelerometer is an accelerometer for measurement of acceleration in the range ± 4g. Since the force required to accelerate a mass is proportional to the acceleration.

9. What is the need of demodulator in variable reluctance accelerometer?

To detect the motion on both sides of zero, a fairly involved phase sensitive demodulator would be required. To eliminate the demodulator, the iron core and springs were adjusted so that core was offset to one side by an amount equal to the spring deflection corresponding to 4 g acceleration.

10. What is the principle of capacitive transducer?

Many industrial variables like displacement, pressure, level, moisture, thickness etc., can be transduced into an electrical variation using capacitance variation as the primary sensing principle.

11. What are the desirable features of capacitive transducer?

The desirable features of capacitive transducer are,

- Its force requirements are very small.

- As the moving plates have very little mass, design of transducer with fast response characteristics is possible.
- There is no physical between moving and stationary parts.
- Does not depend on the conductivity of the metal electrode.
- Shielded against the effect of external electric stray fields.

12. What are the different practical capacitance pickups?

The different capacitance pickups are,

- Equibar differential pressure transducer.
- Feedback type capacitance proximity pickup.
- Condenser microphone.

13. What is Microphone?

Microphone is also a transducer which converts sound energy into electrical energy. Example is condenser microphone.

14. What is the principle of change of capacitance?

The capacitance can be changed by the,

- Change in overlapping area A,
- Change in the distance between the plates, d
- Change in dielectric constant.

15. What are the advantages of capacitive transducers?

The advantages of capacitive transducer are,

(a) They require only small force to operate.

(b) Have a good frequency response.

(c) Extremely sensitive.

(d) High input impedance.

16. What are the disadvantages of capacitive transducers?

The disadvantages of capacitive transducer are,

(a) The metallic parts of the capacitive transducers must be insulated from each other.

(b) Non-linear behaviour.

(c) This leads loading effects.

(d) The cable may be source of loading resulting loss of sensitivity.

17. What are the uses of capacitive transducer?

The uses of capacitive transducer are,

(a) Can be used for measurement of linear and angular displacement.

(b) Can be used for measurement of force and pressure.

(c) It can be used as pressure transducer.

(d) Measurement of humidity in gases.

(e) Commonly used for measurement of level, density, weight.

18. What is the value of capacitance for measurement of level of a non-conducting liquid?

$C = 2\pi\varepsilon_0 \, [\varepsilon_1 h_1 + \varepsilon_2 h_2 / \log_e (r_2/r_1)]$

where,

h_1 – Height of liquid

h_2 – Height of cylinder

ε_1 – Relative permittivity of liquid

ε_2 – Relative permittivity of vapour above liquid

r_2 – Inside radius of outer cylinder

r_1 – Outer radius of inner cylinder

ε_0 – Relative permittivity of free space

19. What is analog transducer?

Analog transducer converts input quantity into an analog output which is a continuous function of time. Thus a strain gauge, LVDT, thermocouple, thermistors may be called as analog transducer.

20. What is digital transducer?

Digital transducer converts input quantity into an electrical output which is in the form of pulses.

UNIT V

Other Transducers

5.1 PIEZOELECTRIC TRANSDUCER

- Certain materials can generate an electrical potential when subjected to mechanical strain, or conversely, can change dimensions when subjected to voltage. This is known as the piezoelectric effect [see fig. 5.1 (a) & (b)].

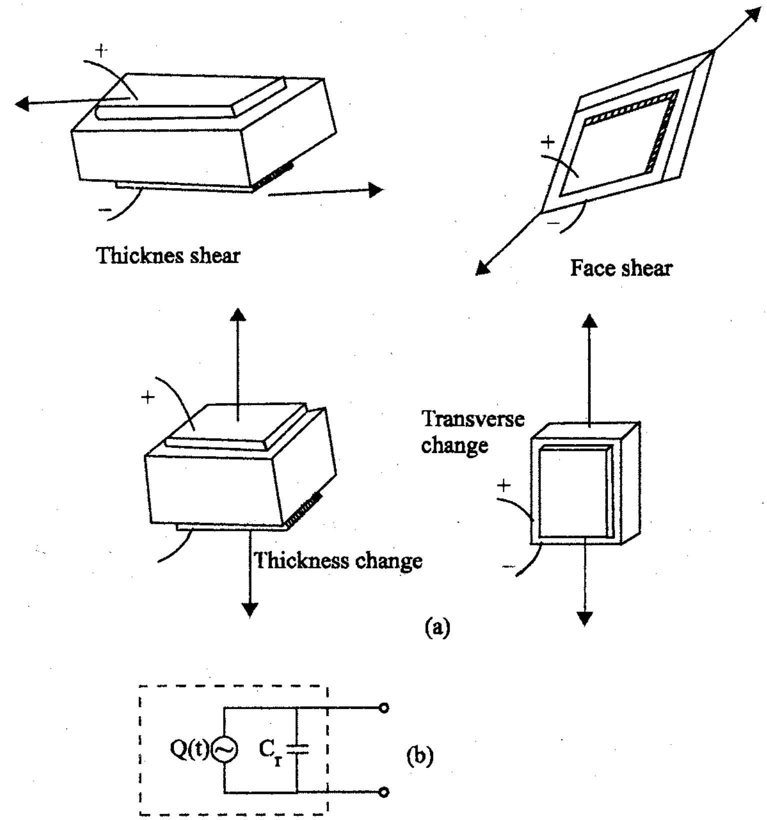


Fig. 5.1 (a) Basic deformation - modes for piezoelectric plates (b) Equivalent circuit for a piezoelectric element.

- Piezoelectric transducers are converters of mechanical energy into electrical energy and are based on the direct piezoelectric effect observed in certain nonmetallic and insulating dielectric compounds.
- Electrical change is developed on the surface of the crystals, when they are under mechanical strain due to application of stress.
- Due to their high mechanical rigidity they are treated as near-ideal transducers of measurement of force and thereby pressure, acceleration, torque strain and amplitudes of vibration.
- They are popular due to their small size, high natural frequency, linearity, high sensitivity, wide measuring range and polarity sensitivity.
- The commonly used materials are stable enough for all applications at temperatures up to 200°C.
- The small capacitance of the transducer and its high insulation resistance cause some problems for measurement of charge developed, and the consequent voltage across the faces.
- The charge leaks away through its insulation resistance, and hence special amplifiers such as charge amplifiers are used to measure the charge.
- The transducer is unsuitable for measurement of steady quantities due to the leakage of charge.
- The "anisotropic effect" noticed in p-n junctions of semiconductor diodes and transistors is allied to the piezoelectric phenomenon.
- The application of localized stress on the upper surface of a semiconductor junction results in a change of current across the junction.
- Such devices are known as piezoelectric transistors and are used for measurement of small pressure and force.
- Conversion of electrical energy into mechanical energy is possible by using the same device.

- The application of electric potential between the surfaces of a crystal results in a change of its physical dimensions.
- This is the reverse effect and is also known as electrostriction.
- The effect is widely applied for generation of ultrasonic waves.

5.1.1 Piezoelectric phenomenon

- Pierre and Jacques Curie are credited with the discovery of piezoelectric effect in 1880.
- Notable among these materials are quartz, Rochelle salt (Potassium sodium tartarate), properly polarized barium titanate, ammonium dihydrogen phosphate, and even ordinary sugar.
- Of all the materials that exhibit the effect, none possesses all the desirable properties, such as stability, high output, insensitivity to temperature extremes and humidity, and the ability to be formed into any desired shape.
- Rochelle salt provides the highest output, but requires protection from moisture in the air and cannot be used above about 45°C (115°F).
- Quartz is the most stable, yet its output is low.
- Because of its stability, quartz is quite commonly used for stabilizing electronic oscillators.
- Quartz is silicon dioxide (SiO_2) and is available as a natural substance.
- The atoms are arranged in the crystal as shown in fig. (5.2), forming a hexagon in the plane of paper while the optical axis (z-axis) is perpendicular to the xy-plane.
- For the three Si atoms, the six oxygen atoms are lumped in pairs, thereby forming a hexagonal crystal.
- The x and y axes are referred to as electrical and mechanical axis respectively.
- Under stress-free conditions, all charges are balanced, but when a force is applied along the x-axis, the balance is disturbed and electrical charge is developed on the two faces A and B as shown in fig. 5.2(b). This is known as "longitudinal effect".

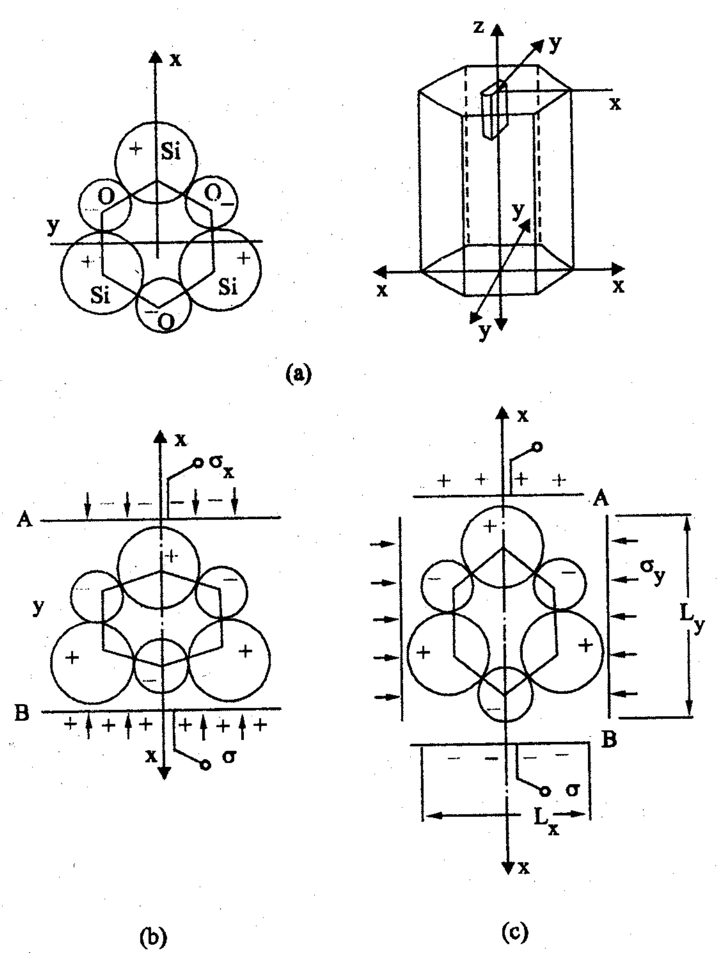


Fig. 5.2 (a) Arrangement of atoms of a piezoelectric crystal and the crystal axes (b) Crystal under longitudinal effect (c) Crystal under transverse effect

- A force along the y-axis also distorts the arrangement of atoms, and charges are developed on the two faces A and B, as shown in fig. (5.2(c)) and is referred to as "transverse effect".
- Due to the symmetry along the optical axis, no effects are noticed when force is applied along the z-axis.
- The characteristic features of the longitudinal effect are that the charge generated is independent of area of the crystal and its thickness in the x-direction.

- The charge developed on a given area of the crystal face is proportional to the area affected by the pressure and thus proportional to the total force applied normal to the surface.
- However, when a force is applied in the transverse (y) direction, the charge generated on A and B depends on the lengths (L_x, L_y) of the faces in the x and y directions.
- Application of shear stress τ about any of the three axes may also yield charge on the faces perpendicular to the x-axis.
- The charge sensitivity or the piezoelectric d-coefficient is the charge developed per unit force.
- The net piezoelectric effect is represented by the vector of electric polarization $\overline{P}$ as

$$\overline{P} = P_{xx} + P_{yy} + P_{zz} \quad \text{... (5.1)}$$

where, x, y and z refer to the conventional orthogonal system related to the crystal axes P_{xx} indicates the net effect on the face perpendicular to the x-axis due to application of axial stresses σ and shear stresses τ to the crystal.

5.1.2 Piezoelectric materials

- The materials exhibiting the piezoelectric phenomenon are divided into two groups: (i) Natural (ii) Synthetic
- The natural group consists of quartz, Rochelle salt and tourmaline.
- The synthetic group consists of ammonium dihydrogen phosphate (ADP), lithium sulphate (LS), and Dipotassium tartarate (DKT).
- Depending on the crystal structure, discs or wafers are cut and used for measurement of force in one or the other of the modes described.
- Quartz is the most stable material and artificially grown quartz is normally preferred as it is purer than the natural quartz.
- Tourmaline is the only material exhibiting a large sensitivity.
- Rochelle salt is the material that is being produced on industrial scale for producing gramophone pick-ups and crystal microphones.
- It has the highest relative permittivity among the natural group.

- ADP crystals possess the lowest resistivity which is also temperature dependent. With temperature compensation they are used in acceleration and pressure transducers.
- Lithium sulphate is highly sensitive.

5.1.3 Ferroelectric Materials

- They are certain polycrystalline ceramic compounds which exhibit the property of retaining electric polarization when exposed to intense electric fields.
- These materials are known as ferroelectric materials (equivalent to ferromagnetic materials), and after polarization, their behavior is similar to the piezoelectric materials.
- Three such common substances which are popularly used for piezoelectric transducers are Barium titanate ($BaTiO_3$), lead zirconate-titanate, and lead metaniobate.

5.1.4 Piezoelectric semiconductors

- A localized stress on the upper surface of the $p-n$ junction of a semiconductor diode caused a very large reversible current change in the current across the junction.

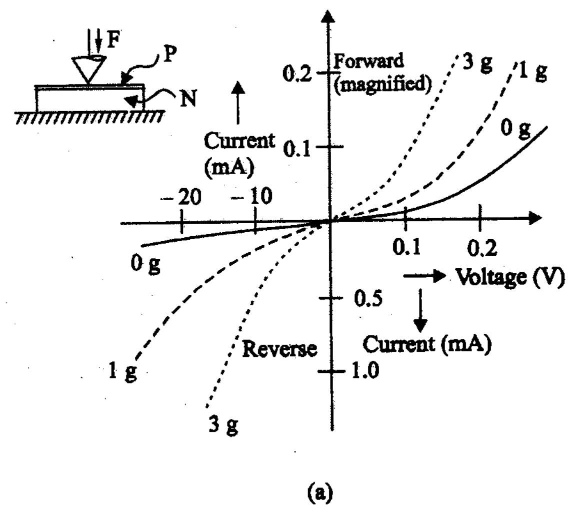


Fig. 5.3 Piezoelectric semiconductor diode and its characteristics

- This phenomenon is due to the anisotropic stress effect in $p-n$ junctions, and devices utilizing this effect are known as piezoelectric diodes and transistors.
- The variation of current across the junction of a Germanium diode for forward and reverse voltages is shown in fig. (5.3).
- It is observed that considerable change in the magnitude of the current results from application of a few grams of localized force.
- Moreover, the change is reversible.
- The behavior of a silicon $n-p-n$ planar transistor is shown in fig.(5.4).
- The force is applied to the surface by means of a pointed stylus.
- The current gain of the transistor decreases with increase of force, and the capacitance between base and collector changes in a similar fashion.

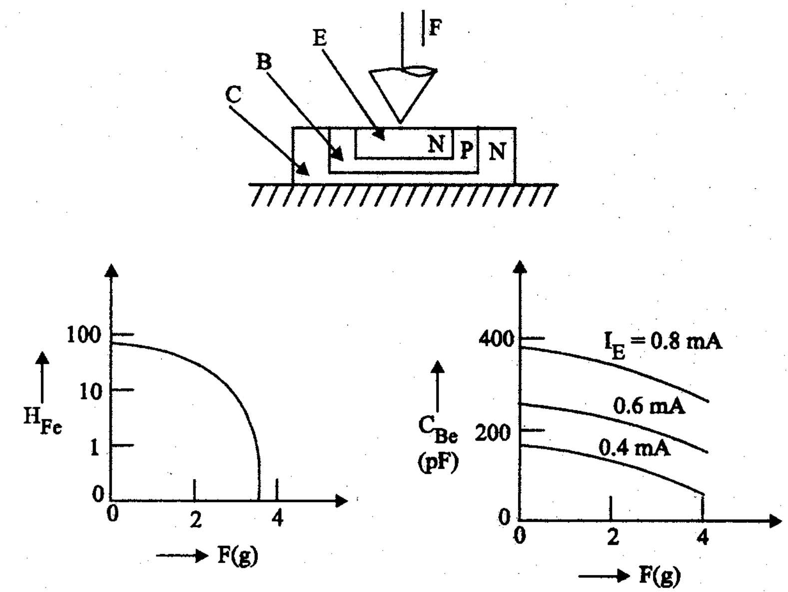


Fig. 5.4 Piezoelectric semiconductor transistor and its characteristics.

5.1.5 Piezoelectric Force Transducer

- Piezoelectric crystal or element primarily responds to force-[illegible] possesses all the desired characteristics of an ideal force tr[illegible]

- The element can be directly stressed by application of force at one point of the surface.
- Multiple forces can also be applied at more than one point of the surface, and summed by using one single crystal.
- To increase the charge sensitivity, more than one element can be used to form a transducer system and such combinations are known as bimorphs or multimorphs (or piezopile), depending on whether they are of two elements or more.
- The series and parallel connected bimorphs are shown in fig. (5.5).
- A multimorph of four elements, which develops four times the charge of a single element, is shown in fig. (5.6).
- The four elements are mechanically in series but electrically in parallel and hence the net capacitance of the transducer increases correspondingly.
- When bimorphs are made up of ceramic elements, the direction of polarization of the two elements should be noted, and then connected so as to develop charges and voltages under stress as shown in fig. (5.6(a)). These are called as Bender-type bimorphs.
- A twister bimorph is shown in fig. (5.6(b)), with the force applied at A, while the remaining three corners B, C and D are held rigidly.

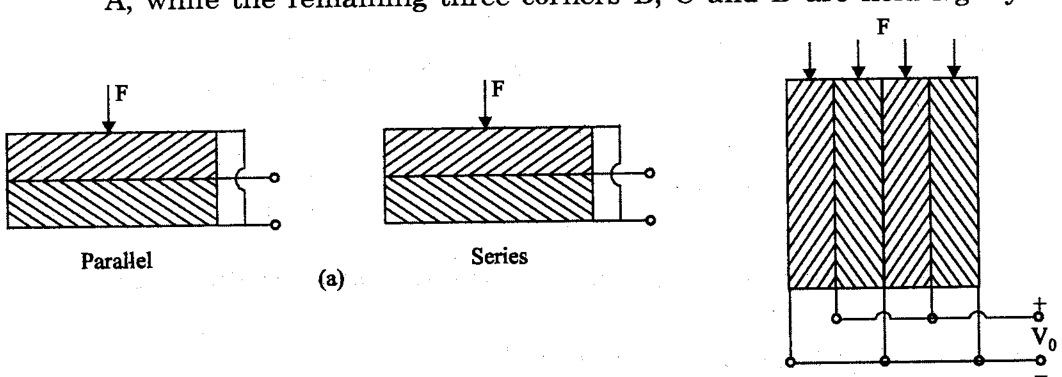


Fig. 5.5 (a) Parallel and Series connected bimorphs
(b) Multimorph of four piezoelectric elements.

- If the four corners can be subjected to concentrated forces as shown in the four-point twister of fig. (5.6 (b)), the expanding diagonals will be perpendicular to each other and on opposite sides of the bimorph.

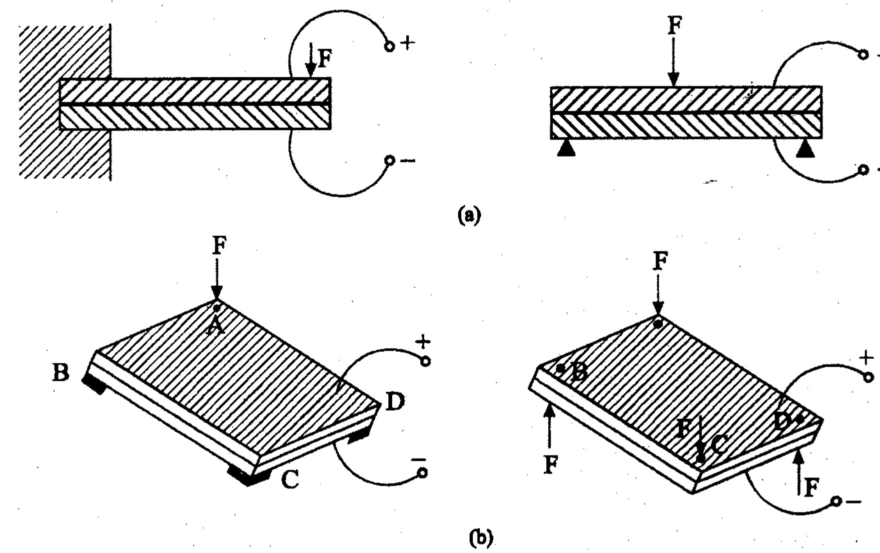


Fig. 5.6 (a) Bender type bimorphs (b) Twister type bimorphs

5.1.6 Piezoelectric Strain Transducer

- Any piezoelectric element cemented to the surface of the structure is under stress, the strain in the structure is transmitted to the element.
- A voltage proportional to strain is directly available from the transducer.
- The output is obtained by using the *h*-coefficient given by

$$V_0 = h\,et \qquad \text{... (5.2)}$$

where $e \rightarrow$ strain

$t \rightarrow$ thickness of the element, *m*

- The sensitivity of the transducer is very high.
- Piezo-resistive strain transducers, though known to be suited for transient strain measurements, are not as sensitive as the piezoelectric type.

- If accuracy and stability are of primary interest, metallic alloy resistive strain gauges are chosen, especially when static strain is monitored over a long period of time.

5.1.7 Piezoelectric Torque Transducer

- A cantilever type bender bimorph can be used as a twister bimorph for the measurement of torque as shown in fig.(5.7).
- The twisting moment may be due to a small force transmitted through a lever or may be obtained directly by connecting it to a driving shafts/spindle as obtained in instrument mechanisms.
- The sensitivity is high and is therefore very much useful for measurement of small driving torques under dynamic conditions.

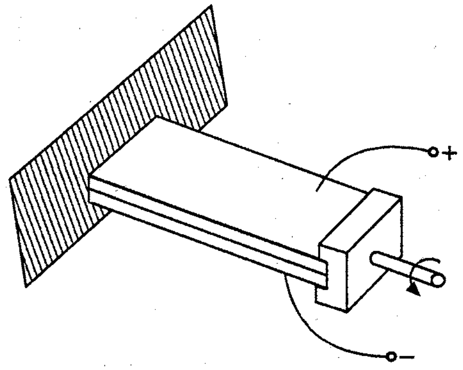


Fig. 5.7 A cantilever type twister bimorph

5.1.8. Piezoelectric Pressure Transducers

- Piezoelectric transducers are more suitable for pressure measurements under dynamic conditions only and are often used as microphones, hydrophones, and engine pressure indicators.
- In the piezoelectric microphone, the diaphragm and the bimorph are connected together by means of a fine needle (spindle) as shown in fig. (5.8).
- The natural frequency of the diaphragm, the bimorph, and the associated system should be made higher than the highest frequency to be responded to (10 KHz normally).
- When used in sound level meters, it is essential for microphone to have flat frequency response upto 10 KHz.

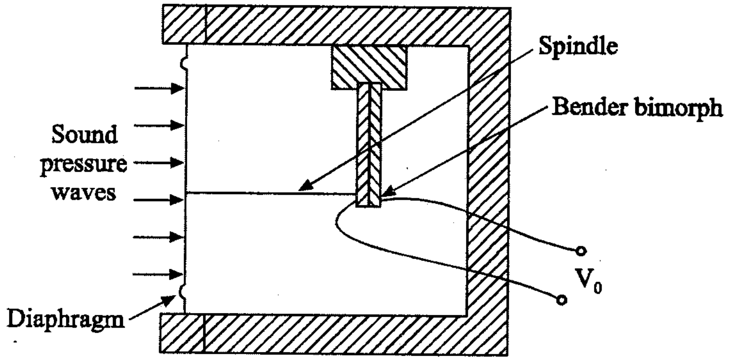


Fig. 5.8 Piezoelectric microphone

- Large pressure variations occurring at frequencies upto 20 KHz in internal combustion engines are measured using multimorphs (piezopile) of quartz elements.
- The surfaces of the elements, connecting electrode surfaces in between and the diaphragm or load plate at the extremes, should be optically flat, and no air should be trapped in between as it would reduce the natural frequency of the system.
- The transducer is prestressed so as to enable pressure fluctuations about a mean value to be measured.
- The prestressing is produced by a thin-walled tube under tension, as shown in fig. 5.9 (a).

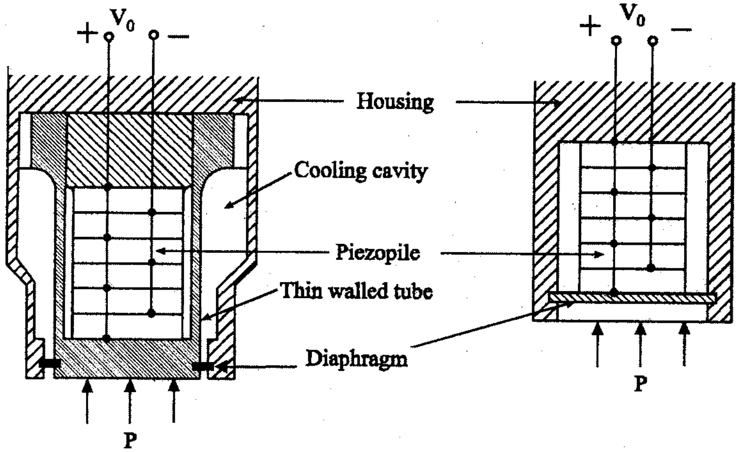


Fig. 5.9 Piezoelectric pressure transducers prestressed by (a) a thin-walled tube (b) a thick diaphragm.

A very thin diaphragm of flexible material is used for sealing.

- The preload may also be developed by a stiff diaphragm as shown in fig. 5.9 (b).
- The net force F_1 to which the piezopile responds is given by

$$\frac{F_1}{F} = \frac{K_1}{K_1 + K_2}$$

where

$F \rightarrow$ the total force acting on the transducer

$K_1 \rightarrow$ spring-rate of piezopile

$K_2 \rightarrow$ spring-rate of the preloading tube or diaphragm.

- For the measurement of air-blast pressures and underwater pressure transients.
- A small hollow cylinder shown in fig. 5.10 is used is most cases.

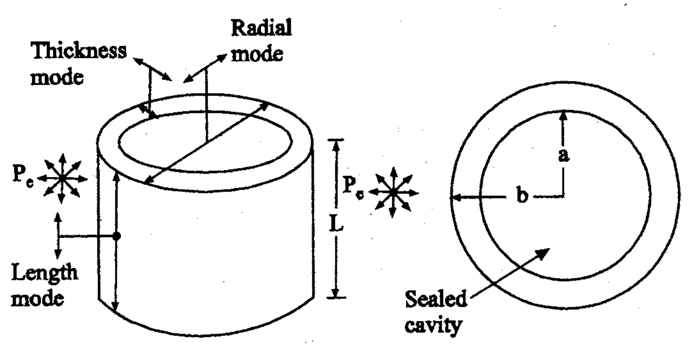


Fig. 5.10 Pressure transducer for under water pressure measurement

- The outer and inner surfaces are metallized and used as electrodes.
- The walls are polarized in a radial direction.
- The tube cavity may be sealed against the external pressure and the blast pressure is applied to the outer surfaces.
- The cylinder responds to the pressure P_e in all the three modes as shown in fig. 5.10.

- For a thin-walled hollow tube, the open circuit voltage generated by the radial stress and tangential stress is given by

$$V_0 = P_e \left(g_{33}\, b \frac{a-b}{a+b} - g_{31}\, b \right) \qquad \text{... (5.3)}$$

where

$g_{33}, g_{31} \rightarrow$ the g-coefficients of the material

$b \rightarrow$ outer radius

$a \rightarrow$ inner radius

5.1.9 Piezoelectric Acceleration Transducer

- The acceleration transducer design is like that of a force transducer except that a proof mass is added to the acceleration transducer for developing force under acceleration inputs.
- The single crystal or the piezopile is prestressed by screwing down the cap on the hemispherical spring shown in fig. 5.11 (a).
- The input-output characteristics of piezoelectric acceleration transducer is shown in fig. 5.11 (b).

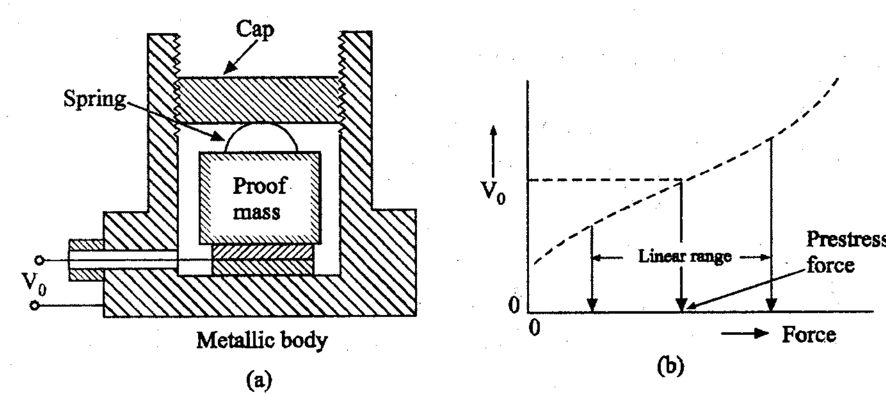


Fig. 5.11 (a) Piezoelectric acceleration transducer (b) Its input-output characteristics

5.2 MAGNETOSTRICTIVE TRANSDUCERS

- Magnetostrictive transducers are similar to piezoelectric **transducers** and are based on the application of the magnetostriction **phenomenon**.

- They are converters of mechanical energy into magnetic energy and are also known as magnetoelastic transducers.
- The phenomenon is reversible and the devices developed convert energy from one form to another.
- The natural frequency of the transducers can be as high as 10 KHz and are very much used as transmitters (senders) and receivers in vibration and acoustic studies.
- The transducers possess very high mechanical input impedance and are suitable for measurement of force and hence acceleration and pressure.
- They can measure large forces, both static and dynamic.
- They are rugged in constructional features and, when used as active transducers, the output impedance is low.
- Nickel and nickel-alloys are mostly used.
- It is the basic non-linearity in the B-H characteristic which is responsible for its limited scope of application, especially when high accuracy is desired.

5.2.1 Magnetostriction Phenomenon

- Certain ferromagnetic materials are considerably affected in their magnetic properties when they are mechanically stressed. This phenomenon is known as "magnetostriction" (Villari effect) and is particularly significant in nickel and nickel-iron alloys.
- The shape and size of the $B-H$ characteristic and the $B-H$ loop is sufficiently altered when the material is subjected to tensile, compressional or shear stress.
- The $B-H$ characteristics of nickel and nickel-iron (Ni, 68%) alloy are presented in fig. (5.12) showing the effect of increasing tensile stress σ on the materials.
- Similarly, the magnetization characteristic is affected and it is observed that the permeability increases with increase in tensile stress in the case of nickel-iron alloys and decreases in the case of pure nickel.

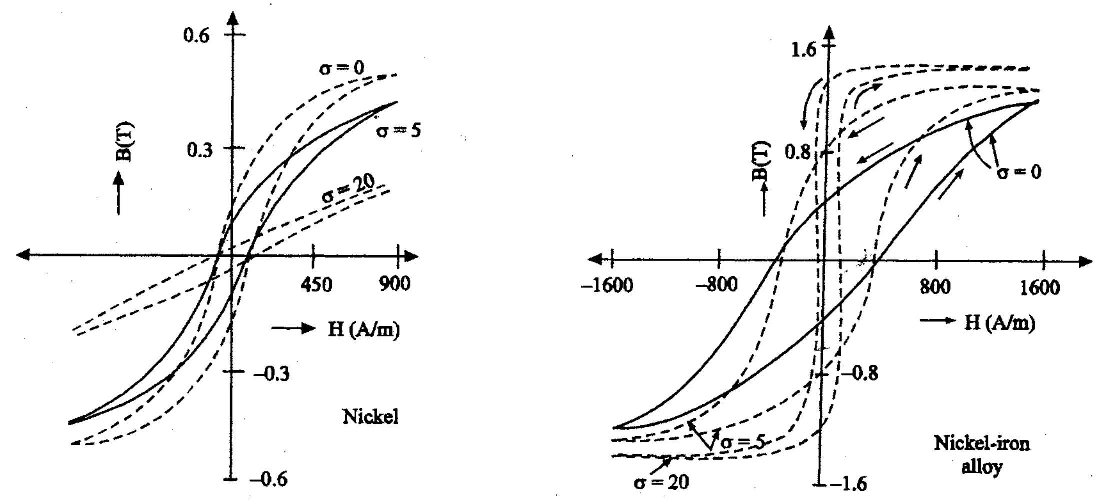


Fig. 5.12 B-H characteristics under different stress values
(a) For nickel (b) For nickel-iron alloy

- The change in the shape of the $B-H$ loop alters the remnance B_r of the material.
- When B_r and permeability decrease with increase in stress, it is known as "negative magnetostriction".

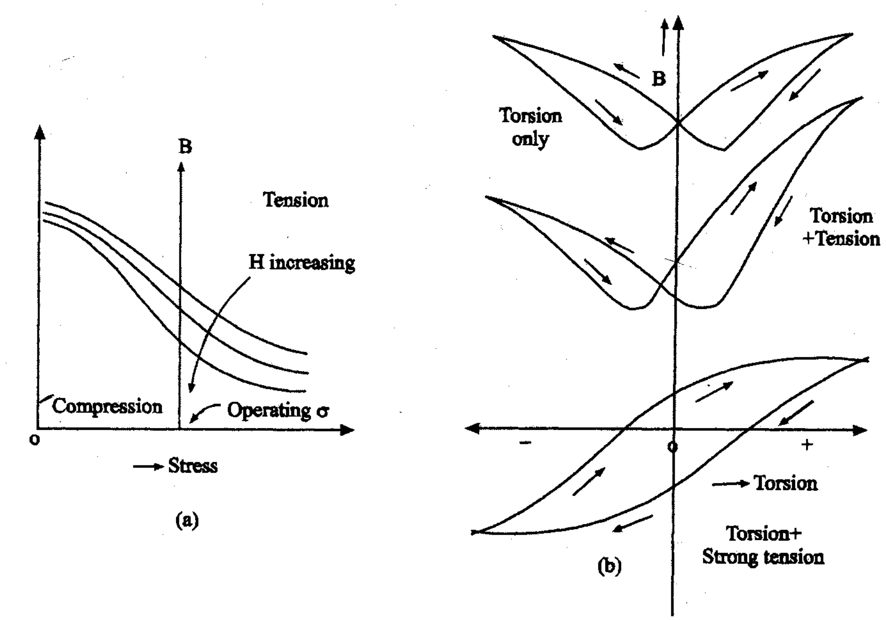


Fig. 5.13 Characteristics of a nickel sample
(a) For H variation (b) For superposed cyclic torsion.

- The percentage of nickel in the nickel-iron alloy has considerable influence on the characteristics.
- The materials are sensitive to the polarity of stress and hence the transducers enable measurement of alternating forces.
- Some ferrite materials such as 'Ferroxcube B' exhibit magnetostriction of considerable degree but due of their brittleness, they are not used.

Fig. 5.13(a) shows the variation of B with stress at different values of H, and fig. 5.13 (b) shows the effect of superposition of cyclic torsion on tensile stress for the case of a nickel sample.

5.2.2 Magnetostrictive Force Transducer

- The self-inductance of an iron-cored coil change if the core characteristic is changed due to application of force.
- It is the mechanical strain that affects the orientation of the magnetic domains, and hence the change in the value of effective permeability.
- The magnetic path should be continuous with no air gap present.
- The core may be laminated.
- The laminations are stacked to form the core, and a coil is provided to enable measurement of its self-inductance.
- The coil current is so adjusted as to make the self-inductance maximum and make it most sensitive to stress.

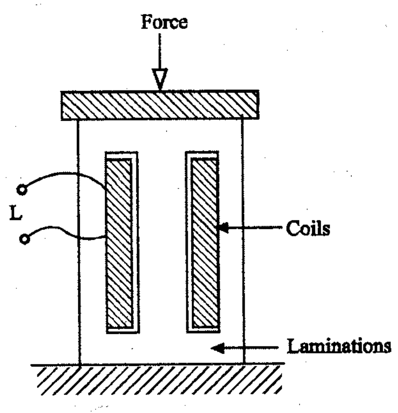


Fig. 5.14 Magnetostrictive force transducer

- One of the simple configurations commonly employed is shown in fig. (5.14).
- The arrangement in fig. (5.14) allows the measurement of large static forces and 10-20 percent change in self-inductance is observed with nickel and nickel-iron alloy transducers.
- Application of stress σ results in a change of B bv $+\Delta B$, depending on the material.
- The sensitivity of the transducer is defined as the ratio of ΔB to σ and is given by

$$S = \left.\frac{\Delta B}{\sigma}\right|_{B=B_0}$$

where B_0 = operating point of flux density

- For small sinusoidally varying σ, corresponding variations of ΔB are assumed to be sinusoidal.
- If a coil is provided on the core, the induced emf will be proportional to σ and sinusoidal.
- The sensitivity is observed to be maximum in the case of nickel-iron (Ni 68%) alloy when B_0 is adjusted to $1/\sqrt{3}$ of saturation flux density. It is approximately equal to 3×10^{-8} T/N.
- The operating flux density B_0 may be chosen as the remnant flux density B_r for reasons of simplicity and stability.
- The sensitivity may be lower but it is preferred since bias winding is not needed.
- The fall in sensitivity can be made up by providing more turns in the pick-up coil, utilizing the window space of the bias winding.
- The emf induced in the winding is given by

$$e(t) = SAN\frac{d\sigma(t)}{dt} \quad \text{... (5.4)}$$

where $A \rightarrow$ area of coil

$N \rightarrow$ number of turns

- Transient forces and stresses can be measured by integrating $e(t)$ before it is displayed on the oscillograph.

5.2.3 Magnetostrictive Acceleration Transducer

- To extend the application of the transducer for measurement of acceleration, addition of proof mass is required.
- The mass of the core itself serves as proof mass to some extent and additional mass is provided by a brass cylinder of at least an equal mass, as shown in fig. 5.15.

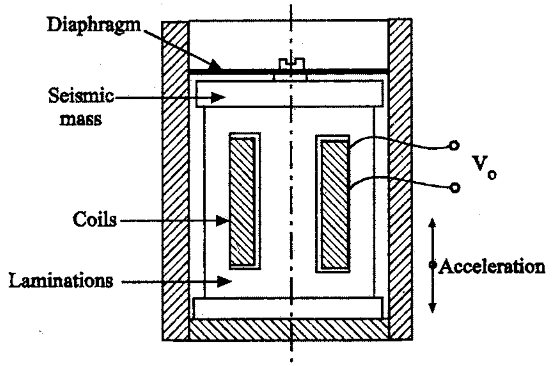


Fig. 5.15 Magnetostrictive acceleration transducer

- To prevent the transducer from responding to transverse accelerations, the brass cylinder is guided by a flexible diaphragm.
- The induced emf of the coil is integrated in such a way as to extend the bandwidth of the system towards the lower frequencies.
- As compared to piezoelectric accelerometers, these transducers are of larger size and mass and are lower in accuracy.
- While measuring acceleration, the variation in the earth's magnetic field affects the sensitivity.
- Laminations and coil should be rightly held in position so as not to be affected under high accelerations.

5.2.4 Magnetostrictive Torsion Transducer

- Magnetostrictive torsion transducer consists of a nickel wire of 0.5 – 1 mm diameter kept stretched between the poles of a permanent magnet and having a small stylus rigidly attached to it at the midpoint.
- The wire is prestressed by twisting it, before being installed into the position.
- Two pick-up coils of fine wire are wound round the wire on either side of the mid-point, as shown in fig. 5.16.
- Any displacement of stylus to one side or the other increases the torsion on one side and decreases it by an equal amount on the other side.
- This results in an increase of magnetic flux in one-half and a decrease in the other half.

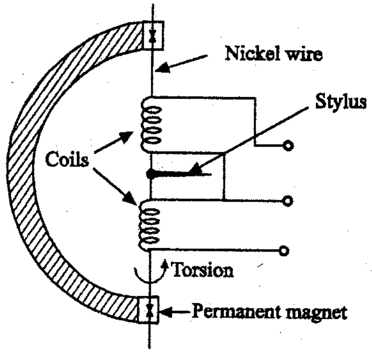


Fig. 5.16 Magnetostrictive torsion transducer

- The corresponding induced emfs are in phase opposition and are processed by suitable networks as in the case of linear variable differential transformer.
- It is used as phonograph pick-up and is designed to have flat frequency response over 150 Hz - 15 KHz frequency range.
- Due to the nonlinearity and hysteresis in the performance, **it is** normally limited for use when time-varying torsions of small **amplitude** are to be measured.

5.2.5 Hall-Effect Transducers

- The Hall-effect is one of the galvanomagnetic phenomena in which the interaction between the magnetic field and moving electrical charges results in the development of forces that alter the motion of the charge.
- The Hall effect is observed in all metals, but is very much prominent in semiconductor materials.

A thin strip of bismuth or n-type germanium is subjected to magnetic field B normal to its surface as shown in fig. 5.17, while it carries a current I along the length of the strip, but normal to B.

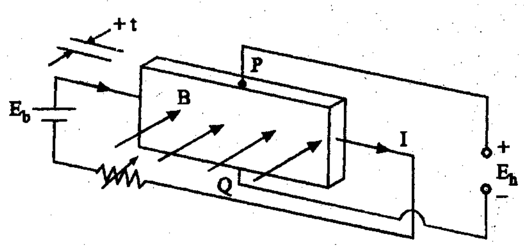


Fig. 5.17 Hall effect transducer

- The magnetic field exerts a force (known as Lorentz force) on the electrons moving at a velocity v, with the result that some of them drift towards the edges of the strip.
- The edge surfaces act like charged electrodes and the potential difference measured between P and Q is known as Hall potential E_H which increases with increase of B and I.
- The build-up of the charge on the edge surfaces will, in turn, develop an electric field (Hall field) of such a polarity that counteracts the collection of charges on the surfaces.
- The force on the electrons due to Hall field and the Lorentz force balance each other finally.
- The time required to reach this equilibrium is about 10^{-14} S.
- If e is the charge of electron, then the Lorentz force Bev and the force due to Hall field are equal to each other. Hence,

$$Bev = e\,E_H/\,b$$

$$E_H = Bbv \text{ (volts)} \qquad \text{... (5.5)}$$

where $B \rightarrow$ the magnetic flux density, T

$v \rightarrow$ velocity, m/s

$b \rightarrow$ width, m

- The electrons and the free charge carriers assume a velocity along the length of the strip, which is proportional to electric field along the direction of motion.
- It the mobility of the charge carriers is represented by χ, then v is given by

$$v = \chi\, E_b / L \qquad \text{... (5.6)}$$

and using $E_b = I\,\rho L/bt$, v is given by $\chi\, I\rho/bt$

- Hence, $E_H = \rho\chi\, BI/t = K_H\, BI/t$ ... (5.7)

where $K_H \rightarrow$ Hall coefficient (or) Hall constant ($= \chi\rho$)

$t \rightarrow$ thickness of the strip, m

$L \rightarrow$ length of the strip, m

5.2.6 Applications of Hall Transducers

- In the field of instrumentation, the Hall element is highly valued for its speed of response in detection changes in the magnetic field to which it is exposed.
- The advantages are its small size and high sensitivity.
- It is used as a proximity detector as it does not require to establish a mechanical link with the test object.
- It is used to measure the change in the strength or direction of the magnetic field due to the displacement or nearness of the test object.

5.2.6.1 Angular displacement transducer and proximity detector

- Fig. (5.18) shows the Hall effect angular displacement transducer and Hall effect proximity transducer.

- As the element can respond to quick changes in the field, it is equally applicable for measurement of amplitudes of vibration of objects and count the number of fast moving objects across the magnetic field.

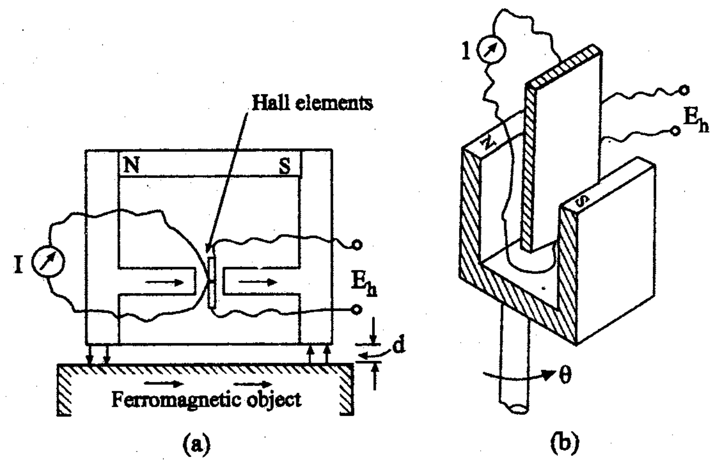


Fig. 5.18 (a) Hall effect displacement transducer (b) Hall effect proximity transducer

- In all the above applications, the current through the element should be held constant at about 5 - 20 mA dc using constant current sources.
- The value of E_H for the case of an *n*-type germanium element, carrying a current of 10 mA is 1.4 mV when exposed to a magnetic field of 0.1 mT.
- The output impedance varies from one element to another and is about 5 - 200 ohms, depending on the material and size of the element.

5.3 IC SENSOR

Although conventional sensors are commercially still very much in use, over the last three decades, the use of solid state sensors also have been increased. In this category, the semiconductor micro and nano-sensors, ceramic and chemical sensors using new materials and technologies such as IC technology, VLSI chips, and micromachining techniques are included.

For semiconductor micro-sensors, the IC technology comprising of photolithographic etching, deposition, metallization, and assembling is essential and this is the basis for thick and thin film, chemical and electrochemical, and biological sensors. IC elements are now extensively used in the measurement of temperature, flow and magnetic field.

5.3.1 Film Sensors

- Basically, such sensors are produced by film deposition of different thickness on appropriate substrates.
- The deposition techniques used are different for the thick and thin film sensors.
- Sensors produced through these techniques have varying electrical and mechanical properties while a variable is being sensed.

5.3.3 Thick Film Sensors

- Thick film process had been in use for producing capacitor, resistor and conductors-and for sensor development.
- The processing of a sensor can be expressed schematically as

Step 1 : Selection and preparation of a substrate.

Step 2 : Preparation of the initial coating material in paste or paint form.

Step 3 : Pasting or painting the substrate by the coating material or screen printing it.

Step 4 : Firing the sample produced in step 3 in an oxidising atmosphere at a programmed temperature format.

- The substrates used for developing thick film over them are alumina (96% or 99.5%) and beryllia (99.5%).
- These are fired at about 625°C.
- Others used are enamelled steel which is low carbon steel coated with low alkali content glass first that are fired at around 850°C.
- Alumina or beryllia have dielectric constants around 9.5 and 7 respectively with dielectric strength around 5600 V/μm.
- Sensors which are produced through thick film deposition (~ 20μ m) are used for sensing temperature, pressure, gas concentration, and humidity.
- Temperature: Thick film sensors such as (i) thermopiles (usually of gold and gold-platinum alloy), (ii) Thermistors (usually with oxides of

manganese, ruthenium, and cobalt), and (iii) temperature dependent resistances based on gold, platinum and nickel are used for temperature sensing.

- Pressure: Sensing pressure is possible by making thick film diaphragms or capacitive devices made with alumina (Al_2O_3) and $Bi_2Ru_2O_7$, or piezoresistive devices made of same materials.
- Concentration of gases: Gases such as methane (CH_4), CO and C_2H_5OH can be checked for concentration using films of SnO_2 + Pd, SnO_2/ThO_2 + hydrophobic SiO_2. H_2, CO, C_2H_5OH, and isobutane are sensed by SnO_2 + Pd, Pt, Ba^-, Sr^- and $CaTiO_3$ (Nasicon). Oxygen and hydrogen gases also are separately sensed by these types of films.
- Humidity:

It is sensed by

(i) resistive films made from RuO_2 (spinel type) / glass and

(ii) Capacitive films made from glass ceramic / Al_2O_3. On the other hand, dew point is sensed by films made from ($BaTiO_3/RuO_2$)-glass.

- Starting from the same basic material, $SnSO_4$, one can produce SnO_2 - based sensors for H_2, CO and NH_3.
- The other thick film variety is the ceramic metal or 'cemet' which consists of gold/silver/ruthenium/palladium based complex oxides in an insulating medium, mainly glass (lead borosilicate).

5.3.3 Thin Film Sensors

- This film sensor processing differs from thick film technology mainly in the film deposition techniques.
- This technology is similar to that used in silicon micromechanics.

A number of techniques are used for thin film deposition such as:

(a) Thermal evaporation
  (i) Resistive heating
  (ii) Electron beam heating

(b) Sputter deposition
  (i) DC with magnetron
  (ii) RF with magnetron

(c) Chemical vapor deposition (CVD)

(d) Plasma enhanced chemical vapor deposition (PECVD)

(e) Metallo-organic deposition (MOD)

(f) Langmuir - Blodgett technique of monolayer deposition.

5.3.3.1 *Plasma enhanced chemical vapor deposition*

- Plasma enhanced chemical vapor deposition (PECVD) has been found to be particularly suitable for sensor fabrication.
- This is a low temperature process in which plasma is introduced into the deposition chamber to enhance the pyrolytic process which in normal CVD process is performed by thermal decomposition that requires high temperature.
- In this process, the volatile compound of the material to be deposited is thus vaporized, decomposed, and made to react with gaseous species over the substrate to produce a nonvolatile amorphous product on the surface of the substrate.
- The deposition level is controlled by controlling the flow rates of the vapors.
- A parallel plate, radial flow type PECVD processing chamber is shown in fig. 5.19.

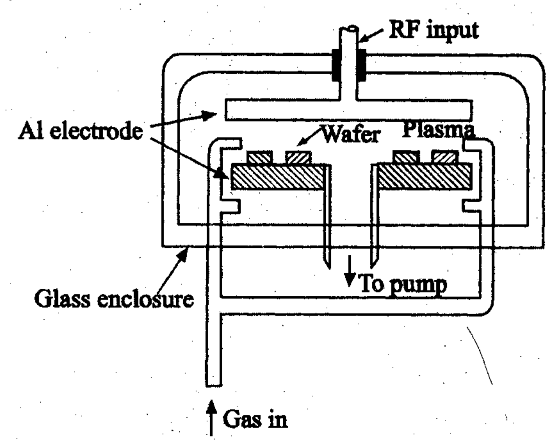


Fig. 5.19 A PECVD processing system

5.3.3.2 *Metallo-organic deposition (MOD)*

- This is another versatile technique which can be used both for thick and thin film sensor fabrication.
- It consists of applying ink of metallo-organic compound to the silicon substrate consisting of silicon wafer coated with silica, then spinning the assembly at about 3000 rpm and finally heat treating the deposit.
- Metallo-organic compounds consists of a central metal ion bonded with a ligand through a heterobridge containing oxygen, sulphur, nitrogen, phosphorus, arsenic, ad so on.
- It is prepared by dissolving the compound in organic solvent.
- Specially prepared thin films, by this technique are barium titanates ($BaTiO_3$) and their derivatives that are mostly used in pyroelectric measurement, tin-oxides for gas sensors, superconducting oxides such as Yttrium-barium-copper oxides ($YBa_x\,Cu_y\,O_z$) for high temperature and ZnO_2, TiO_2 stabilized by Yttrium for oxygen sensors.

This film sensors measure the same variables as done by thick film counterparts with variations in principles and materials. Table (5.1) shows the variable, sensing element, and principle of sensing for certain different variables.

Table 5.1 Working principles of the materials

Variable	Material	Principle
Flow	Au	Thermoanemometry
Humidity	Ta_2O_5	Capacitance change
Magnetic field	$Ni_{81}Fe_{19}$, NiCo, $CO_{72}Fe_8B_{20}$	Magnetoresistive effect
Oxygen	ZnO	Variation in electrical conductivity
Pressure	Polysilicon	Piezoresistive effect (Diaphragm)
Radiation	Au	Bolometry
Strain	CrNi	Piezoresistive effect
Temperature	Pt	Resistance variation

5.3.4 Standard Methods of Semiconductor IC Technology

- The solid state sensors (semiconductor micro-and nano-sensors, ceramic and chemical sensors) are developed through standard IC technology as used in VLSI design and micromachining techniques.
- The necessary steps in the processing of sensors in semiconductor sensor fabrication using IC technology are shown in fig.(5.20)
- Starting with a polished Si, Ge, or, GaAs wafer on which film is deposited by

  (a) Epitaxial growth, or

  (b) Oxidation, or

  (c) Polysilicon and dielectric deposition, or

  (d) Metallization

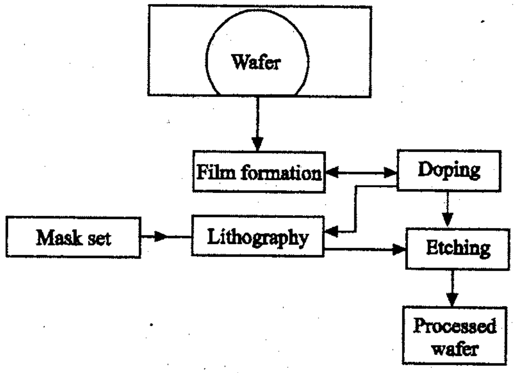


Fig. 5.20 Processing steps in semiconductor technology

- 'Doping' (imparting impurity) is done usually by ion implantation or diffusion.
- At this stage, the mask patterns are transferred to the film surface by lithographic process.
- The unwanted film and, substrate parts are then removed by 'etching'.
- The process may be repeated for n number of times for transfer of n mask patterns.
- A finished wafer would contain thousands of identical chips (features) which are then separated by diamond sawing or laser cutting.

- Single crystal and polycrystalline silicon have been grown on insulator surfaces such as sapphire (silicon-on-sapphire (SOS)) and SiO_2.
- GaAs can be grown on silicon by epitaxy.
- This process is important as optical sensors can be developed in this way.
- Oxidation of Si wafers can also be employed as it passivates the wafer surface and serves as diffusion and ion implantation masks.
- 'Oxidation' can be dry (in dry oxygen) or wet (in steam vapor).
- 'Lithography' transfers the pattern desired to a layer of resist which transfers the pattern to the films or substrates through etching.
- Resist is the radiation sensitive material.

Lithography can be classified as

(i) Photolithography (with optical radiation)

(ii) X-ray lithography (with X-radiation)

(iii) E-beam lithography (with electron beam), and

(iv) Ion-beam lithography (with ion-beam as radiation).

Etching

It is essential for surface polishing, removing contamination, drawing pattern, and opening windows in the in-between insulator (SiO_2, say) and fabrication, specifically three dimensional features by micromachining techniques.

Substrates used for etching are Si, GaAs, metals and insulators. Etching is of two types: Wet and dry.

Diffusion and ion implantation

These are the two very important processes by which dopant impurity atoms are introduced in controlled quantities into the selected regions of the wafer, to make the semiconductor substrate regions n or p-type. Selectivity is ensured by masking the top surface of the wafer impurities.

5.3.2 Microelectromechanical Systems (MEMS)

- MEMS are basically miniature devices on a silicon chip which have found a major use in sensors.
- In UK and the European continent, these are often referred as microsystem technology (MST).
- This is termed as micro engineering and the terms micro machining and micromechanics are associated with it.

5.3.6 Micromachining: (See Fig. 5.21)

Micromachining can be done in many ways. More important ones include:

(a) Bulk micromachining

- There are differences in etch rates between the crystallographic directions of silicon with particular etchants.
- Using this property, features can be fabricated in particular crystal planes.
- The substrate is masked by SiO_2 when ethylene diamine pyrocatechol is used as etchant, or Si_3N_2 is used for KOH as the etchant.

(b) Surface micromachining

- Differences between the etch properties of polysilicon and SiO_2 are used for feature development.
- The process is based on CMOS technology.
- Polysilicon layer is deposited on top of SiO_2 and then etched.
- The thickness of the deposited layer is limited to a few microns only.

(c) LIGA

- A process known as LIGA from the words **LI**thographic, Galvanoformung, Abformung, is an alternative to the process of surface micromachining.
- It uses the lithographic exposure of thick photoresist and then electroplating is carried out for building mechanical parts.

- This process fabricates thicker structures than that by surface micromachining.
- Lasers and UV sources have been used when the penetration depths are limited to 200 μm and 20 μm respectively.

(d) DRIE of BSOI

- A new process in development is based on bonded silicon-on-insulator (BSOI) where siliconwafer is thermally bonded to an oxidized silicon (SiO_2) substrate.
- The bonded wafer is polished to the desired thickness, between 5 μm and 200 μm, and the etching is done by Deep Reactive Ion Etching (DRIE).

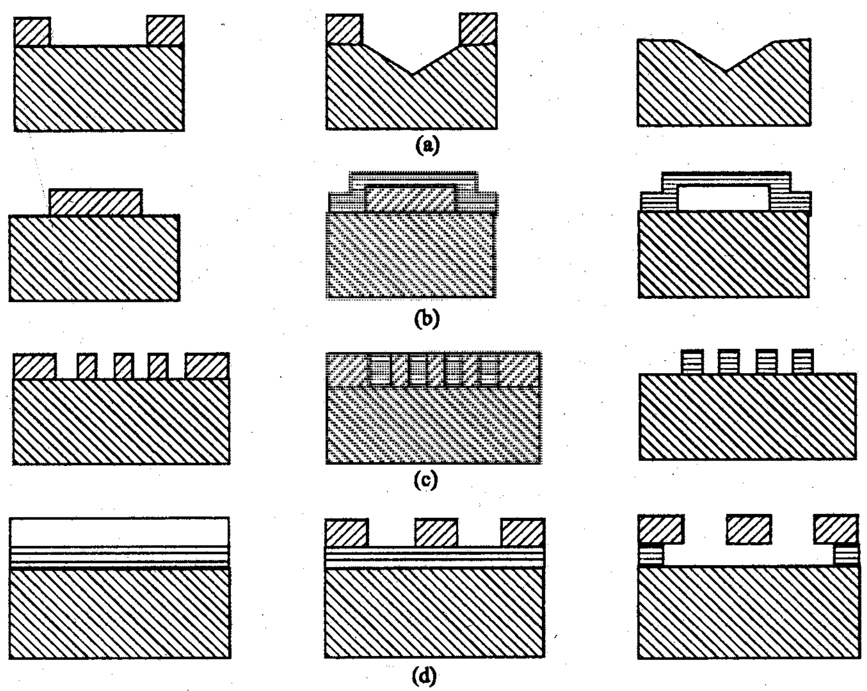


Fig.(5.21) (a) Bulk micromachining, (b) Surface micromachining, (c) LIGA, and (d) DRIE of BSOI

5.3.7 Nano-Sensors

- Microelectronics naturally leads to nanoelectrons for realizing nano-devices which are expected to create an impact in the enhancement of energy conversion, control of pollution, production of food, and improvement in the conditions of human health and longevity.
- While progressing towards the development of fast and miniaturized memory structures, giant magnetoresistance structures have been produced using Thomson effect.
- These giant magnetoresistance (GMR) structures consist of layers of magnetic and nonmagnetic metal films where in the critical layers have thickness of the order of nanometers.
- They are used as extremely sensitive magnetic field sensors.
- Organic nanostructures have been developed combining chemical self-assembly, with a mechanical device.
- The organic sample is reduced to a size that consists of a single molecule and this is connected by two gold leads.
- This structure has been successfully used to measure the electrical conductivity of a single molecule.
- Fig. 5.22 (a) shows the microstructure, while Fig. 5.22 (b) shows the operation mechanism of a GMR.

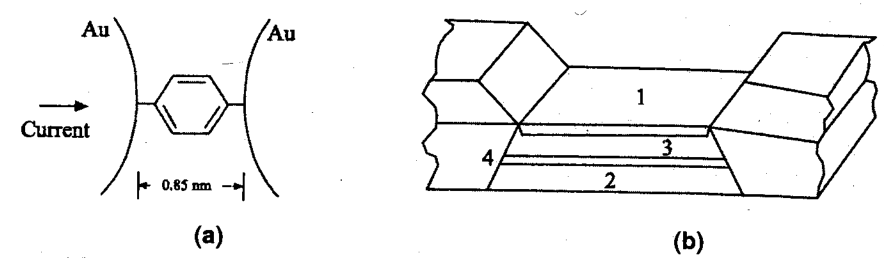


Fig. 5.22 (a) The microstructure (b) Operation scheme of a GMR

1. Antiferromagnetic exchange film
2. Ni-Fe GMR free film
3. Co-GMR pinned film
4. Cu-Spacer

5.4 DIGITAL TRANSDUCERS

Transducers dealt with so far are analog transducers whose output signals are in analog form. The ease and versatility provided by digital signal processing circuits and digital computers necessitates the development of digital transducers providing digital output signals directly. As there are only a few such digital transducers, the analog outputs of analog transducers are converted into digital signals using analog-to-digital converters. With the increasing application of digital computers, digital transducers that are compatible with the digital nature of the computer are under development. Direct digital transducers provide output signals in the form of rectangular pulses of constant duration and amplitude, the presence or absence of which in its time slot is taken to stand for either 1's or 0's. However, transducers are treated as digital type, if they provide pulses whose pulse rate is counted.

Similarly, transducers whose output signals are sinusoidal and the frequency of which is related to measurand are considered to be digital type when working in combination with digital frequency measuring system. Such transducer systems may be treated as indirect digital type.

5.4.1 Digital Displacement Transducers

- One of the direct digital transducers is the digital encoder for linear and angular displacements. It is also known as linear or angular digital encoder (LDE or ADE).
- Such transducers are available in different sizes with differing resolution and accuracy.
- Basically they are divided into two types
    1. Incremental encoder
    2. Absolute encoder

1. Incremental encoders

These encoders require a counting system which adds increments of pulses generated by an encoder, a sensing system and some datum from which increments are added or subtracted.

2. Absolute encoder

These encoders present a digital readout for each angular position and do not require a datum.

All encoders require a sensing system of either the contacting type using brushes or the noncontacting optical technique.

The encoders shown in figs. (5.23) and (5.24) consist of two distinct regions signifying the two logic level signals, 0 and 1.

- The linear encoder of fig. (5.23) for the contacting type has a pattern of metallic areas on a matrix of nonconducting areas.
- All the metallic areas get connected together and energized through a fixed brush that rests on a continuous track and is in contact for all positions.

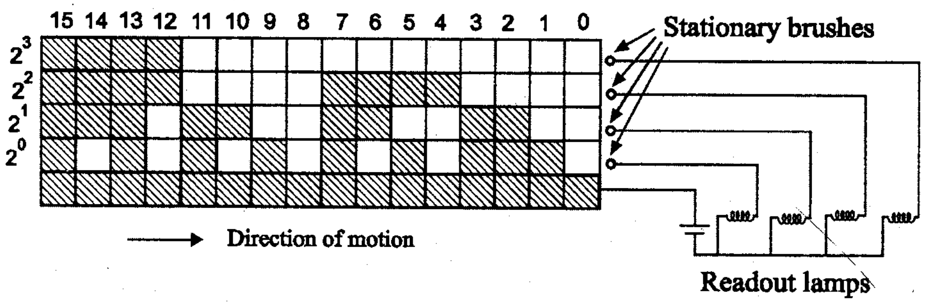


Fig. 5.23 Linear digital encoder (LDE)

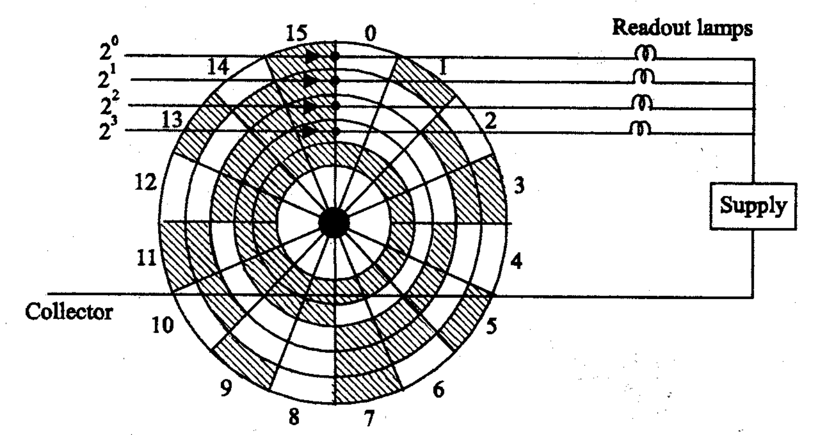


Fig. 5.24 Angular digital encoder (ADE)

- The encoder shown has four tracks, resulting in digital output in four bits.
- The scales and discs shown in figs. (5.23 and 5.24) are encoders providing digital outputs in four bits.
- The angular digital encoder of fig. 5.24 is also known as shaft angle encoder and is normally meant for a total angular displacement of 360°.
- Both the encoders shown are absolute encoders.
- Linear digital encoders (LDE) may also be obtained by converting linear motion into rotary motion through a rack and pinion or some such arrangement and using the shaft-angle encoders.
- Simple arrangements using a pulley or a cable are shown in fig. 5.25.

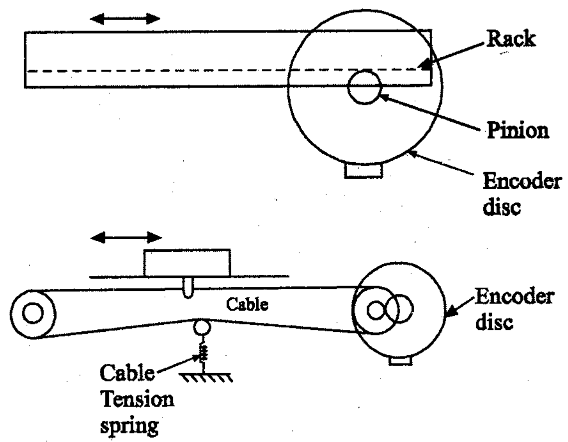


Fig. 5.25 Linear digital encoders using ADE.

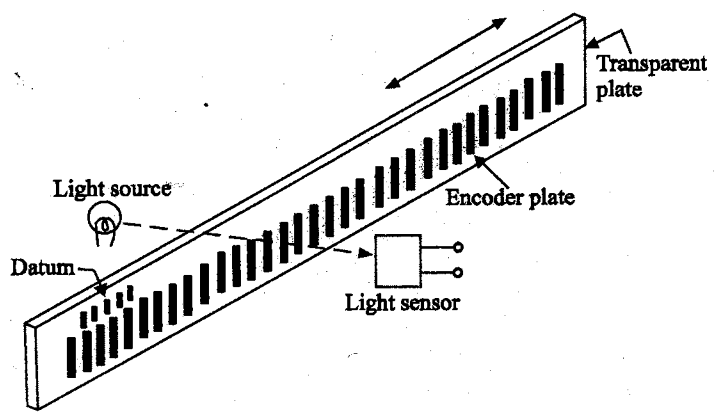


Fig. 5.26 Incremental digital encoder

Incremental encoders are single track discs or scales provided with alternating conducting and nonconducting areas as shown in fig. 5.26.

- All incremental encoders are designed to generate a fixed number of pulses for each unit of angular or linear displacement of the encoder.

5.4.1.1 *Optical encoders*

- The majority of shaft-angle encoders use noncontacting type sensing systems so as to make the measurements free from the problems of the brush contact.
- Optical encoders use optical and photoelectric sensing systems.
- The linear and angular encoders have a pattern of transparent and opaque areas corresponding to the conducting and nonconducting areas respectively of the contacting brush type.

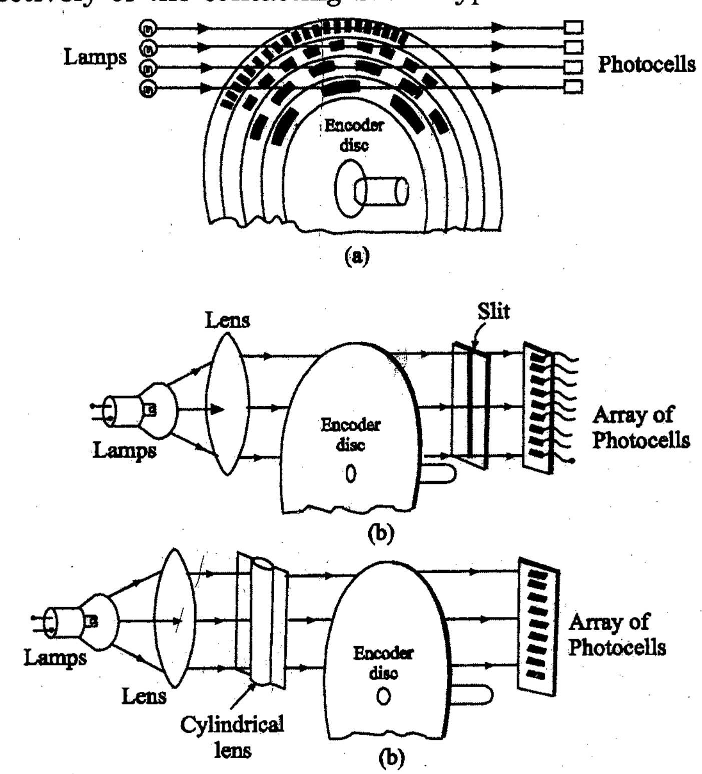


Fig. 5.27 (a) Optical encoder; (b) Arrangement of light sources and photosensors

- The sensing system consists of light sources, each provided with a focusing lens and an equal number of photoelectric devices, and receiving the light beam from its corresponding light source.
- The light sources are kept on one side and the photosensors on the other side of the encoder as shown in fig. 5.27 (a).

Instead of having a large number of light sources, a single lamp and a lens is used as shown in fig. 5.27 (b) to flood the encoder on one side, while the sensors receive light through a narrow slit located accurately with respect to the reference line.

- Alternatively, a cylindrical lens produces a single line beam which is so projected on to the reference line of the disc as to be incident on the sensors, after passing through the disc.

5.4.1.2 Magnetic encoders

- In this type of encoders, magnetic tape with magnetised portions and non magnetised portions is moved over sensing heads.
- The sensing heads have toroidal cores.
- Each toroidal core has two coils namely reading coil and interrogate coil.
- The interrogate coil is energised with a constant voltage of 200 KHz signal.
- The reading coil develops output signal due to transformer action only when the toroidal core is against the nonmagnetised portion.
- When the core is against the magnetised portion no voltage is developed because the core is saturated.
- A schematic diagram of the arrangement is given in fig. 5.28.

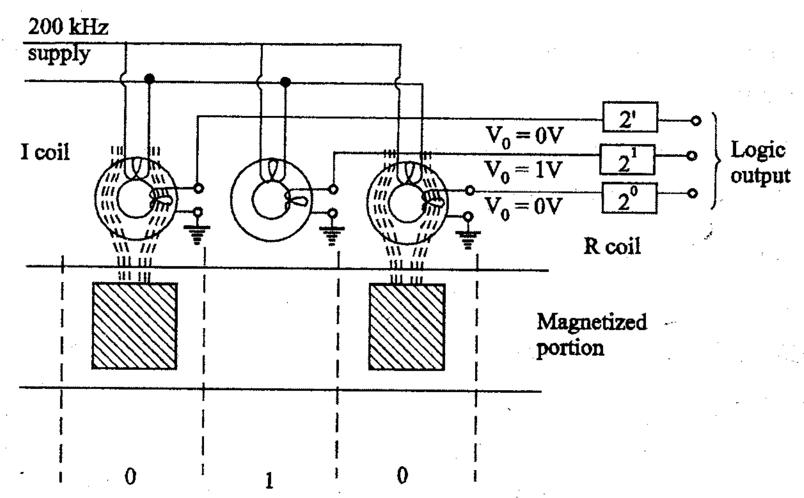


Fig. 5.28 Magnetic encoder

5.4.2 Digital Speed Transducers

(a) Variable reluctance type

(b) Variable capacitance type

(a) Variable reluctance type

In variable reluctance type of transducer, a rotating shaft is attached with a toothed rotor, which provides a variable reluctance in a magnetic circuit.

- This trandcuer is shown in fig. 5.29

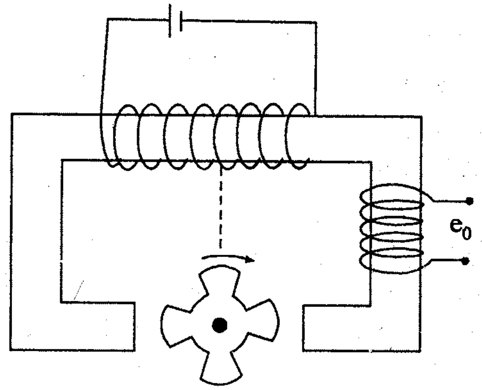


Fig. 5.29 Variable reluctance speed sensor

- When the teeth are against the stator poles the reluctance is less and hence e_0 is more.
- When the slot of the toothed shaft comes against the stator pole, the reluctance is high and hence the voltage induced across e_0 is small.
- Whenever the teeth crosses the pole a voltage pulse appears across e_0.
- By counting the number of pulses per second, we can determine the speed.
- The output of the transducer is a series of pulses, this can be interfaced with any digital equipment.

(b) Variable capacitance type

- The variation of the capacitance between a probe plate and a toothed rotor may be used to generate pulses.
- The number of pulses per second is equal to the rotor speed and the number of teeth in the serrated rotor.
- By counting the number of pulses by suitable counters, a digital readout proportional to the speed can be designed.

5.5 SMART SENSORS

A sensor producing an electrical output when combined with interface electronic circuits is said to be an intelligent sensor if the interfacing circuits can perform (i) ranging (ii) calibration and (iii) decision making for communication and utilization of data.

Both sensors and actuators are used as intelligent components of instrumentation systems. In fact they are used as field devices. The block diagram of one such intelligent equipment is shown in fig. 5.30.

Fig. 5.30 shows the simplified version with facilities of processing that can be incorporated.

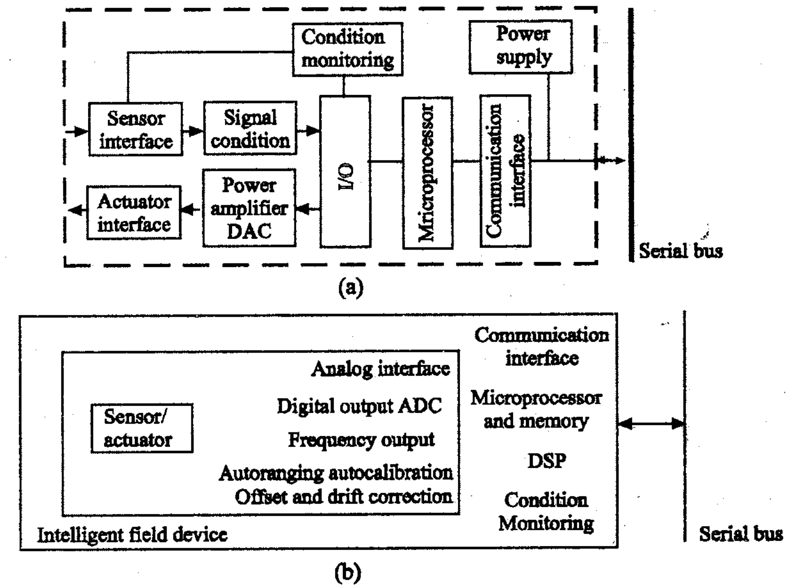


Fig. 5.30 (a) Typical intelligent sensor and actuator and (b) Simplified version of (a)

Properties of intelligent field device

1. Automatic ranging and calibration through a built in digital system.
2. Auto-acquisition and storage of calibration constants in local memory of the field device.
3. Autocorrection of offsets, time and temperature drifts.
4. Autoconfiguration and verification of hardware for correct operation following internal checks.
5. Auto linearization of nonlinear transfer characteristics.
6. Self-tuning control algorithms, fuzzy logic control is being increasingly used now.
7. Control programme may be locally stored or downloaded from a host system and dynamic reconfiguration performed.
8. Control is implementable through signal bus and a host system.
9. Condition monitoring is also used for fault diagnosis which, in turn, may involve additional sensors, digital signal processing and data analysis software.
10. Communication through a serial bus.

Intelligent sensors are also called smart sensors. The initial motivation behind the development of smart sensors include processing and bus interfacing for communication.

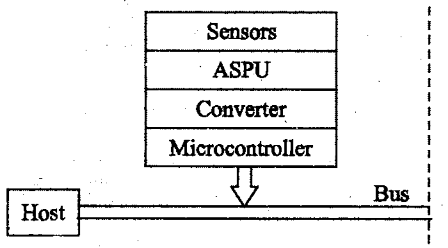


Fig. 5.31 A sensor interfaced with a host system

Fig. 5.31 shows a sensor interfaced with a host system.

5.5.1 Primary sensors

Existing sensors of all kinds with a cascaded block for providing electrical output in the form of voltage or current can be adapted to an integrated processing system but the system can then be called a smart sensor.

External stimuli such as strain/stress, thermal/optical agitation, and electric/magnetic field change the behavior of materials at atomic/molecular level or in crystalline state. This concept is utilized in designing a primary sensing element for particular stimulus or a specific physical variable.

5.5.2 Excitation

Excitation is a generalized term used for supply to the primary sensors and the processing units.

(a) Compensation for the non-ideal behavior of the sensors and (b) Provision for communication of the process data with the host system. Traditional sensors that are being used, have varying requirements of compensation and signal processing objectives.

Thus, for each type of variable a different kind of processing is required. The smart sensor is intended to sense as well as do the sensing-related processing within itself. Further, it communicates the response to the host system so that the efficiency and accuracy of information distribution are enhanced with cost reduction.

Certain sensors require supply, constant voltage or constant current along with comparison capabilities; the feature is included in sensor subsystem. Amplification is necessary which usually analog, may also be controlled digitally.

Analog filters were employed which have now been replaced by digital counterparts.

These three systems, namely the supply, amplification, and filters, comprise the Analog signal processing unit (ASPU). Smart sensor also requires a data conversion module either from analog to digital (A/D) or from frequency to digital (F/D) which interfaces with the microprocessors for information. This supply may be required to provide different output to different stages of the system. In the thermocouple form of sensors, no excitation to the sensors is needed while for resistive bridge, an extremely stable supply is required. In stages of electronic processing units, ac supply or else pulsed form supply may be required for phase sensitive detection in the processor unit.

5.5.3 Amplification

As the output of the sensors are small, amplification is essential in all smart sensors. If the gain requirement is very high, noise becomes a problem. However, stage wise approach with adequate compensation realizes the requirement, the design and layout being critical.

5.5.4 Filters

Analog filters are often used as the digital type consume large real time processing power.

5.5.5 Converters

- Conversion is the stage of internal interfacing between the continuous and the discrete processing units. Often, controlled conversion through software is provided with range selection and so on.
- Data conversion from analog amplitude to frequency is often done for convenience of signal transmission, internally or externally and/or for subsequent digital conversion.

- Voltage-controlled oscillators are used for these purposes. One such converter is a multivibrator shown in fig.(5.32). Analysis shows that the time period of the generated square wave is given by

$$T = 2\,RC\,\ln\left(1 + 2\frac{R_2}{R_1}\right) \quad \text{... (5.8)}$$

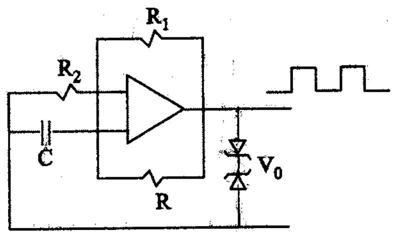


Fig. 5.32 A multivibrator

The parameters R and C can be related to the input voltage. Fixing R_2/R_1 at 0.859, T is obtained as

$$T = 2\,RC \quad \text{... (5.9)}$$

or, frequency f is given by

$$f = \frac{1}{2\,RC} \quad \text{... (5.10)}$$

- The capacitance or resistance may be the sensed instead of the input voltage or measurand/sensor output voltage.

5.5.6 $V-f$ Converter

Ring oscillator realized with MOS technology is one popular $V-f$ converter (or signal to frequency converter).

- A $V-f$ converter which consists of an odd number of cascaded NOT, NOR, or NAND gates with its last gate-output fed back to the first stage to form the ring.
- With the gain of each stage greater than one, the circuit is self-oscillatory with the frequency determined by the number of gates and their delays.
- Supply frequency and chip temperature need be controlled on which also depends the frequency.

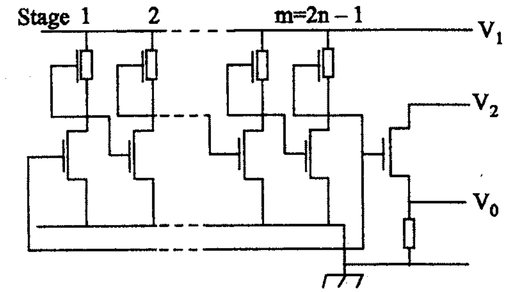


Fig. 5.33 An integrated ring oscillator

- If the MOS channel resistance is a piezoresistance whose value may be made dependent on the pressure exerted on it; this would change the gate delay and there is a frequency change.
- Supply frequency and temperature changes are usually compensated by using two ring oscillators and the ratio of-two frequencies is taken as the output.

5.5.7 Frequency to digital conversion

- In digital conversion, frequency from the sensor oscillator is counted by actually counting clock pulses in a pulse-width of the oscillator.
- Typical digital conversion is shown in fig.(5.34).
- Over the time period $T_x = 1/f_x, f_{ref}$ would be counted; dividing f_x by a suitable factor n, this time interval is suitably increased to obtain a better resolution.
- The resolution, R_n is given by

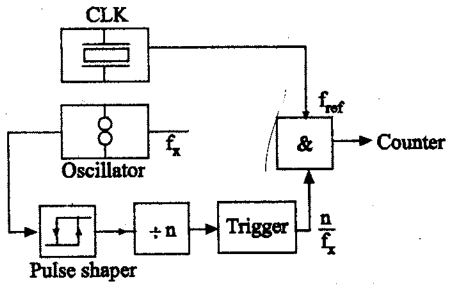


Fig. 5.34 A typical digital conversion method

$$R_n = \frac{1}{n}\left(\frac{f_x}{f_{ref}}\right) \qquad \text{... (5.11)}$$

where $1/R_n$ is the actual count.

5.5.8 Compensation

Compensation is an attempt to counter all sorts of nonideality in the primary sensor characteristics as well as environment of measurement. The common defects of sensor are:

1. Non linearity
2. Noise
3. Response time
4. Drift
5. Cross sensitivity and
6. Interference
7. Data communication

1. Non linearity

- Analog processing shows serious nonlinearity which at one time, was solved by piecewise linear segment approach modelled by linear electronic circuits.
- A very common technique in use is to refer the look-up tables while other are polygon interpolation, polynomial interpolation, and cubic splines interpolation techniques of curve fitting.

2. Noise and Interference

- Thermal noise is important in almost all sensors.
- Besides, there are other unwanted signals that may be picked up due to external magnetic fields (sort of an interference) when the structure is not adequately screened.
- Noise is also introduced at different stages of signal processing such as data conversion, analog to digital interfacing by stray effects.

- The methods of minimization of noise are appropriate signal conditioning techniques that include filtering, signal averaging, and correlation among others.
- If the signal is periodic as in the case of the output of the frequency converter, the correlation technique improves the signal-to-noise ratio by a large value, the ratio by a large value. This is due to the superposition property of autocorrelation.
- Again, if the input is corrupted at any stage by noise, specifically white noise, a cross correlation technique can be used to obtain the system response/function without this corruption.

3. Response time

Because of the presence of storage and dissipative elements, a sensor is likely to have quite inferior time response characteristics and the dynamic correction of sensor becomes necessary.

This is possible with the use of microprocessors/micro computers with suitable algorithm if the dynamic parameters are known through solving the convolution integral.

4. Drift

- Drift appears in a sensor because of slow changes in its physical parameters either due to ageing or deterioration in ways of oxidation, sulphation, and so on.
- Drift is a kind of noise and should be counteracted.
- As drift tends to change the sensor characteristics, the reference points for polynomial interpolation also tend to drift.

5. Cross sensitivity

- A sensor, while responding to a specific variable, responds to others as well, may be, with much less sensitivity.
- It is therefore necessary to maximize the sensitivity for the desired measurand and minimize that for the others.
- The compensation is made through devising algorithm by monitoring the change in response characteristics because of any interfering quantity, is quite common as it is possible to develop the algorithm

from measured data. Such a compensation is called as monitored compensation.

- Other compensations are tailored compensation and deductive compensation.

6. *Information coding / Processing*

- The signal from a sensor is processed providing correction, compensation, linearization, freedom from cross-sensitivity and drift.
- Such a processed signal is finally made available in digital form and perhaps in a serial form.
- The smart sensors are generally multi-sensor systems and a number of signals are available for either display or further processing.
- Information, the state of the process in the form of a processed signal through sensor and signal processing systems, is first received by the information coding system.
- Some of these signals are released, some stored and some destroyed.
- For indication purposes only, the signals are coded and displayed over appropriate display modules as is done in digital meters, indicators & recorders.

The fig. 5.35 shows a typical IC temperature sensor-based smart sensor.

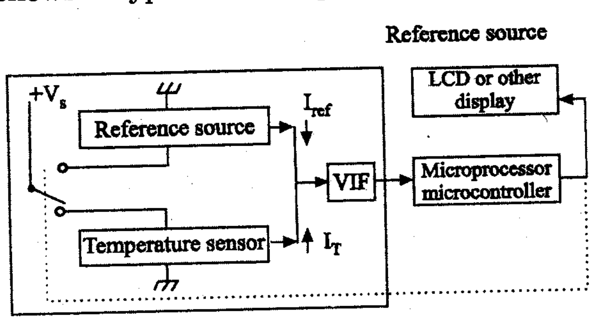


Fig. 5.35 A typical IC-temperature based smart sensor.

- Information processing assembly in a smart sensor is basically an encoder, the encoded data from this are fed to the communication unit.
- The conventional signal processing provides an output of 4 – 20mA.

- Voltage to frequency converter is another kind which is quite extensively used (see fig. 5.33), then using a reference frequency generator, frequency difference encoding is employed.

7. *Data communication*

- Data communication is essential in smart transmitters where the sensor outputs are communicated with the host through bus system.
- Coded data are processed for communication by a software processor and a suitable interface system communicates between the processor and the bus.
- Each smart sensor/transmitter has always been provided with a local operating system in a ROM, that consists of an application programme and library modules, for ADC and DAC hardwares, bus driving hardware, local interface hardware and LCD/keyboard hardware.
- A typical transmitter with HART protocol appears as the one shown in fig. 5.36.
- Some other protocols that find use are High Level Data Link Control (HDLC), Synchronous Data Link Control (SDLC), Factory Instrumentation Protocol (FIP).

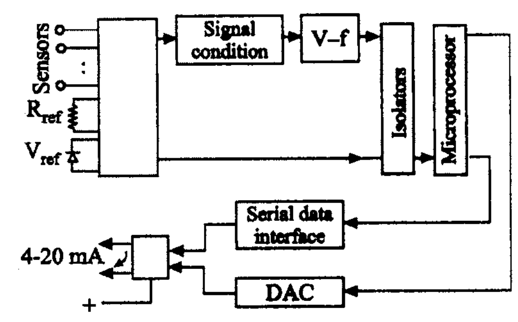


Fig. 5.36 A smart transmitter

- The HART protocol has been designed for direct use of 4 – 20 mA output device having facilities of digital communication with superimposed modulation between the field device and a host system.
- Such devices can be connected in parallel.
- The addressing procedure allows each unit to set its output for power supply at 4 mA and the device is forced to communicate only digitally.

- The basic multiloop connection method is presented in fig. 5.37 & fig. 5.38 shows the hardware requirements for microprocessor-based field devices.

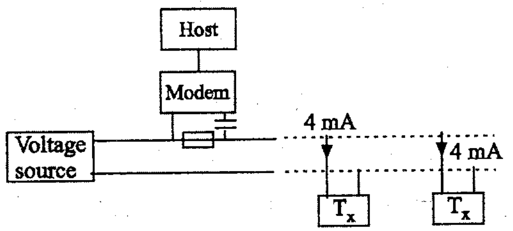


Fig. 5.37 The basic multiloop connection

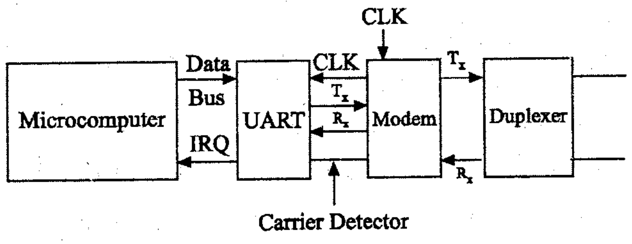


Fig. 5.38 Demonstration of hardware requirement of an intelligent field device

- Frequency shift keying (FSK) is used for coding digital information.
- Logic 1 is represented by 1200 Hz and 0 by 2200 Hz both with sine wave of amplitude 0.5 mA.
- Data rate is 1.2 Kb/s. The implementation of this digitally signalling technique can be done by using a modem of telephony standard.

5.6 FIBRE OPTIC TRANSDUCERS

- Fibre optic sensors could be classified as a separate group of sensors.
- They are considered for sensing different types of variables such as temperature, liquid level, fluid flow, magnetic field, acoustic parameters, and so on.
- However, optical radiation happens to be the energy source in these applications with the fibre acting as medium as well as a sensor.
- Optical fibres are basically considered as communication channels.
- Optical transmission is affected by external parameters/stimuli such as temperature, acoustic vibration magnetic field and many more.
- Fibre has been divided into two groups:

1. Active

The fibre is exposed to the energy source that affects the measurand and a consequent change in the optical propagation in the fibre is detected and related to the measurand.

2. Passive

Light transmitted through a fibre, called input fibre, is first modulated by a conventional optical sensor and this intensity-modulated light is propagated through a second fibre called the output fibre and detected and corrected with the measurand.

5.6.1 Temperature measurement

- Two identical optical fibres are used to propagate radiation from a source (a laser source)
- If one of these fibres is in a medium with temperature different than that of the other, the optical outputs from the two fibres would have a phase difference which is a function of the difference of temperature.
- This phase difference is due to optical path length variations in the two paths occurring due to temperature difference.
- This phase difference is so small that it can only be measured by producing interference patterns.

(a) Phase difference method

- He-Ne laser is the source.
- The detector is Mach-Zender interferometer.
- Beam-splitter (BS) and mirrors (Mi) are used.
- Two identical optical fibres (Reference path fibre and measuring path fibre) are used to propagate radiation from a He-Ne laser source.
- The laser beam is split by Beam splitter (BS) and made to travel through both reference path fibre and measuring path fibre.
- The measuring path fibre is exposed to the temperature to be measured.

- Due to the difference in temperature, the optical outputs from these two fibres would have a phase difference which is a function of the temperature difference.
- The detector will detect the phase difference of the optical outputs from these two fibres.

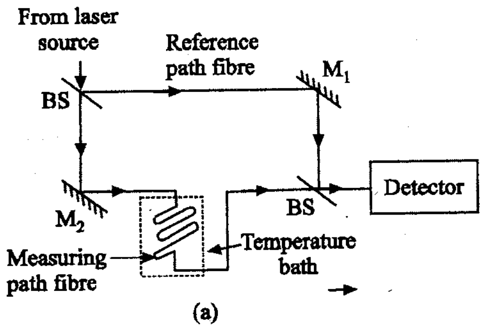


Fig. 5.39 Temperature measurement using optical fibres (a) Phase difference method

Technique using fibre couplers (avoiding beam splitter and mirror)

- He-Ne laser is the source.
- The detector is Michelson interferometer.
- Instead of Beam splitters (BS) and mirrors (Mi), 3 dB-fibre couplers are used.
- The reference path fibre and measuring path fibre are coupled by 3 dB fibre couplers.
- The He-Ne laser beam is propagated by both the fibres.
- As the measuring path fibre is exposed to temperature to be measured, the phase difference in the optical outputs due to temperature difference is detected by a detector system.

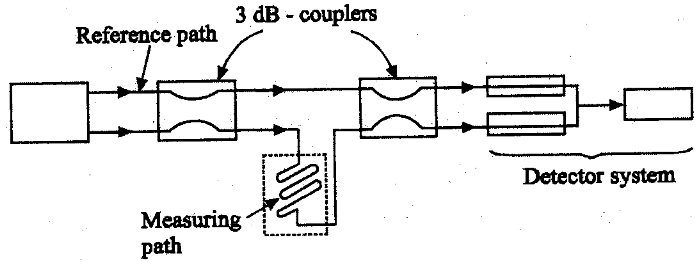


Fig. 5.40 Temperature measurement using optical fibres (b) Using fibre couplers

(c) *Black body method*

- This method of temperature measurement is based on the principle that a black body cavity changes radiance with varying temperature.
- Thus, at the end of a fibre a black body cavity is formed.
- The fibre is a high temperature fibre, usually a sapphire fibre, of diameter 0.25 – 1.25 mm.
- A thin film of iridium is sputtered onto the end-surface and a protective cover of Aluminium oxide (Al_2O_3) is then provided.
- This measuring fibre has a length usually within 0.3 m and not less than 5 cm.
- This propagates the radiation from the formed cavity which is being heated by heat of the process.
- At the propagation end, another fibre, a low temperature fibre made of glass of about 0.6 mm diameter is coupled that has a length usually within 10 m.
- The detector system consists of one lens and two narrow band filters of close range middle wavelengths, two photomultiplier tubes in two measuring channels fed by a beam-splitter and a mirror.

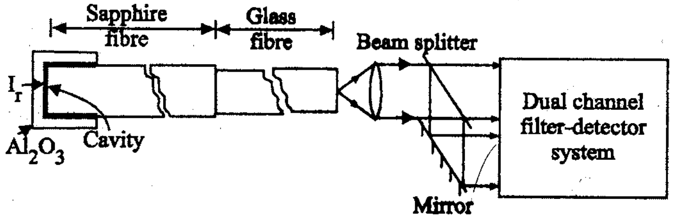


Fig. 5.41 Temperature sensor fibre black body cavity

- The filters have wavelengths of 600 and 700 nm respectively with a spread at the centre of 0.1 μm.
- The two channels are used to measure temperature by comparison over a range 500 – 2000°C.
- With an input power of 0.1 μw, for 1°C change there occurs 20% optical flux change and the system has a resolution of 1 in 10^8.

- This system is now being used as a temperature standard between 630.74 and 1769°C which are aluminium and platinum points respectively.

(d) Temperature measurement using backscatter in optical fibre

- Optical fibre can be used for distributed temperature sensing.
- Optical pulse from a pulsed laser source is sent along a fibre over a distance convering a few kilometres.
- Any localized change in temperature somewhere along the fibre changes its backscattered intensity ratio (Stokes/anti-stokes Raman).
- This backscattered light is filtered and Raman components are detected by photodetectors from which the temperature can be known.
- From the pulse delay time, the location can also be identified.
- Resolution of 1° K and 2-3 metres can be obtained in this system.

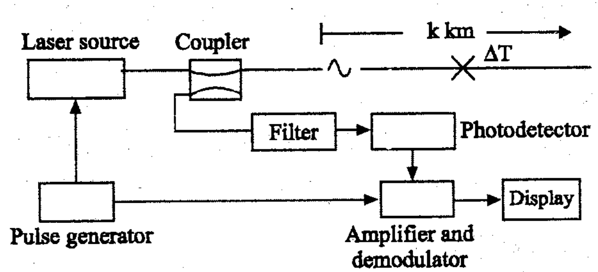


Fig. 5.42 Temperature sensing using backscatter in optical fibre.

5.6.2 Liquid Level Measurement

- Usually, light propagates through a fibre by total internal reflection with appropriate cladding or even without that, if the light incidence angle is properly chosen.
- This is because the refractive index of air is such, with respect to that of the fibre, that no refraction can take place.
- If, however, the fibre is placed in a liquid medium of a different refractive index, it is possible that light refracts into the liquid and total internal reflection inside the fibre stops, stopping light propagation in it.

This principle is utilized in measuring liquid level at specific values as shown in fig. 5.43.

Single position level detection

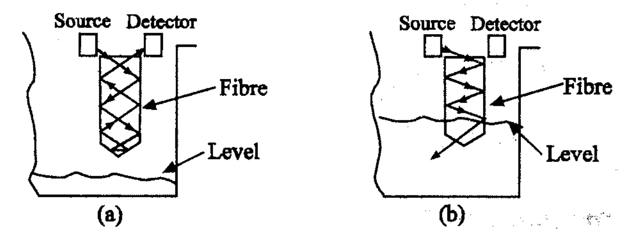


Fig. 5.43 Level detector using optical fibre
(a) Level below sensor and (b) Level covering sensor

- The bottom end of the fibre is shaped like a prism so that with large difference in refractive indices of the fibre and the medium like air, there is internal reflection and the light travels to be detected as shown in fig. 5.43 (a).

When liquid level rises to cover the bottom of the fibre, light refracts into the liquid and the detector fails to show any output, as shown in fig. 5.44 (b).

Multistep level detection

- This single position level detection has been extended for discrete multistep detection covering the entire height of the tank.
- In this, a step-index multimode fibre is used and the fibre goes down carrying the light but in the return upward path, its cladding is exposed and the fibre is also given a zig-zag rise with small bend radius at regular intervals in length.
- When no liquid is there, cladding mode operation continues and a detector at the end of the return path of the fibre shows full intensity.
- But with liquid rising in the tank, refraction of light into liquid occurs at each bend and the intensity detected by the detector becomes less.
- Thus, for n bends there would be n-stepped intensity of signal, reducing in steps with rising liquid.

Fig.(5.45(a)) shows the system and fig.(5.45(b)) depicts the intensity versus height plot.

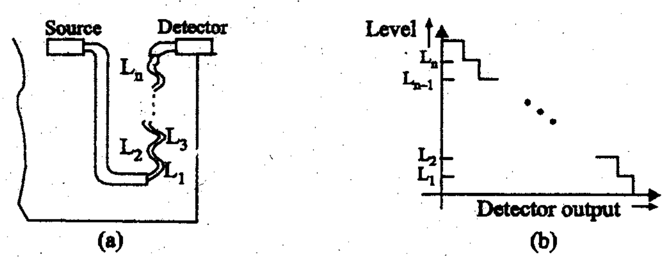


Fig. 5.45 Liquid level sensing in steps

5.6.3 Fluid flow measurement

- Fluid flow rate has been sensed by an optical fibre mounted transversely in a pipeline through which it flows.
- Because of the fibre, mounted across the flow, vortex shedding occurs in the channel and the fibre vibrates, which in turn, causes phase modulation of the optical carrier wave propagating through the fibre.

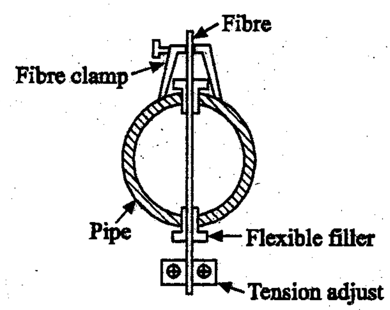


Fig. 5.46 Fluid flow sensing using fibre optics

- The vibration frequency is proportional to the flow rate.
- Using multimode fibres of core diameter 0.2 – 0.3 mm and special detecting techniques, flow rates over a range of 0.2 – 3 m/s can be measured.
- Fig.(5.46) shows the scheme to sense fluid flow.
- The fibre is kept under tension by a tension adjusting system and a fibre clamp.
- Flexible fillers are often used for small adjustment of tension.

5.6.4 Acoustic Pressure Transducer

- Acoustic pressure sensing can be done by the microbending of a multimode fibre.
- Fig. 5.47 (a) and (b) show how light loss occurs in microbends of a fibre.
- The technique is utilized as shown in fig.(5.48)

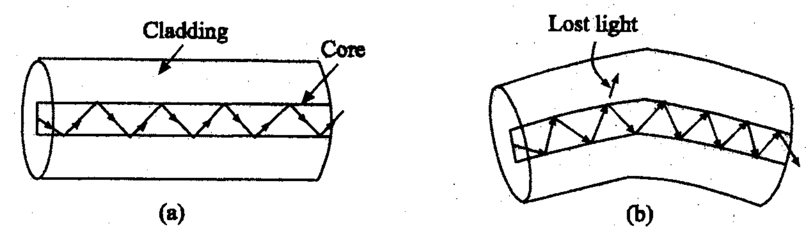


Fig. 5.47 Microbend sensors (a) Normal condition (no loss of light)
(b) Bent condition (Partial loss of light)

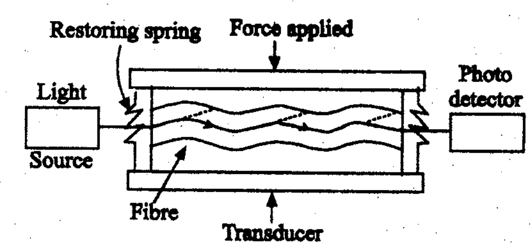


Fig. 5.48 Microbend force sensor using optical fibre

- Optical fibre is placed in two corrugated plates to form a transducer as shown.
- Applied force causes microbending in the fibre.
- Consequently, more light is lost and the receiver detector indicates less intensity.
- A calibration of force in terms of the intensity of detected light may also be made.

TWO MARK QUESTIONS AND ANSWERS

1. **What is piezoelectric transducer?**

   Piezoelectric converts pressure or force into electrical charge. These transducers are based upon the natural phenomenon of certain non-metal and dielectric components.

2. **What are the suitable materials for piezoelectric transducer?**

   Primary quartz, Rochelle salt, ammonium dihydrogen phosphate (ADP), and ceramics with barium titanate, dipotassium tartrate, potassium dihydrogen phosphate and lithium sulphate are the suitable materials for piezoelectric transducer.

3. **What is 'd' coefficient?**

   'd' coefficient gives the charge output per unit force input (or charge density per unit pressure) under short circuit condition. It is measured in Coulomb / Newton.

4. **What is 'g' coefficient?**

   'g' coefficient represents the generated emf gradient per unit pressure input.

   Its unit is $\dfrac{\text{V/m}}{\text{Newton/m}^2}$

5. **What is 'h' coefficient?**

   'h' coefficient is obtained by multiplying the 'g' coefficient by Young's modulus valid for the appropriate crystal orientation of the material, and thus measures the e.m.f gradient per unit mechanical deformation, or $(V/m)/(m/m)$

6. **What are the suitable materials for magnetostrictive transducer?**

   Iron, nickel, 68 permalloy, ferroxcube material are used in magnetostrictive transducer.

7. **What is magnetostrictive transducer?**

   The permeability can increase or decrease depending upon the material, type of stress and the magnetic flux density in the sample.

8. **What are the different magnetostrictive transducer?**

   The various types of magnetostrictive transducer are,

   - Magnetostrictive load cell.
   - Magnetostrictive accelerometer.
   - Magnetostrictive phonographic pickup.
   - Magnetostrictive torque transducer.

9. **What are the errors in magnetostrictive transducer?**

   The errors caused in magnetostrictive transducer are,

   - Hysteresis
   - Temperature
   - Eddy current
   - Input impedance.

10. **What are the special features of magnetostrictive transducer?**

    The special features of magnetostrictive transducer are,

    - It is used to measure large force.
    - It is used to measure several thousand 'g'.
    - Its characteristics depend upon temperature.

11. **Compare digital transducer with analog.**

    Digital transducer gives digital outputs. Analog transducer outputs are continuous functions of time. If these analog transducers are to be interfaced with digital devices, then one has to use analog to digital converters.

12. **How will you achieve high resolution in digital transducer?**

    In digital transducer, to achieve high resolution, the number of tracks must be increased and the length of each coded should be reduced, which would require fine brushes.

13. **What are the different digital transducers available?**

    The various digital transducers are,

    - Digital displacement transducer
    - Shaft angle encoder

- Optical encoder
- Magnetic encoder.

14. What is piezoelectric effect?

A piezoelectric material is one in which an electric potential appears across certain surfaces of a crystal if the dimensions of the crystal are changed by the application of the mechanical force.

15. Give the applications of piezoelectric transducer.

The applications of piezoelectric transducer are,

(a) Insensitive to temperature variation and high stability output. So piezoelectric materials are used in electronic oscillators.

(b) The use of piezoelectric transducer elements is confined primarily to dynamic measurements. The voltage developed by applications of strain is not held under static conditions. Hence, the elements are primarily used in the measurements of such quantities as surface roughness and in accelerometers and vibration pickups.

(c) Ultrasonic titanate, generator elements also use barium titanate, a piezoelectric material. Such elements are used in industrial cleansing apparatus and also in under water direction system known as SONAR.

16. List the modes of operation of piezoelectric crystals.

Piezoelectric crystals are operated in the following modes.

(a) Thickness shear

(b) Thickness expansion

(c) Face shear

(d) Transverse expansion.

17. List the applications of strain gauge.

The applications of strain gauge are,

(a) Primary application is stress strain analysis of structure.

(b) Fabrication of various types of transducers such as force, pressure, torque, load (weight) etc.

18. What are the advantages of semiconductor strain gauge?

The advantages of semiconductor strain gauge are,

(a) Semiconductor strain gauges have the advantages that they have a high gauge factor of about ± 130. This allows measurement of very small strains of the order of 0.01 micro strains.

(b) Hysteresis characteristics of semiconductor strain gauges are excellent. Some units maintain it to less than 0.05%.

(c) Fatigue life is in excess of 10×10^6 operations and the frequency response is upto 10^{12} Hz.

(d) Semiconductor strain gauges can be very small ranging in length from 0.7 to 7 mm. They are very useful for measurement of local strain.

19. Write notes on optical fiber.

An optical fiber is a hair line thin strand of glass or plastic having two or more layers, called core, cladding and insulators. This cables can transmit a wave of light of different colours without any loss using the principle of total internal reflection. The refractive index of core is much greater than cladding.

20. Write note on micro bend displacement sensor.

When a fiber is deformed into a convoluted shape, part of the light travelling through the fiber is lost to radiation. The loss of light is maximum when the convolution have a spacing given by,

$\lambda = \frac{2\pi}{\Delta \beta}$, where $\Delta \beta$ is the difference in propagation constants between propagating and radiation modes. For the aluminium coated multimode with 120 μ diameter, the optimum spacing was found to be 3 mm. In this sensor, the convolution spacing depends on the pressure. The light received by the detector varies according to the convolution spacing which is proportional to the pressure.

21. Write notes on fiber optic displacement sensor.

The set of fiber cables on used to measure the displacement of a target from one end of the cable. There are a set of transmitting and receiving cables. The light is send through one end of the transmitting cables which are opened at another end and face the target. These are reflected by the target and received and sensed through receiving cables. The intensity of the received light depends or inversely proportional to the displacement of target (distance).

22. Write short notes an optical encoders.

This transducer is used to measure the displacement of angular motion. The output obtained by four bits. A rotatable disc consists of conducting and insulating paths on which the numbers [from 1...16] encoded. When angular displacement applied that can measure by the output binary bit. This transducer can measure (0 – 360)°

23. List few IC sensors.

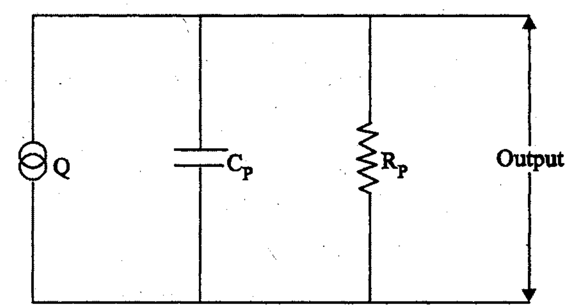
AD 592, AD 590, LM 335, LM 34 are some of IC sensors.

24. Explain about AD 592 IC sensor.

In case, the signal is to be transmitted over a large distance,

AD 592 is a better choice as its output is current signal which is not affected by the resistance of wiring.

25. Draw the equivalent circuit of piezoelectric crystal.



Equivalent circuit of piezoelectric crystal

26. What is meant by bimorphs bender?

Bimorphs bender consists of two transverse expanding plates cemented together in such a manner that one plate contracts and the other expands when a voltage is applied. If the element is free to move, then it will bend. Thus bimorphs can be used to transduce force into a voltage by using as a simply supported beam or cantilever beam. These bimorph elements has got a higher sensitivity and permits larger deflection than a single solid one.

27. What is meant by bimorph twisters?

Two face shear plates are cemented together to have a series connection so that their expanding diagonals are perpendicular. If a voltage is applied and if both plates are free to move then it will bend. For transducing torque, the bimorph twisters can be used.

28. Write short notes on magnetostrictive accelerometer.

This transducer used for the measurement of several thousands of grams which is applied on seismic mass. This force which is on the magneto elastic element changes the dimension, and change in permeability which causes the magnetization change and change voltage drop.

29. A platinum resistance thermometer has a resistance of 150 Ω at 0°C. When a thermometer has a resistance of 400 Ω, What is the value of temperature? The resistance temperature co-efficient of platinum is 0.0039/°C?

$R_0 = 150$; $\alpha = 0.0039/°C$; $R = 400\ \Omega$; $t_0 = 0°C$

$$R = R_0\,(1 + \alpha\,\Delta\,T)$$

$$400 = 150\,(1 + 0.0039\,(t - 0))$$

$$t = 427°C$$

30. A barium titanate crystal has a thickness of 2 mm. Its voltage sensitivity is 12×10^{-3} Vm/N. It is subjected to a pressure of 0.5 MN/m². Calculate the voltage generated.

$E = g.p.t$

$$= 12 \times 10^{-3} \times 0.5 \times 10^{6} \times 2 \times 10^{-3}$$

$$= 12\ V$$

31. What is digitiser?

Digital encoding transducer or **digitiser** enables a linear or rotary displacement to be directly converted **into digital** form without intermediate form of analog to digital (A/D) conversion.

32. What are the classifications of encoder?

Encoder is classified as,

(a) Tachometer transducer

(b) Incremental transducer

(c) Absolute transducer.

33. What are the input characteristics of the transducer?

The input characteristics of the transducer are,

- Type of input and operating range.
- Loading effect.

34. What is zero error of the transducer?

In this case, output deviates from the correct value by a constant factor over the entire range of transducer.

35. What are the different transfer characteristics of the transducer?

The transfer characteristics of transducer are,

(a) Transfer function

(b) Error

(i) Scale error

(ii) Zero error

(iii) Sensitivity error

(iv) Non-conformity

(v) Hysteresis

(c) Transducer response

36. What is magnetostrictive effect?

The permeability of a magnetic material changes when it is subjected to mechanical stress. This effect is called Villari effect.

When a magnetic field linked with a conductor changes, its permeability changes due to that dimensions of the crystal changes. This effect is called as magnetostrictive effect.

37. List few magnetostrictive materials.

Some of the magnetostrictive materials are,

- Nickel
- Permalloy - (Nickel alloy with 68% Nickel)
- Ferroxcube B (This is highly brittle).

38. Write brief notes on magnetostrictive load cell.

Load cell uses the principle of effect of magnetostrictive and uses the measurement of strain or force from several grams upto several tonnes directly. The displacement at the input of the transducer is very small (= micrometer). When a force of several grams applied, permeability of the material changes which increase the magnetic flux. This changes are directly calibrated in terms of strain.

39. List the applications of magnetostrictive transducer.

The applications of magnetostrictive transducer are,

- These transducers can be built to measure large forces upto several tonnes and for fast transient phenomena where frequency is of the order of several thousand cycles per second.
- The accelerometers can be built to measure several thousand grams.
- Can also used for the measurement of torque.